April 3th 2026

P.hD. Dissertation Defense
University of Texas at Austin
Austin, TX

Vestigia Cosmica:

Tracing Fundamental Physics Across Cosmic History

Gabriele Montefalcone

PhD Candidate

Weinberg Institute for Theoretical Physics, University of Texas at Austin

Dissertation Commitee:

Prof. Katherine Freese (Supervisor)
Prof. Kimberly Boddy (Co-supervisor)
Prof. Nicholas Galitzki
Prof. Julian Muñoz



Vestigia Humana



Prof. Katherine Freese



Prof. Kimberly Boddy



Prof. Julian Muñoz

+

Prof. Dan Hooper

Prof. Luca Visinelli

Prof. Ely Kovetz

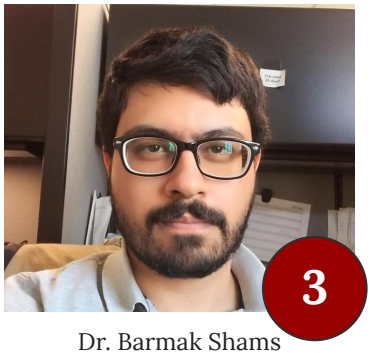
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Dr. Hector Cruz

Dr. Gilly Elor

Dr. Michael Toomey

Dr. Joshua Ziegler



Dr. Barmak Shams



Dr. Ben Wallisch

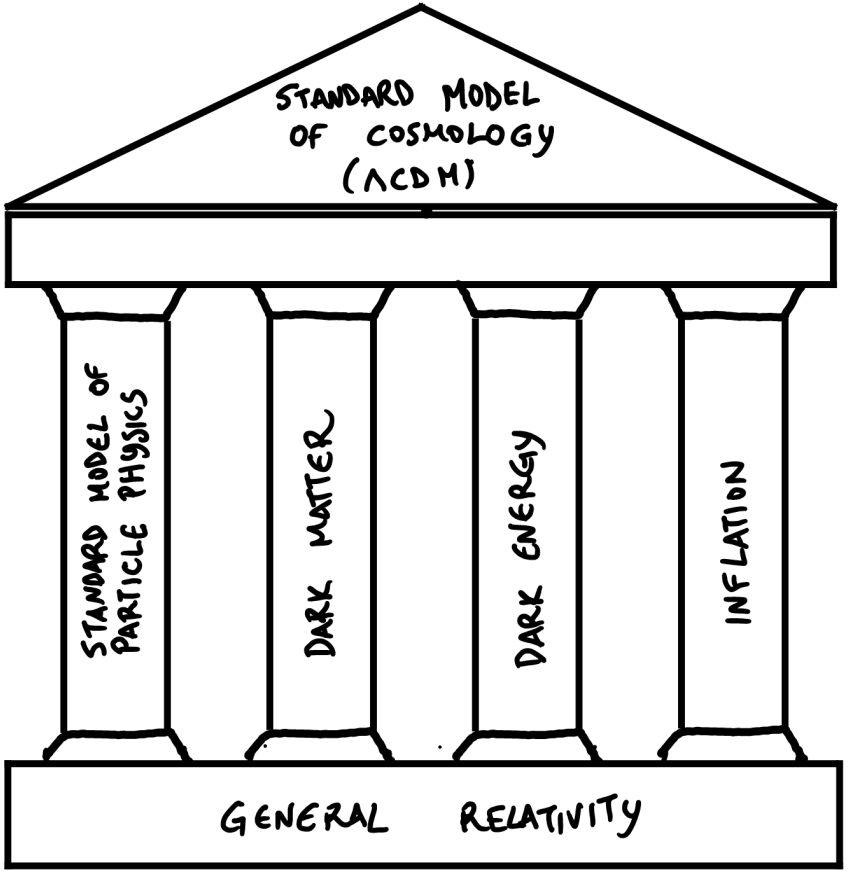


Dr. Nicola Bellomo



Dr. Subhajit Ghosh

Introduction



Introduction

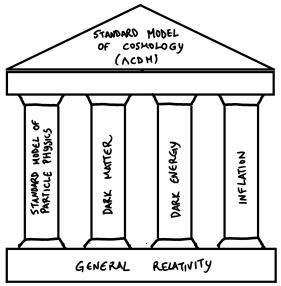
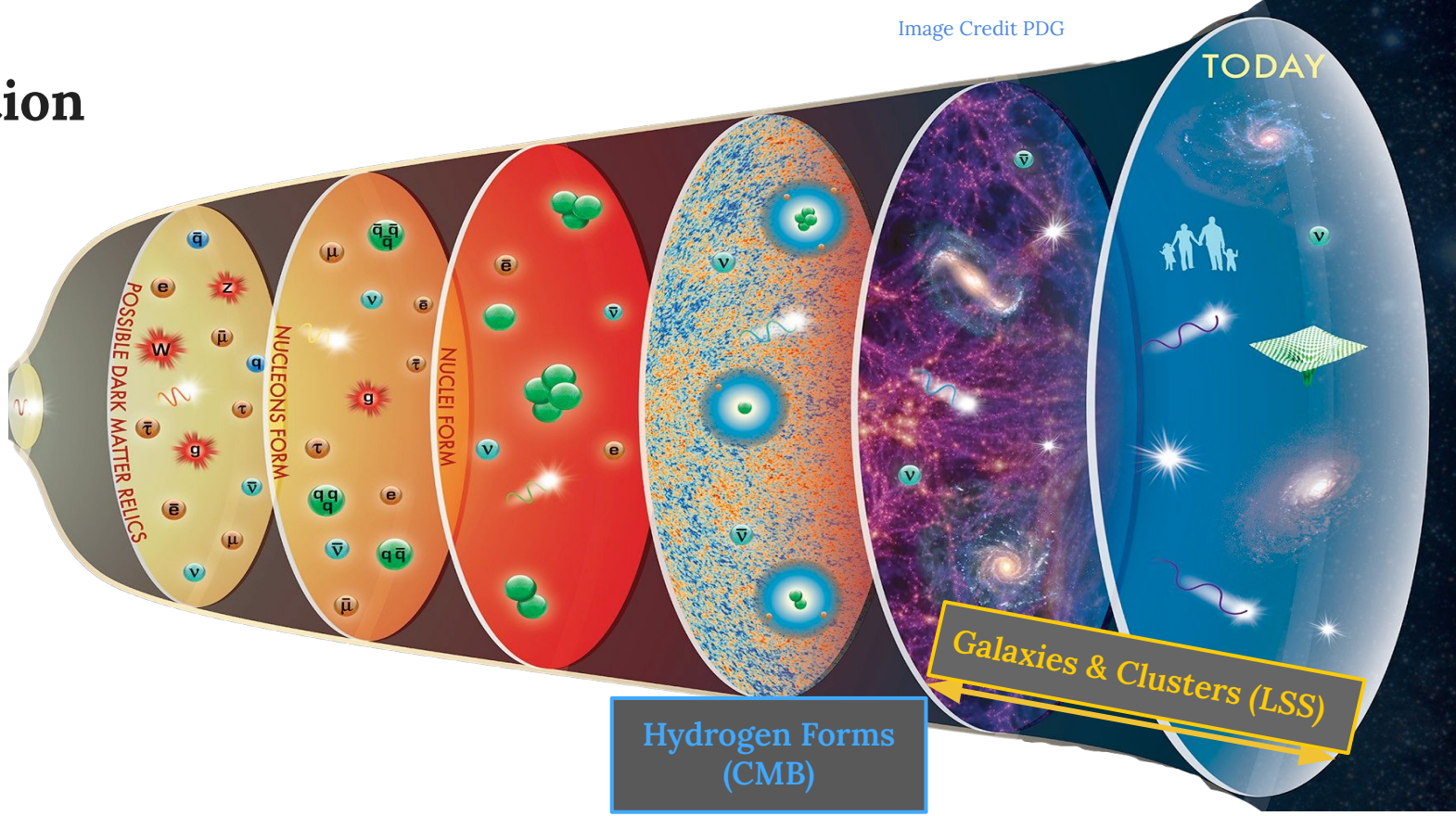
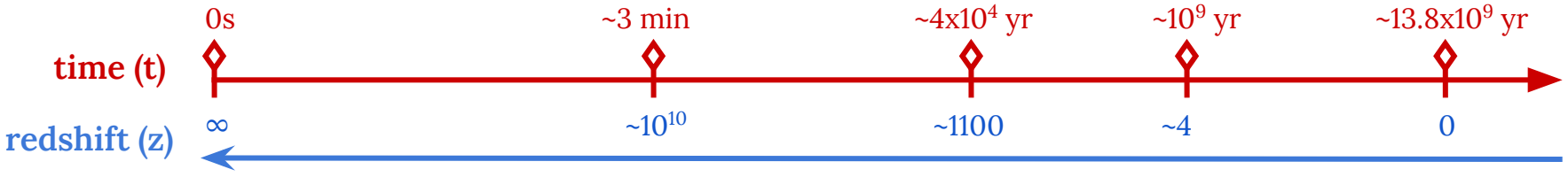
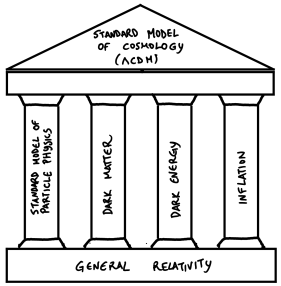


Image Credit PDG



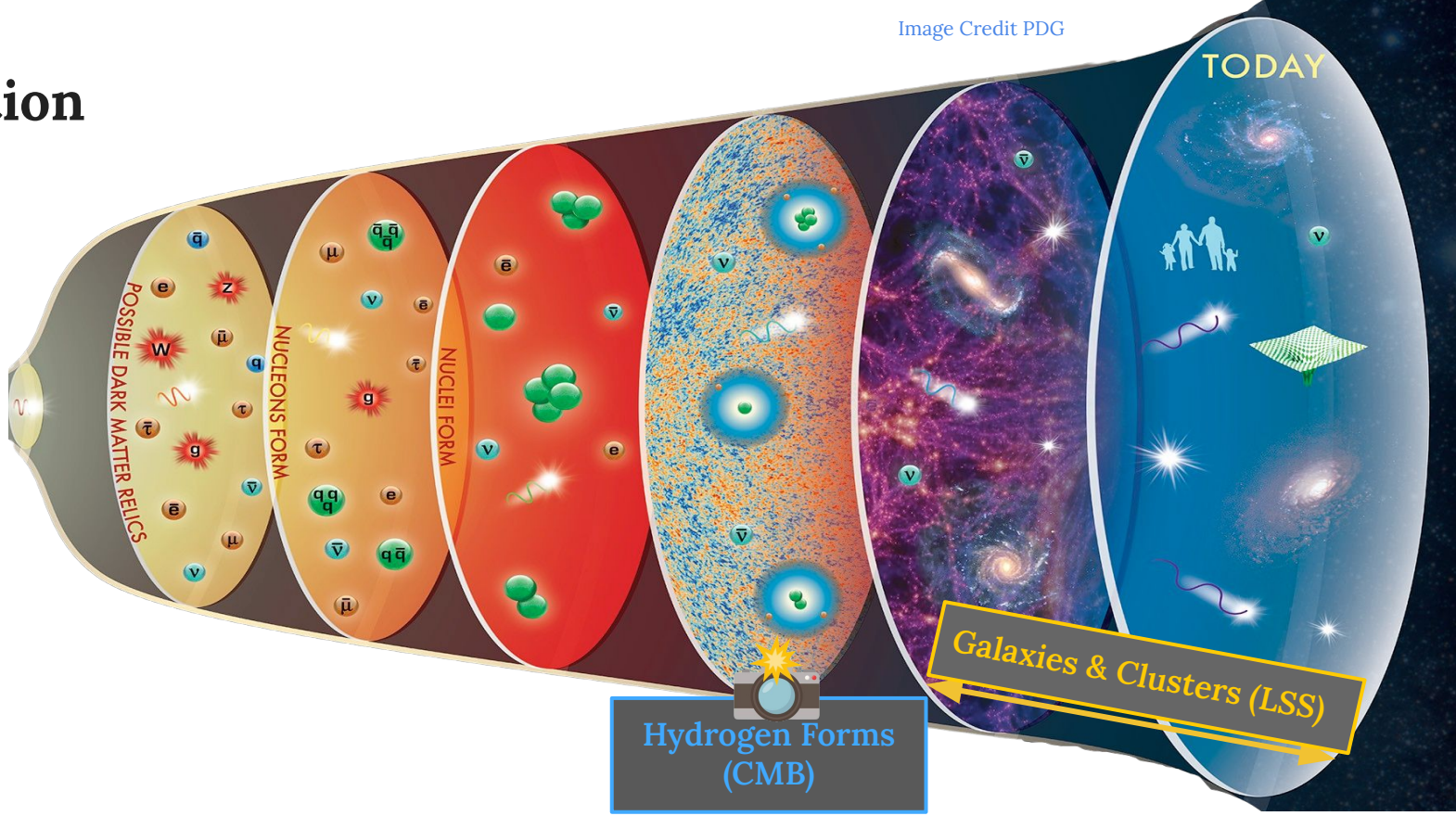


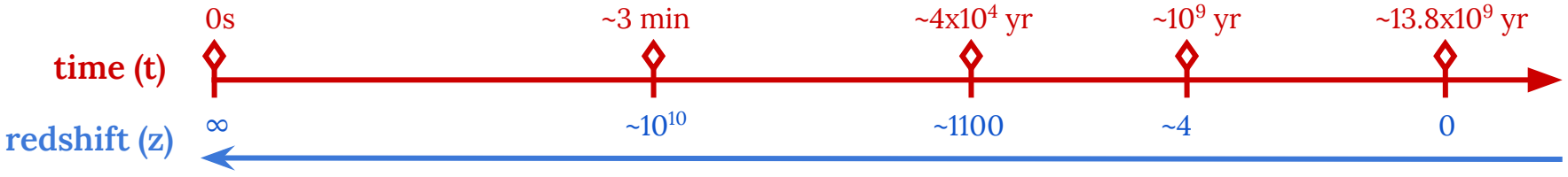
Introduction



Inflation

Image Credit PDG





Introduction

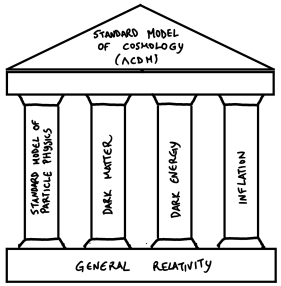
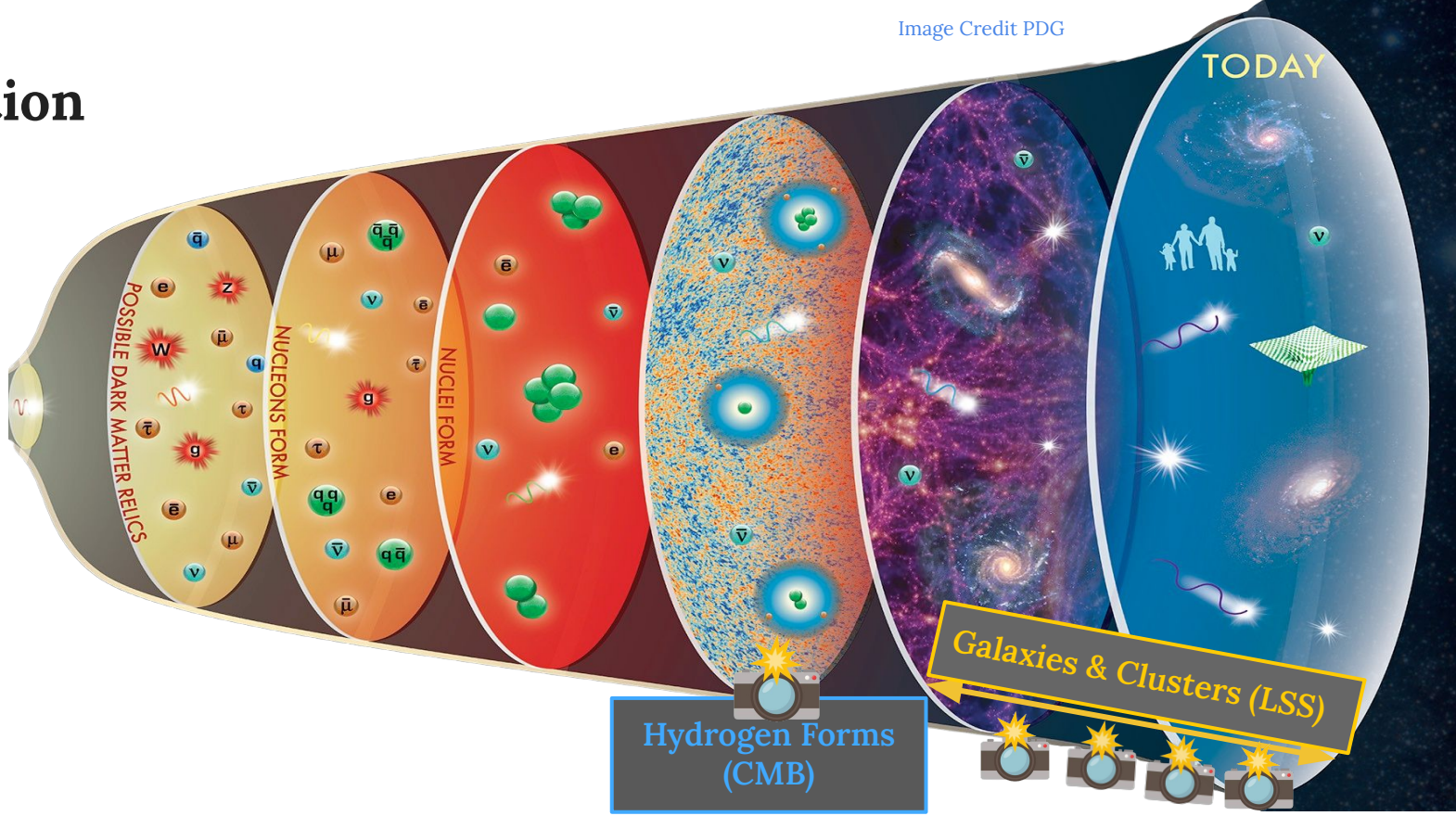
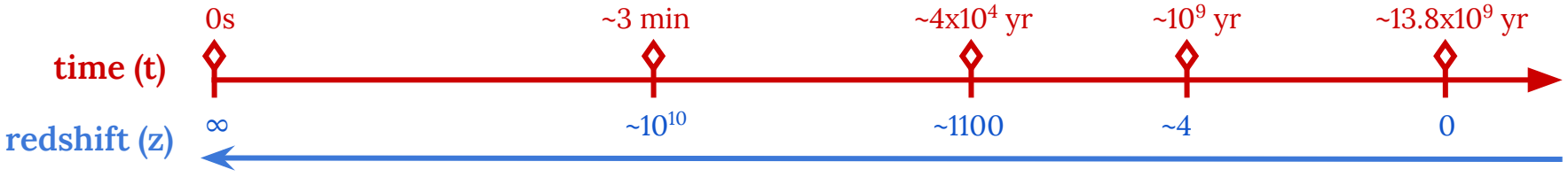
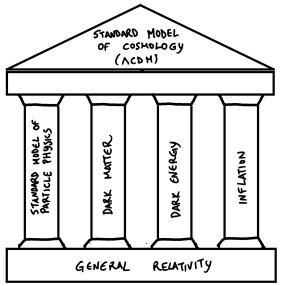


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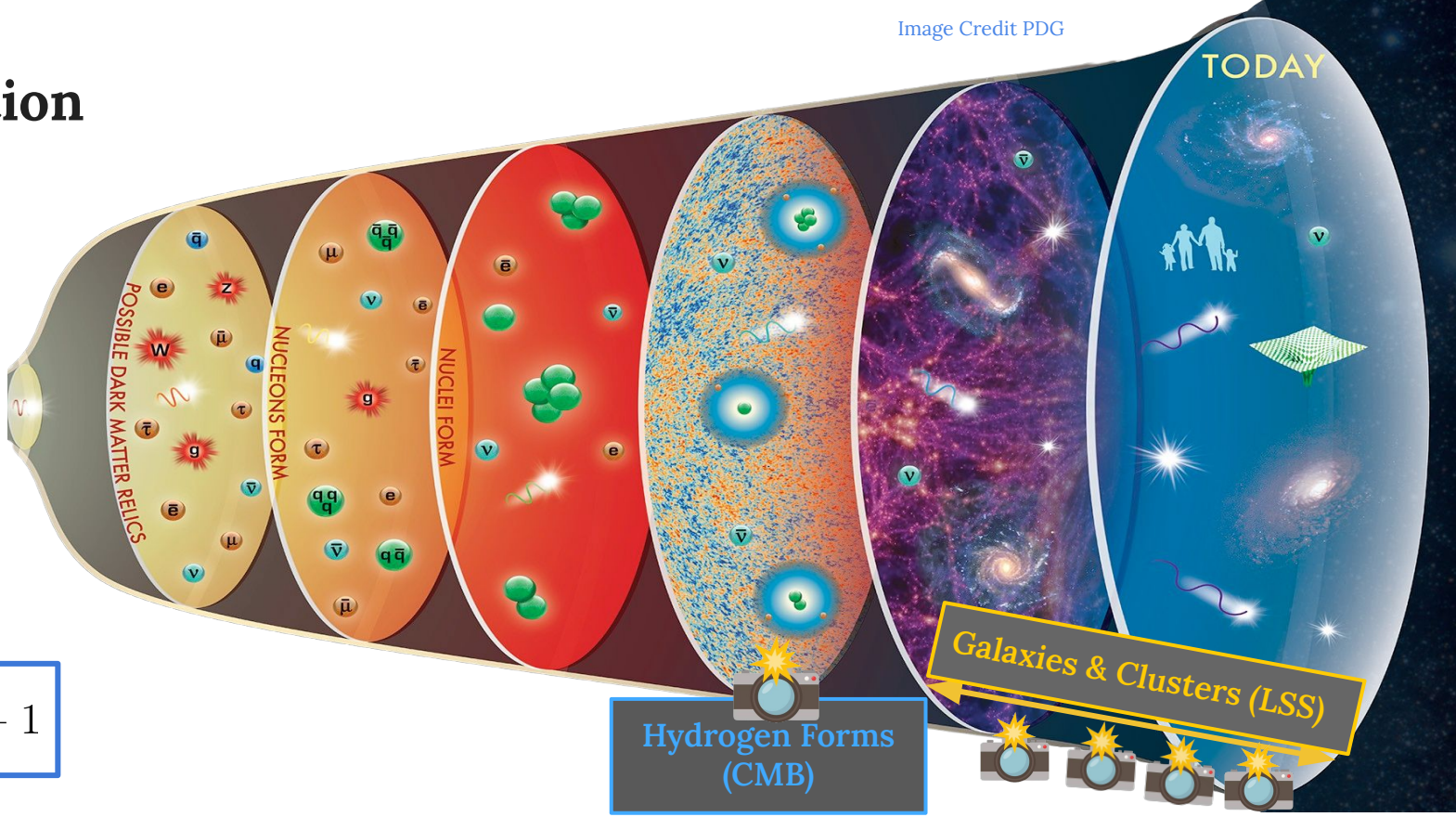


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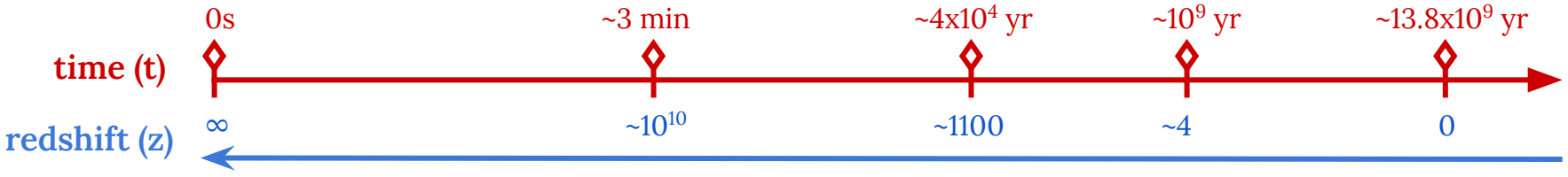


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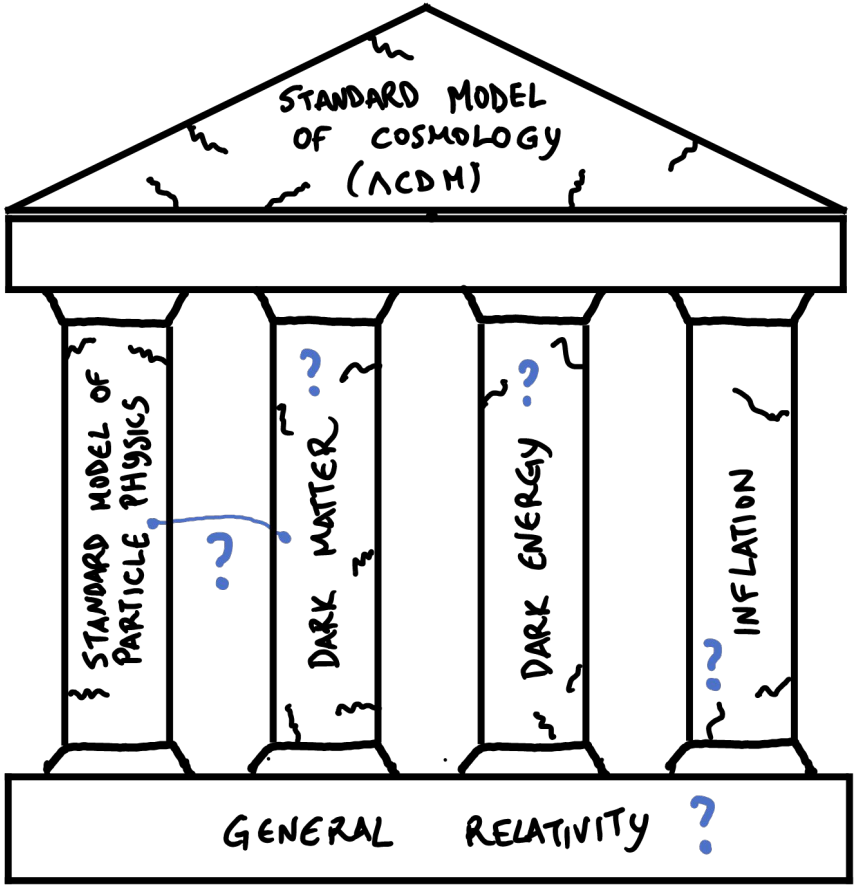


$$z = \frac{R_{\text{today}}}{R} - 1$$



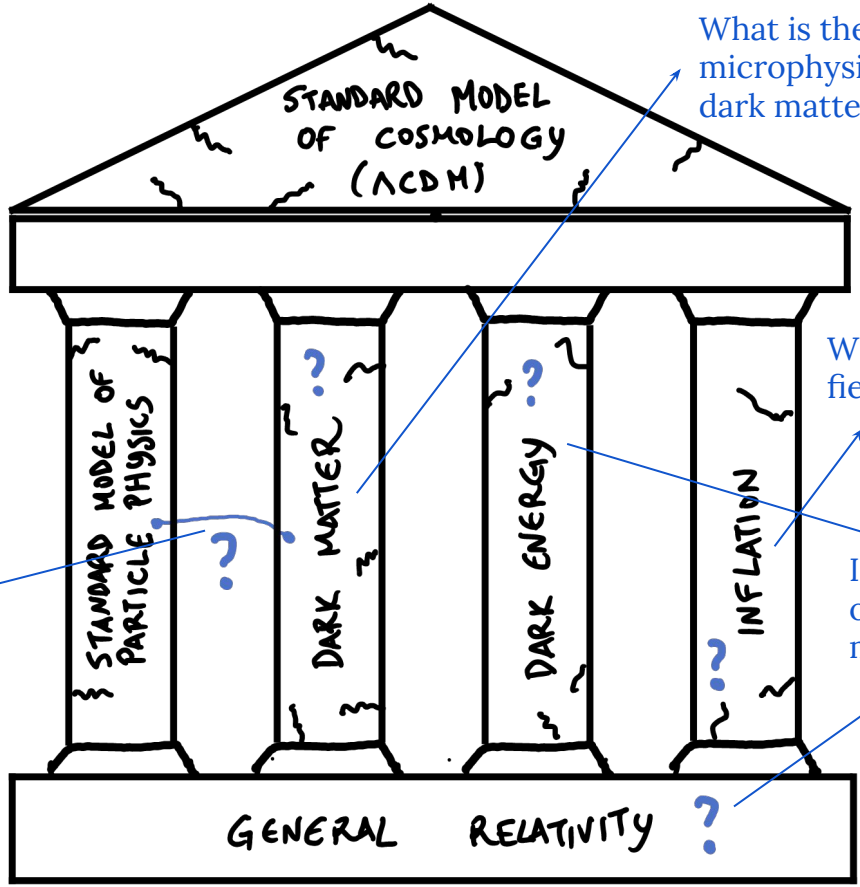
Introduction

The ΛCDM model is far for complete



Introduction

The ΛCDM model is far for complete



What is the microphysics of dark matter?

What is the fundamental field driving inflation?

Is dark energy dynamic or is this a sign of modified gravity?

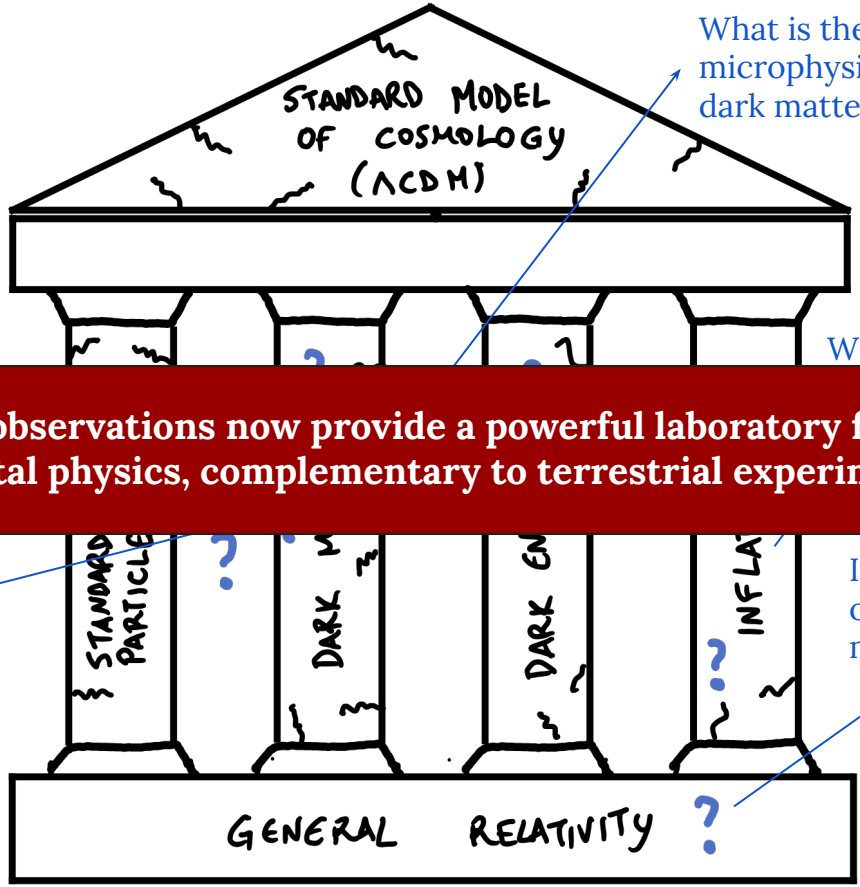
Does dark matter interact what the Standard Model?

... and many more

Introduction

The ΛCDM
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CMB and LSS observations now provide a powerful laboratory for testing fundamental physics, complementary to terrestrial experiments.



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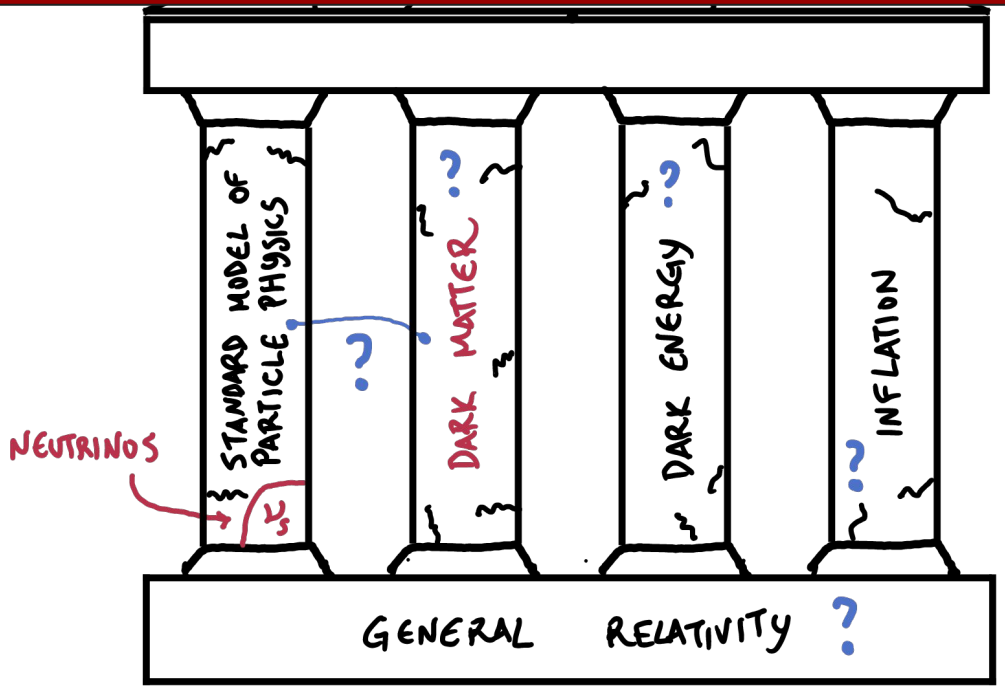
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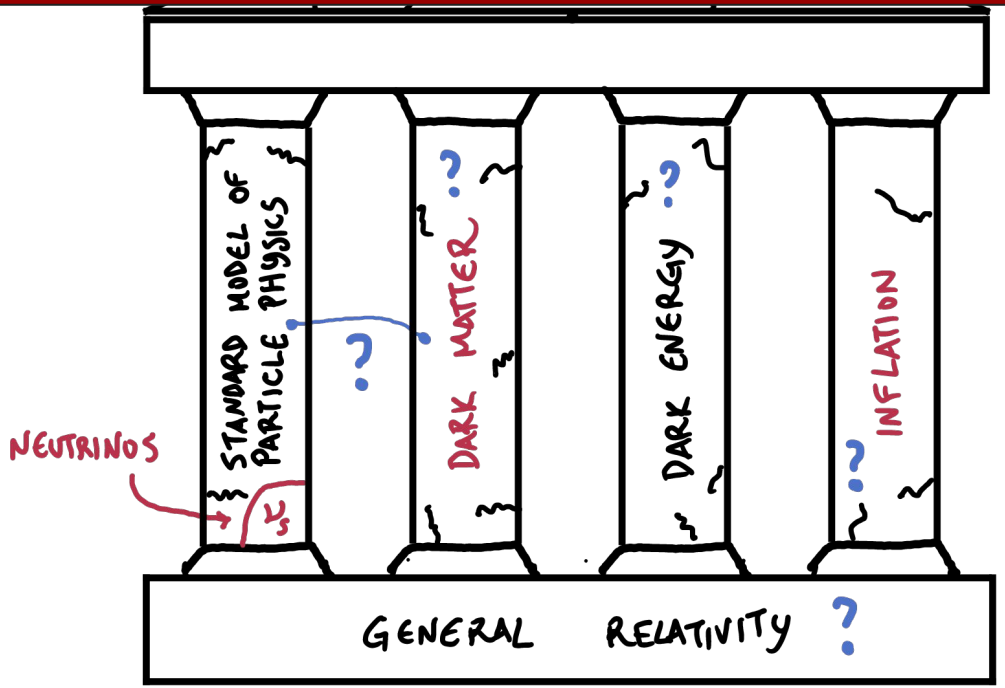
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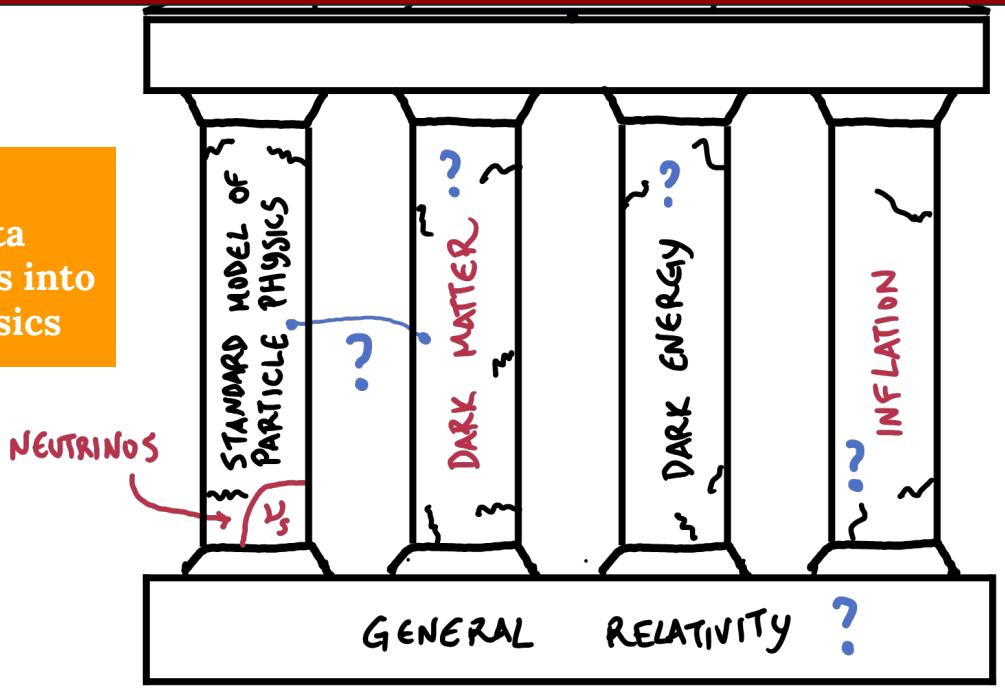
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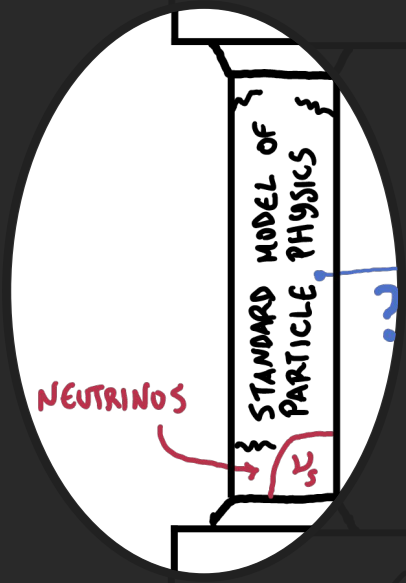
My work:
Connect theory and data to turn cosmological probes into tests of fundamental physics

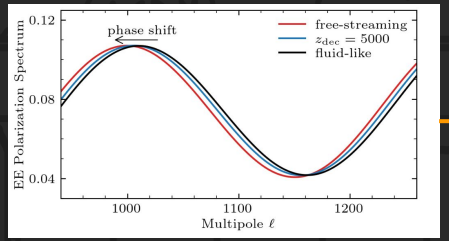


Introduction

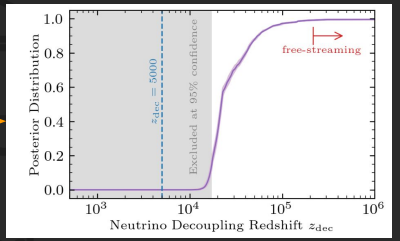
My work:
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The neutrino phase shift as a precision probe of free-streaming and BSM interactions

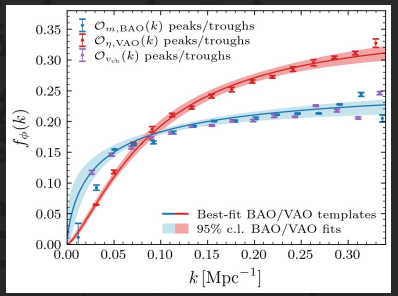




GM, Wallisch & Freese 2025



GM, Ghosh, Boddy, Ho & Tsai 2025

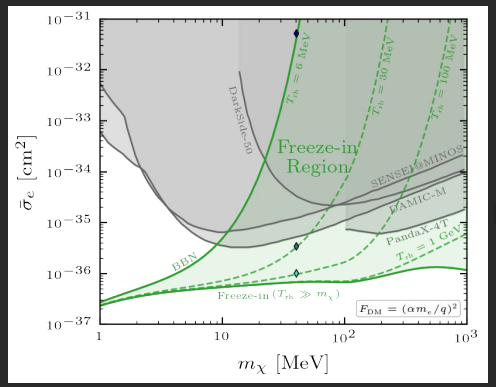


GM, Cruz, Muñoz, Kovetz & Kamionkowski (2025)

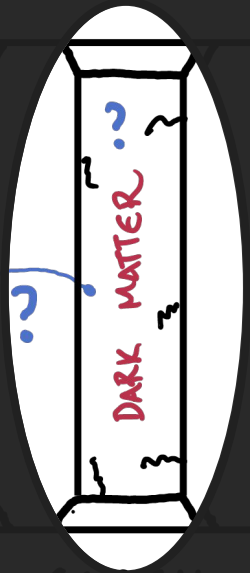
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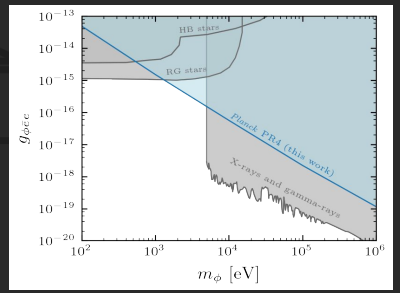
Connecting **cosmological production** to direct detection prospects



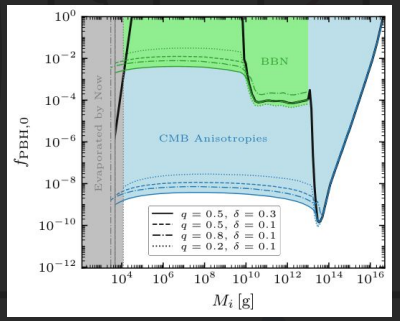
Boddy, Freese, **GM**, Shams 2024



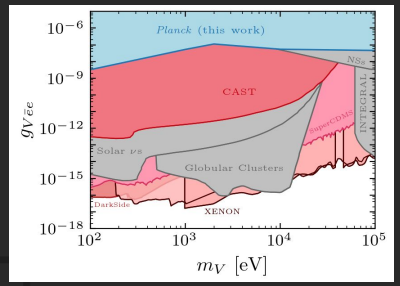
Energy injection signatures as a probe of DM properties and candidates



GM, Elor, Boddy & Bellomo 2025



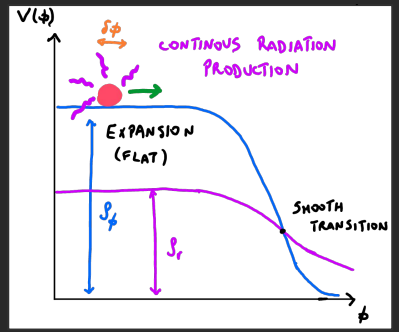
GM, Hooper, Freese, Kelso, Kuhnel & Sandick 2025



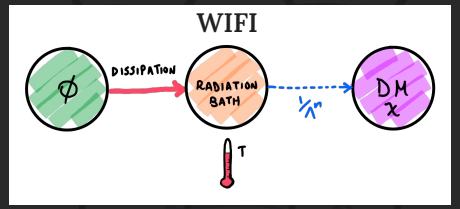
GM, Bellomo & Boddy 2026

Introduction

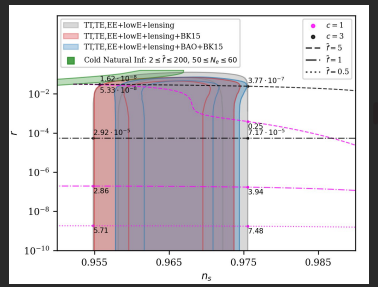
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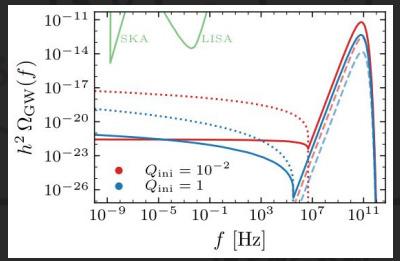
GM, Aragam, Visinelli & Freese 2022a
GM, Aragam, Visinelli & Freese 2023
GM, Ramos, Vicente & Freese 2023



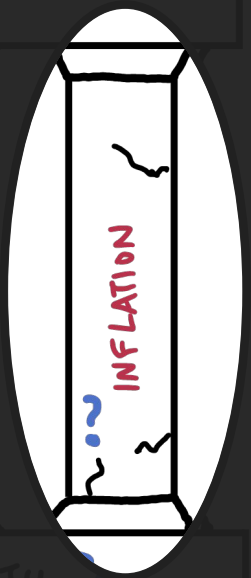
Freese, **GM** & Shams (2024 - PRL)



GM, Aragam, Visinelli & Freese 2022b



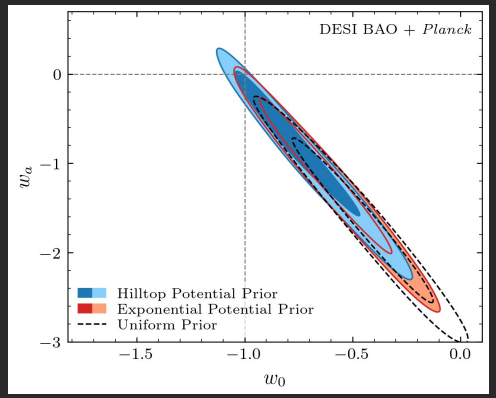
GM, Shams, Xu & Freese 2025



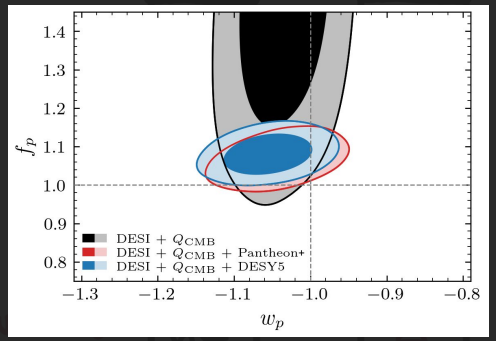
Warm inflation as a distinct inflationary framework: viability, predictions, and novel signatures

Introduction

My work:
Connect theory and data to turn cosmological probes into tests of fundamental physics



Toomey, **GM**, McDonough & Freese 2025



GM & Stiskalek 2026

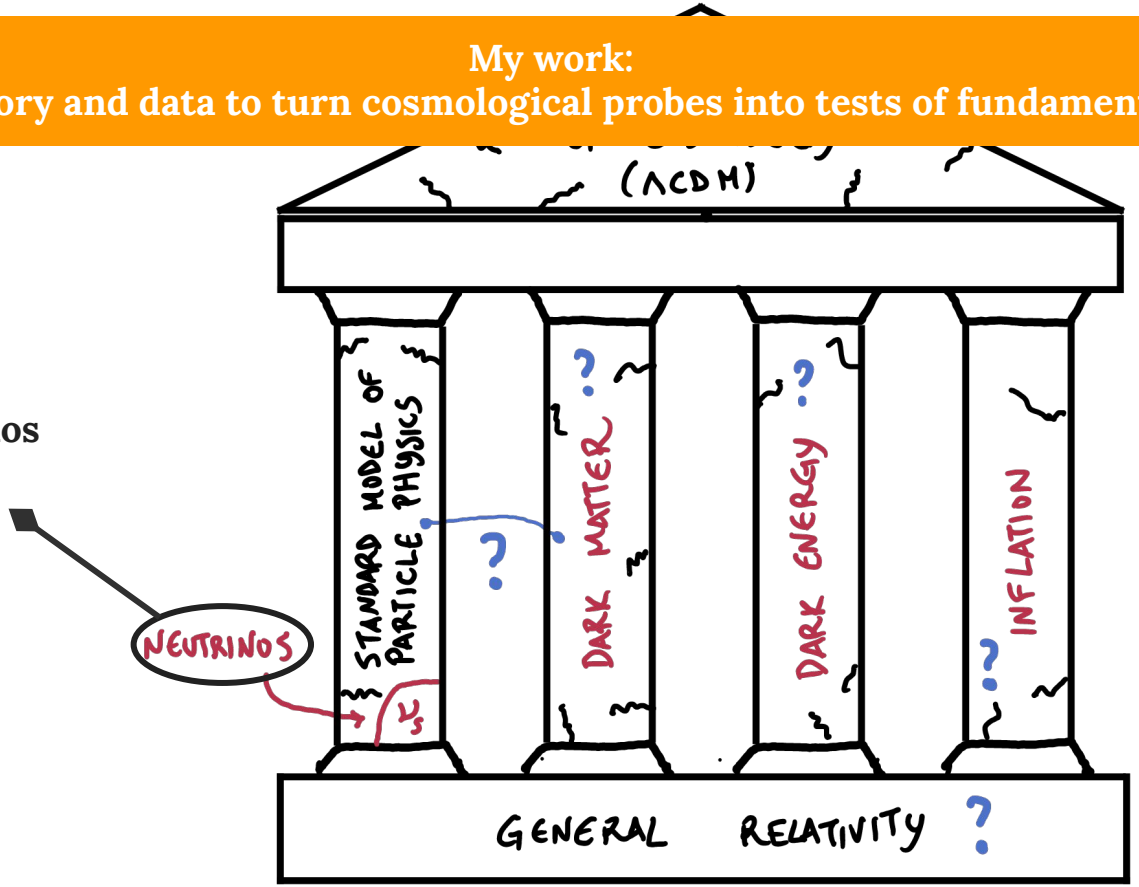


Critically reassessing the claimed evidence for evolving dark energy and developing new tools to test it

Introduction

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Probing neutrinos with the CMB



Probing Neutrinos through CMB Phase Shift Measurements
Outline

1. CMB Physics in a nutshell
2. Cosmic Neutrinos in the CMB
3. The Phase Shift in the CMB and Perturbation Spectra
4. CMB Data Analysis
5. From Phase Shift Signature to Neutrino Decoupling
6. Conclusions

Based on:

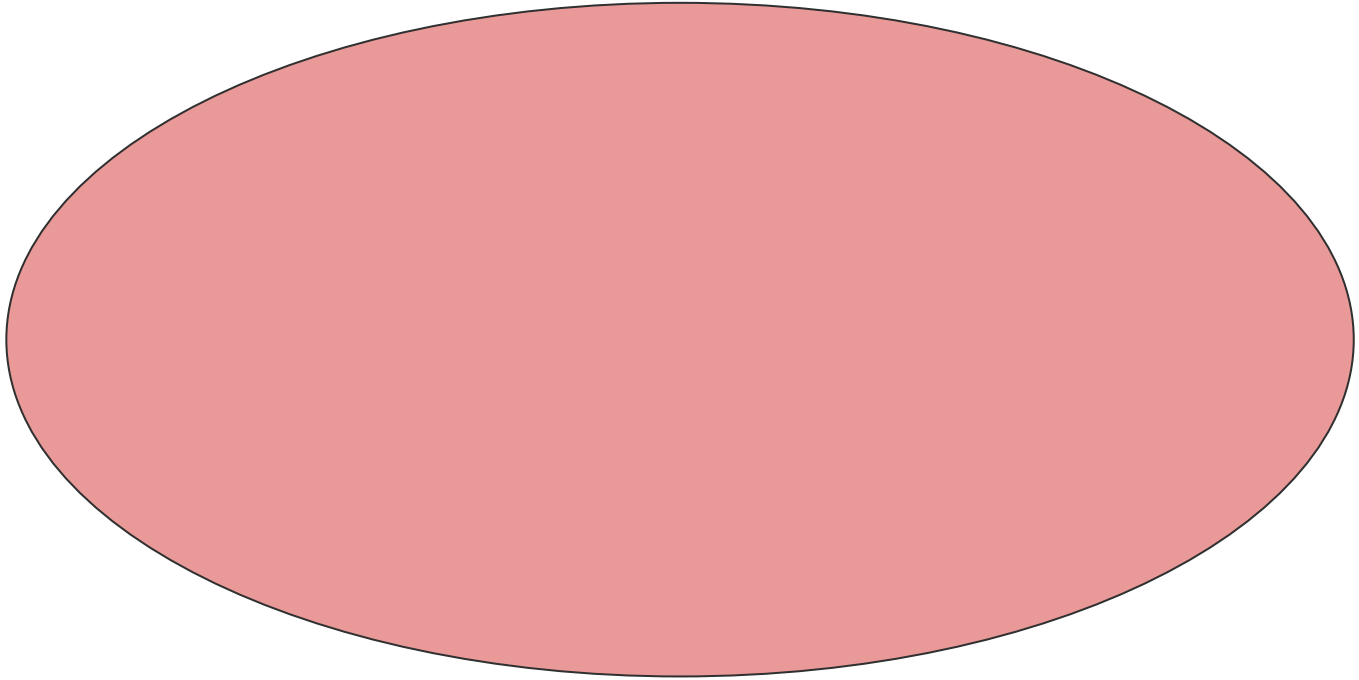
JCAP 08 (2025) 051 with Benjamin Wallisch and Katherine Freese;

PRD 113 (2026) 2, 2 with Subhajit Ghosh, Kimberly Boddy, Daven Wei Ren Ho and Yuhsin Tsai;

The Cosmic Microwave Background

The background radiation associated with the **decoupling** of photons from the primordial plasma

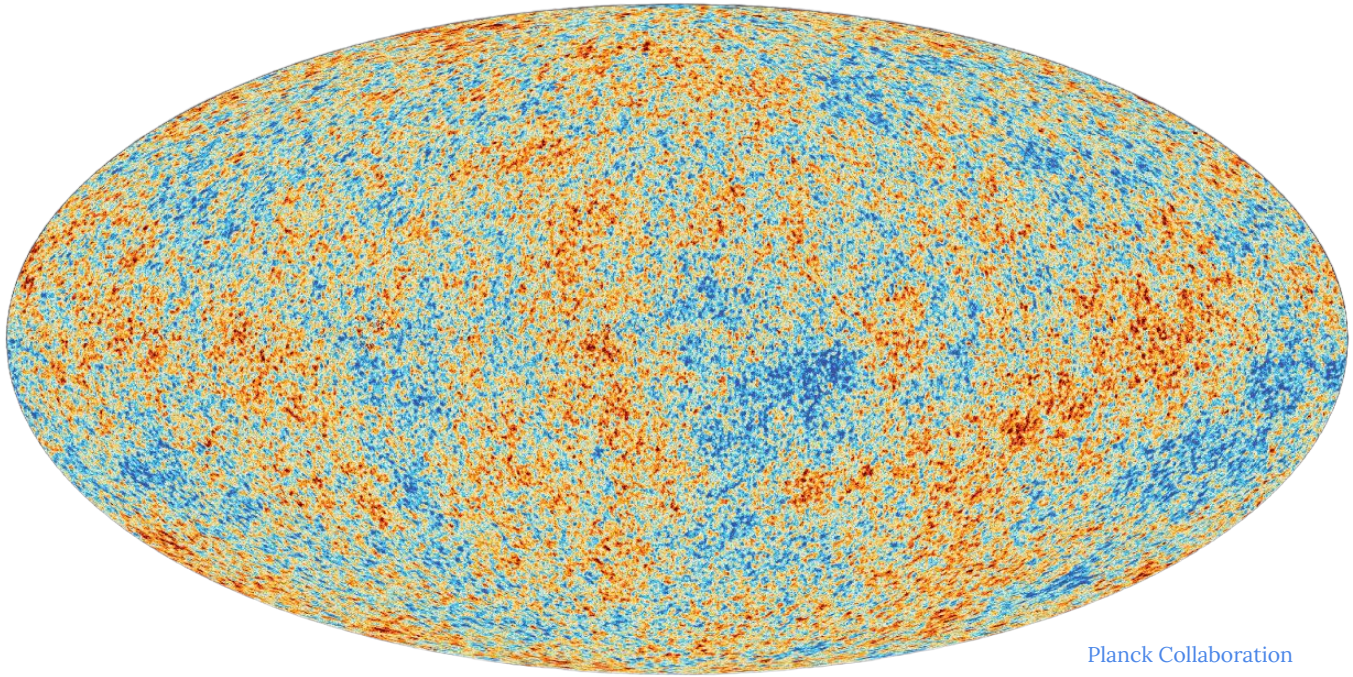
T = 2.7255 K



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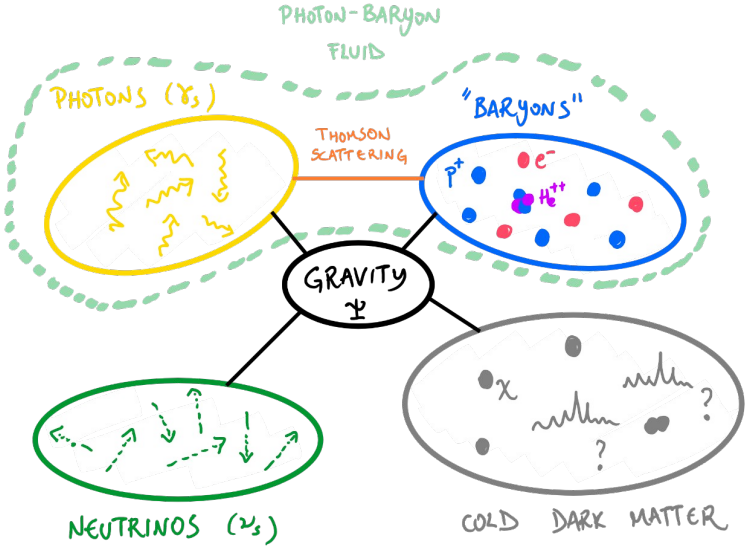
$\Delta T/T \sim 10^{-4}$



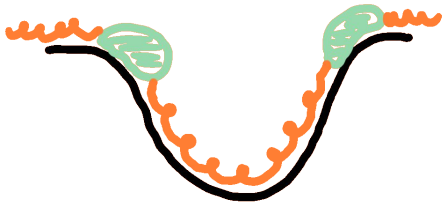
Planck Collaboration

Cosmic Sound Waves

- Photons and baryons are **tightly coupled**
- **Initial fluctuations** excited sound waves in the primordial plasma
- **Gravity** sources the fluctuations in the photon-baryon fluid



$$\ddot{\delta}_\gamma - \underbrace{c_\gamma^2 \nabla^2 \delta_\gamma}_{\text{Photon Pressure}} = \underbrace{\nabla^2 \Psi_+}_{\text{Gravity}}$$



$$\delta_\gamma \sim \underbrace{A_{\vec{k}}}_{\text{Initial condition (inflation)}} \cos(c_s k \tau),$$

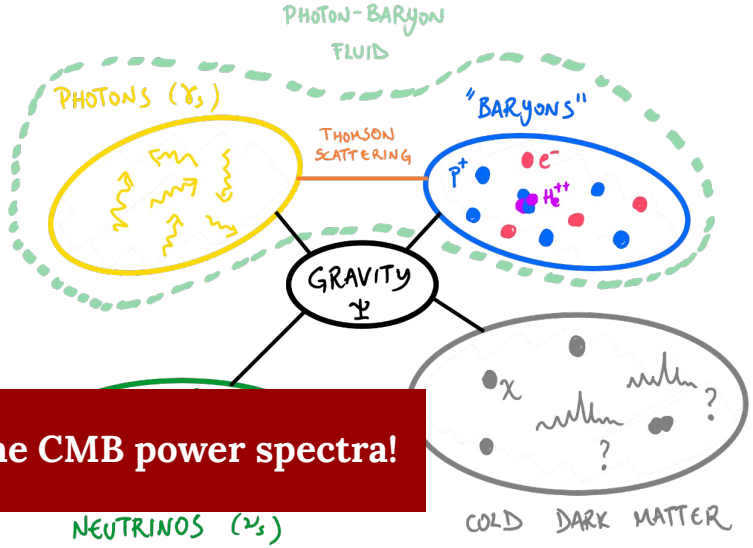
$$c_s^2 \sim \frac{c^2}{3(1+R_b)} \quad R_b \equiv 3\bar{\rho}_b/(4\bar{\rho}_\gamma)$$

Baryons add inertia to the fluid

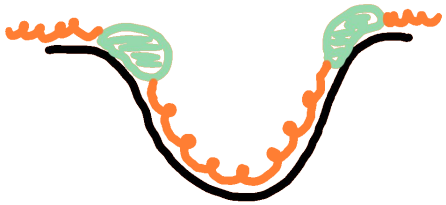
Cosmic Sound Waves

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  photon-

We observe these acoustic oscillations in the CMB power spectra!



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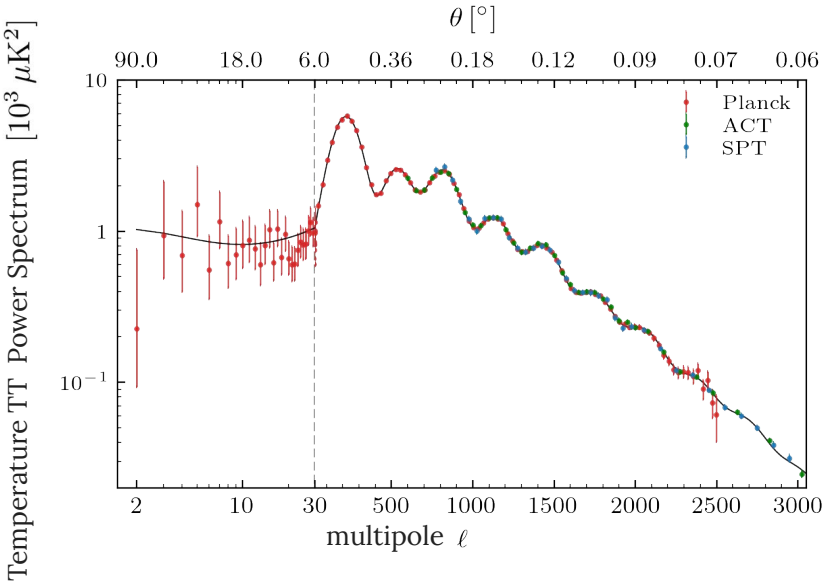
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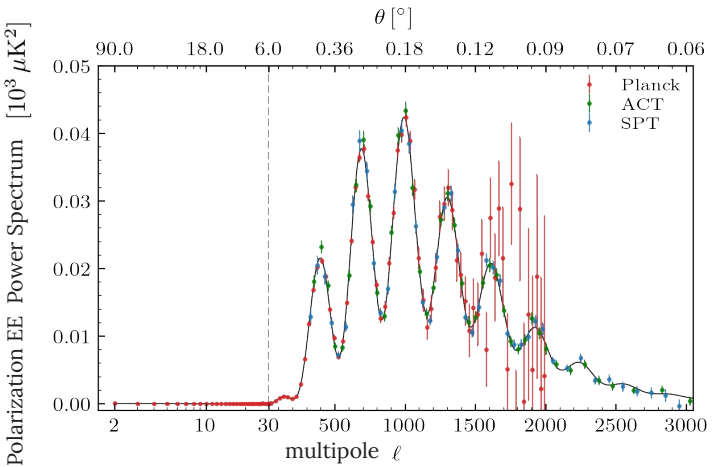
The CMB Power Spectra

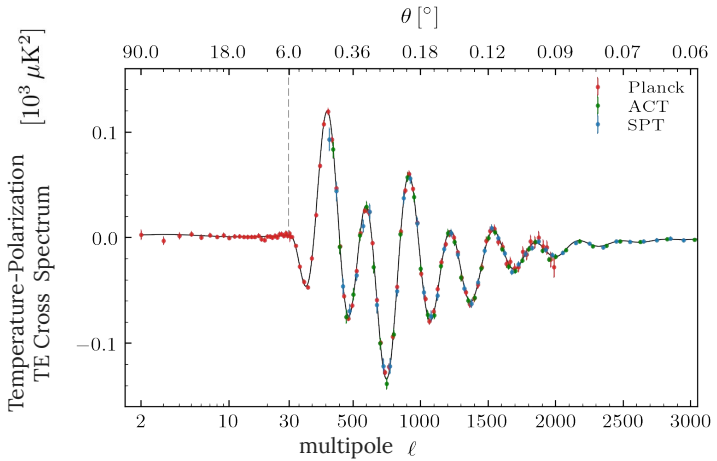
Temperature traces **density** perturbations



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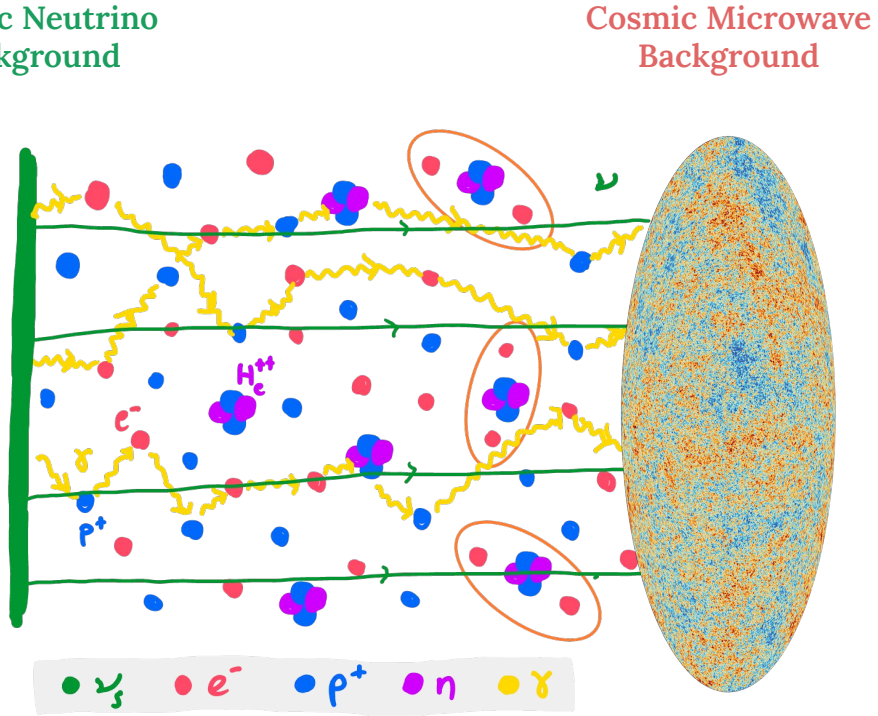
Polarization traces **velocity** perturbations





Cosmic Neutrinos in the CMB

- Significant fraction of **energy density** in neutrinos
    - Parametrized by $\mathbf{N_{eff}}$, known as *the effective number of relativistic species*
    - In the SM, $\mathbf{N_{eff} = 3.044}$ Akita, Yamaguchi (2020)
- **Free-streaming** since their decoupling at **t~1 s**



Cosmic Neutrinos in the CMB

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- **Free-streaming** since their decoupling at **t~1 s**
- CMB measurements are **consistent with SM value** of $\mathbf{N_{eff}}$

$N_{\rm eff} = 2.92 \pm 0.18$

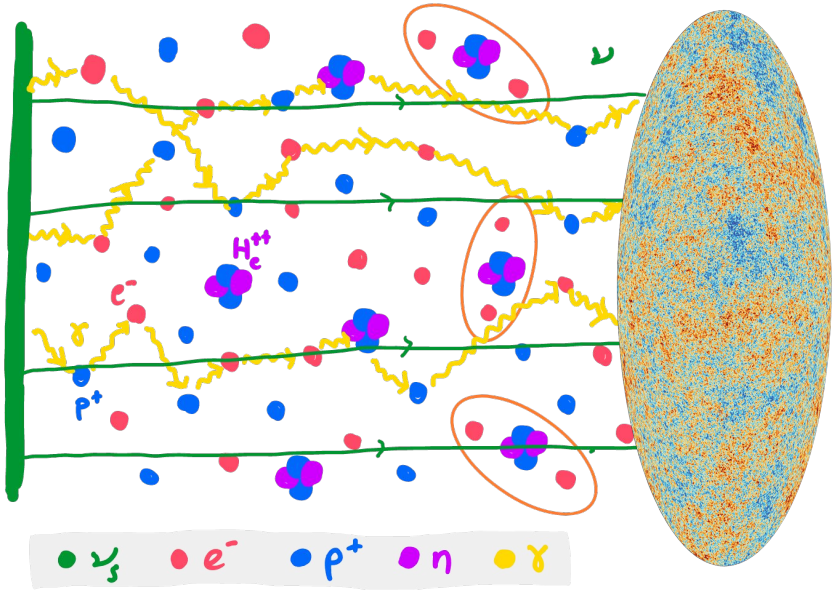
Planck 2018

$N_{\rm eff} = 2.81 \pm 0.12$

Planck 2018 + ACT DR6 + SPT 3G D1

Cosmic Neutrino Background

Cosmic Microwave Background



Cosmic Neutrinos in the CMB

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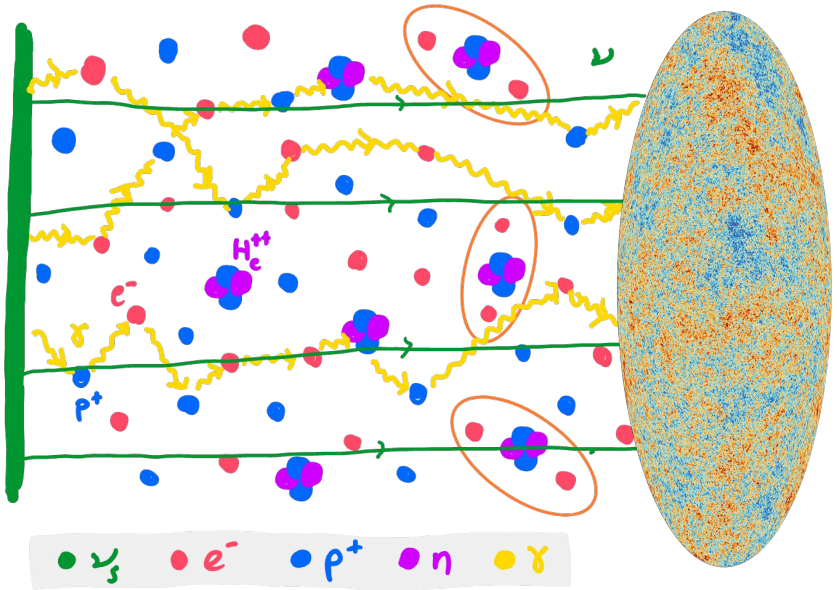
Can we detect this?

$N_{\rm eff} = 2.92 \pm 0.18$

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Can we detect this?

- Neutrino masses $\mathbf{m_\nu}$? no detection and **unimportant** in the early universe

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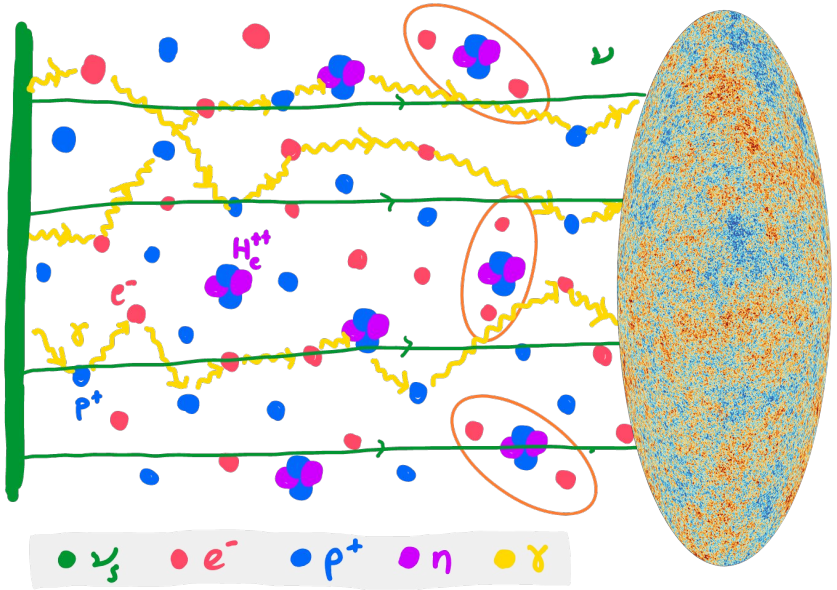
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Cosmic Neutrino Background

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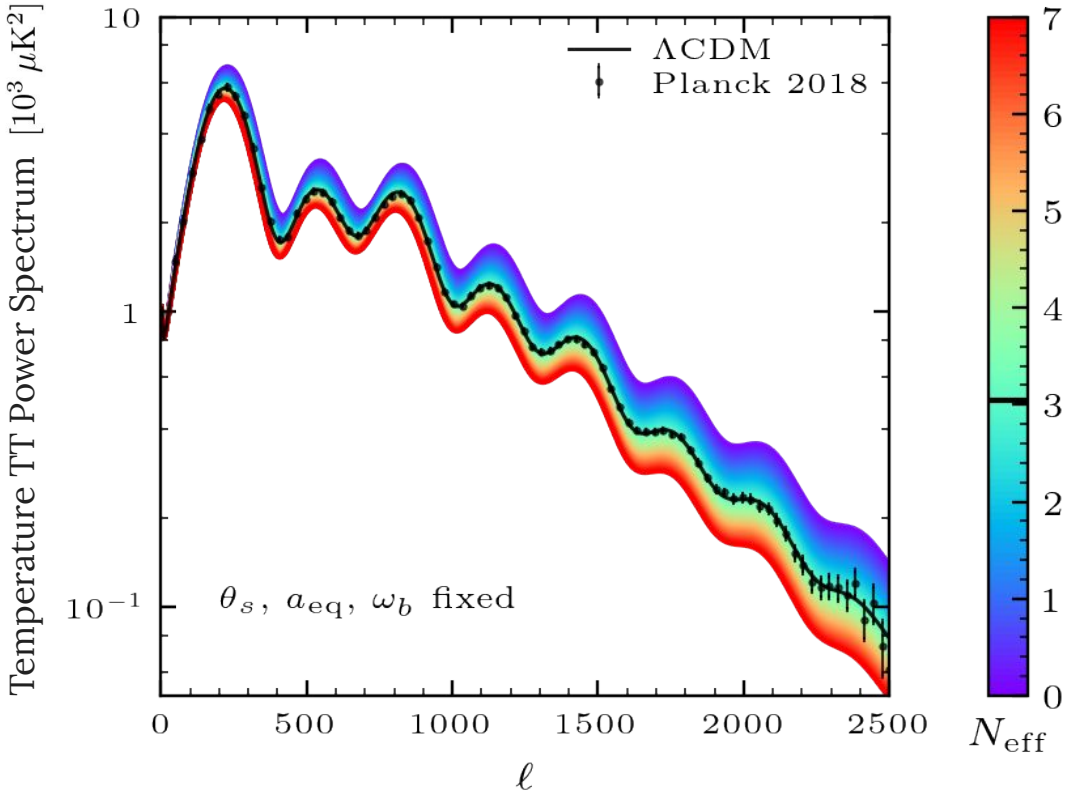


Cosmic Neutrinos in the CMB
Energy Density

- Main effect in the damping tail of the CMB TT power spectrum, via their effect on the expansion rate

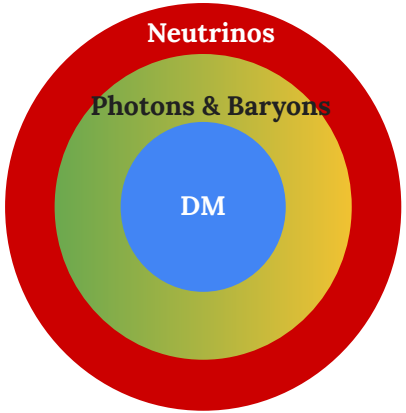
$$\theta_d \propto (H/n_e)^{1/2}\theta_s$$

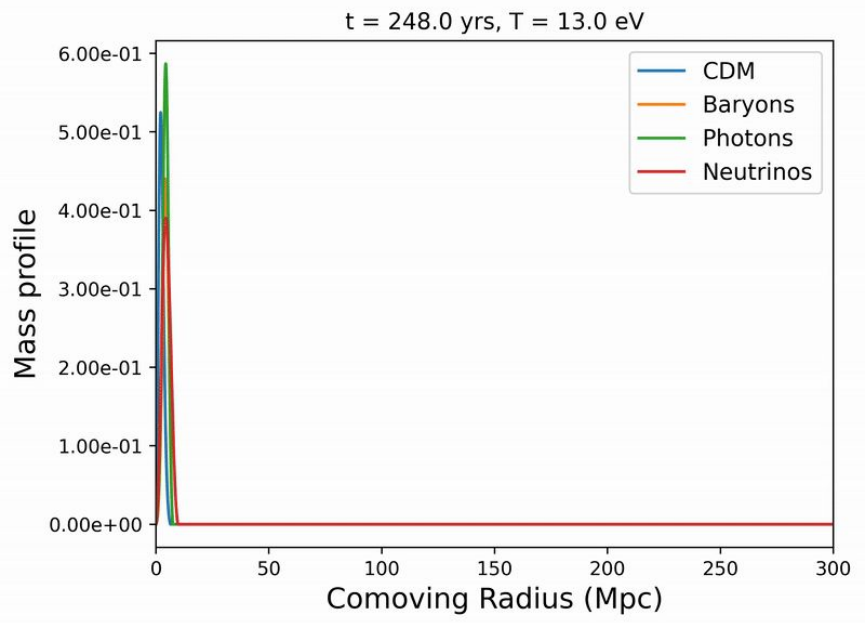
  - For fixed a_{eq} and θ_s, **larger N_{eff} means more damping**



Cosmic Neutrinos in the CMB
Free-Streaming Nature

- Perturbations from free-streaming neutrinos induce **metric perturbations ahead of the sound horizon**
- The photon-baryon fluid is **pulled** by such perturbations, shifting their perturbations peaks to larger radii.





Credit: Murali Saravanan

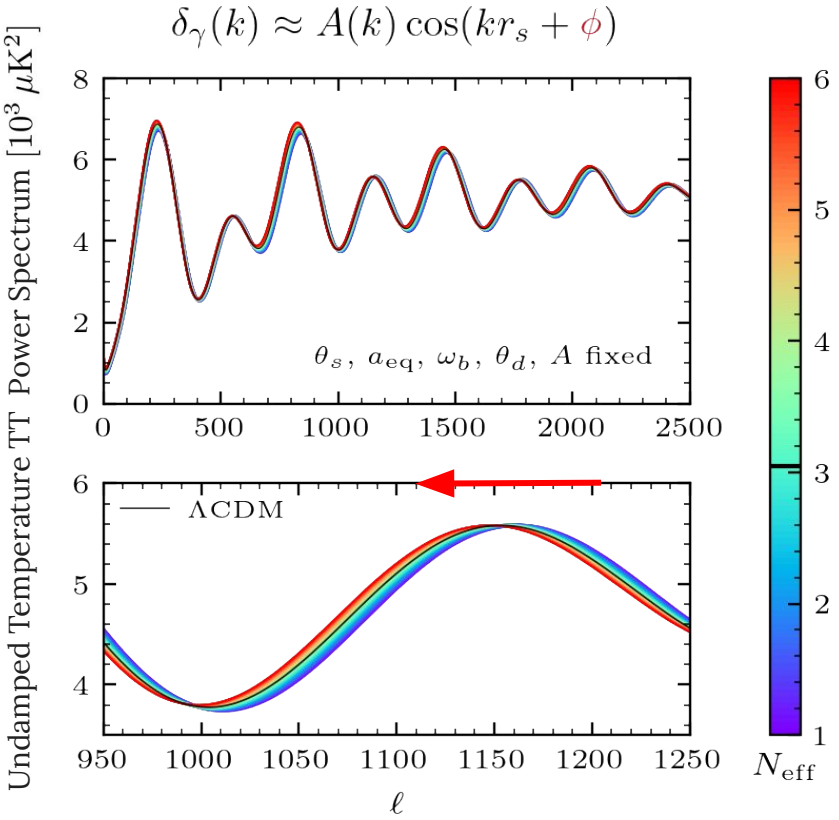
Cosmic Neutrinos in the CMB
Free-Streaming Nature

- This results in a **phase shift in the acoustic peaks of the CMB**

  Bashinsky & Seljak (2003)

  - Larger radii → smaller multipoles
  - Small effect, prop. to: $\epsilon_\nu \equiv \rho_\nu / \rho_r$

- Difficult to reproduce in the absence of **free-streaming**
  - Either free-streaming or non-adiabatic fluctuations Baumann, Green, Meyers & Wallisch (2015)



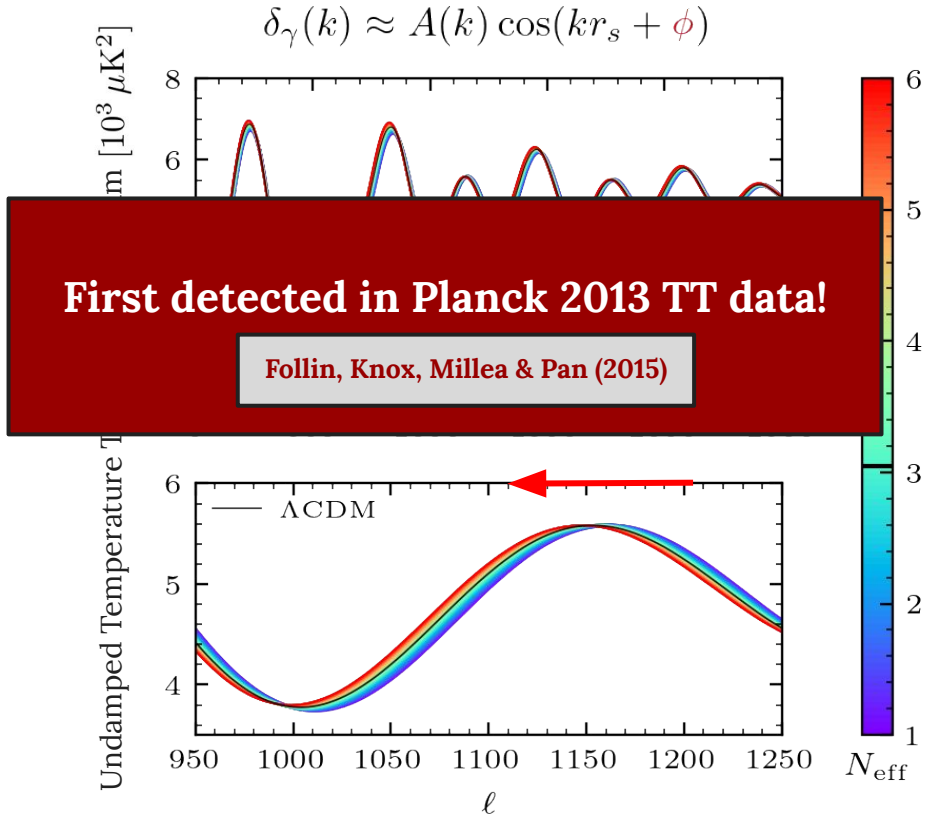
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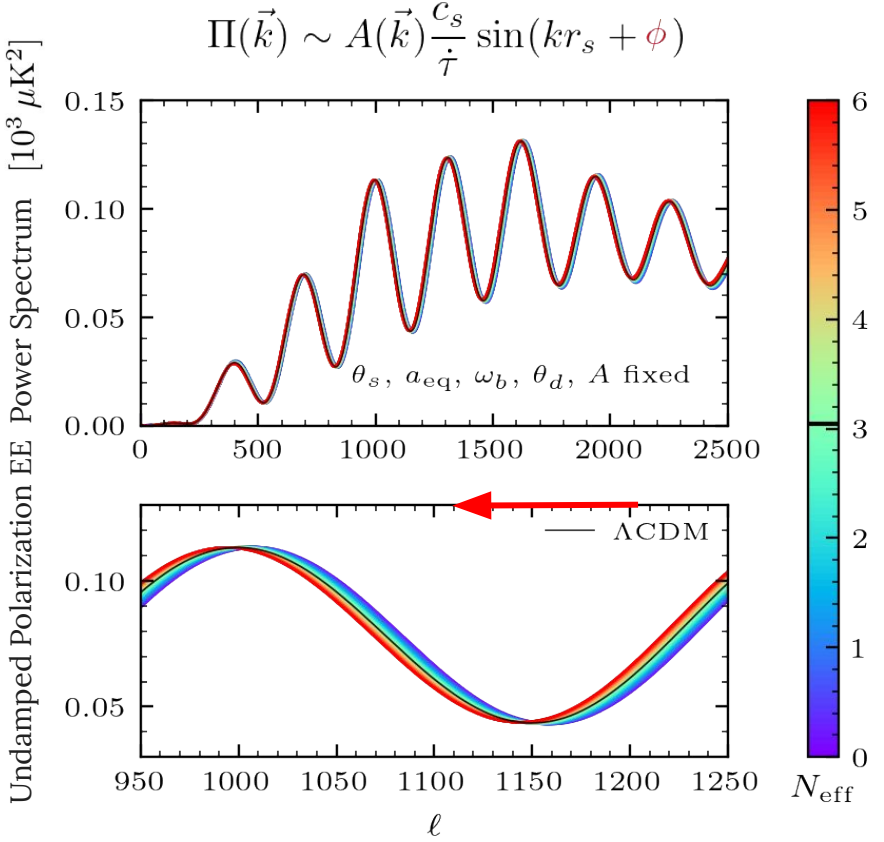


Cosmic Neutrinos in the CMB
Free-Streaming Nature

- Same shift both in temperature and polarization spectrum

$$\Pi(k,\eta) \sim \frac{d}{d\eta}\delta_\gamma(k,\eta)$$

  - polarization provides cleaner signal



The Multipole Shift in the CMB Power Spectra

$$\delta\ell(N_{\rm eff}) = A(N_{\rm eff})\, f_\ell$$

Normalized Amplitude

$$A(N_{\rm eff}) \equiv \frac{\epsilon(N_{\rm eff}) - \epsilon(3.044)}{\epsilon(1) - \epsilon(3.044)}$$

Multipole Dependence

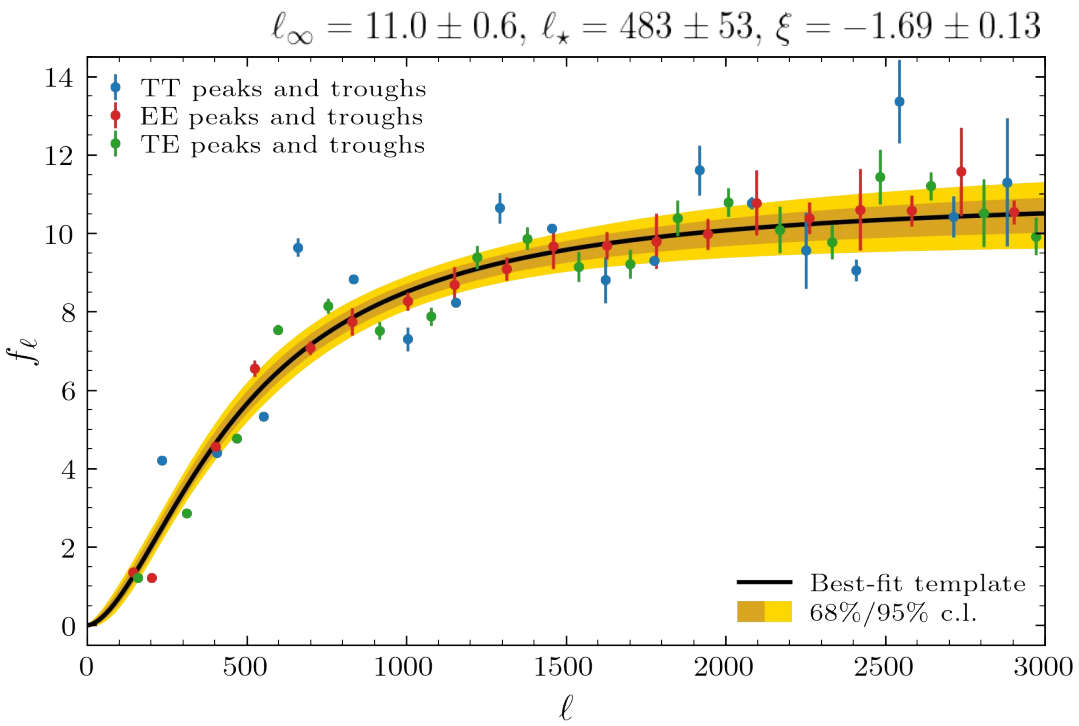
$$f_\ell = \frac{\ell_\infty}{1 + (\ell/\ell_\star)^\xi}$$

- A new parameter to control the shift

$$N_{\rm eff}^{\delta\ell}$$

$$\Delta\ell(N_{\rm eff}^{\delta\ell}, N_{\rm eff}) = \left[A(N_{\rm eff}^{\delta\ell}) - A(N_{\rm eff})\right] f_\ell$$

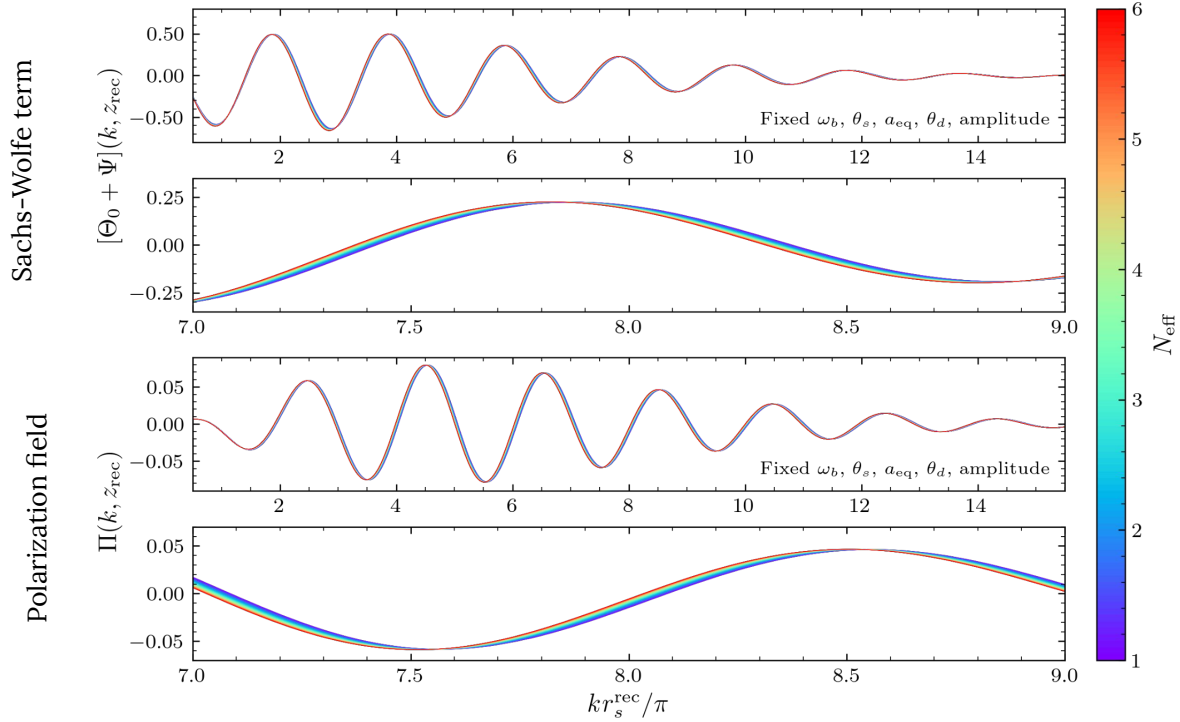
$$\mathcal{C}_\ell \rightarrow \mathcal{C}_{\ell+\Delta\ell}$$



Following Follin, Knox, Millea & Pan (2015)

The Phase Shift in the Perturbations

- Extract the **phase shift** at the **perturbations** level, instead of the CMB power spectra
  - Cleaner signature: avoids projection and smearing effects



$$\delta_\gamma(k) \approx A(k)\cos(kr_s + \phi)$$

$$\Pi(\vec{k}) \sim A(\vec{k})\frac{c_s}{\dot{\tau}}\sin(kr_s + \phi)$$

GM, Wallisch & Freese (2025)

The Phase Shift in the Perturbations

$$\delta\phi(k, z; N_{\rm eff}) = A(N_{\rm eff})\, f_\phi(k, z)$$

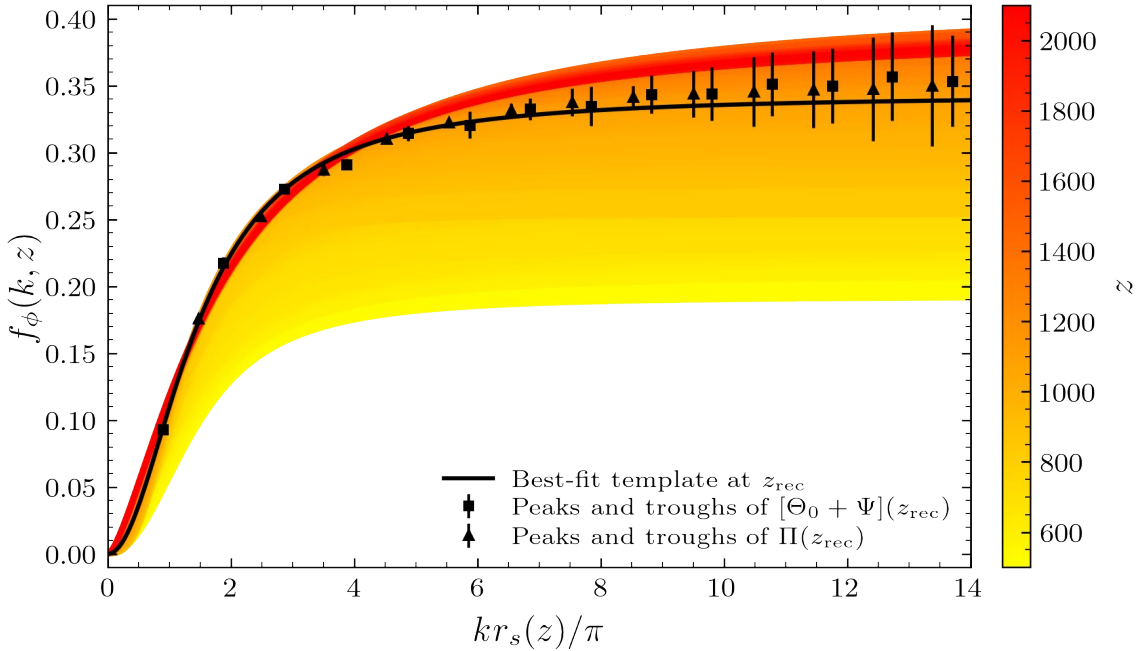
Normalized Amplitude

$$A(N_{\rm eff}) \equiv \frac{\epsilon(N_{\rm eff}) - \epsilon(3.044)}{\epsilon(1) - \epsilon(3.044)}$$

Wavenumber and **Redshift** Dependence

$$f_\phi(k, z) = \frac{\phi_\infty}{1 + [kr_s/(kr_s)_\star]^\xi}$$

Determine the best-fit parameters in relevant redshift range.



GM, Wallisch & Freese (2025)

The Phase Shift in the Perturbations

$$A(N_{\rm eff}) \equiv \frac{\epsilon(N_{\rm eff}) - \epsilon(3.044)}{\epsilon(1) - \epsilon(3.044)}$$

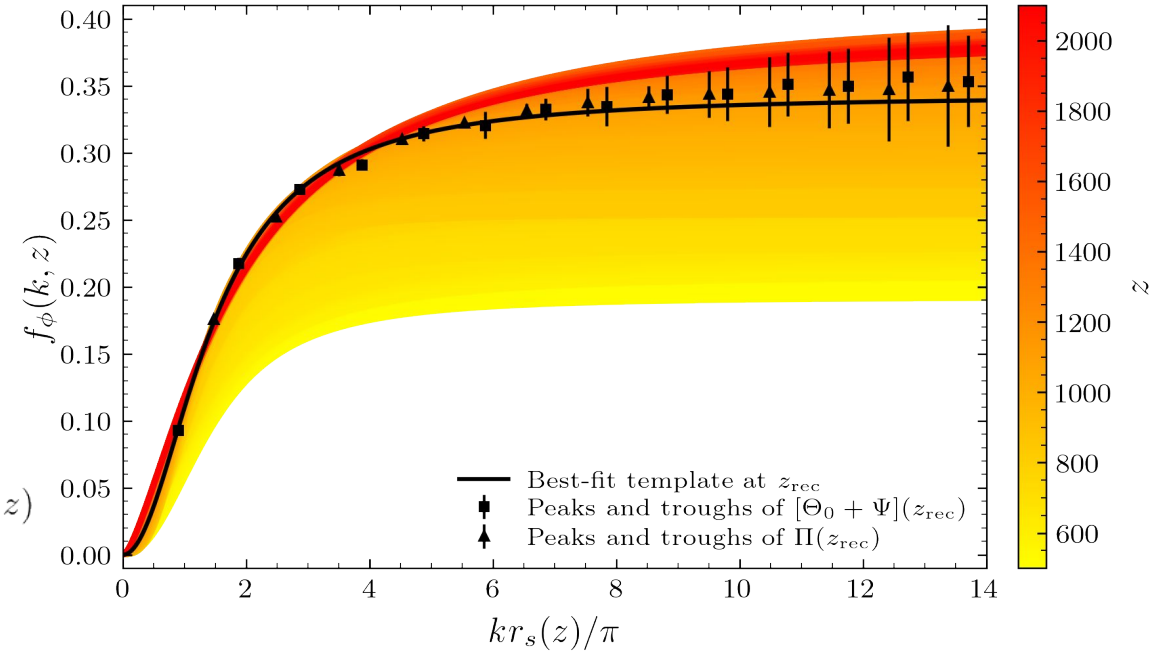
$$f_\phi(k,z) = \frac{\phi_\infty}{1 + [kr_s/(kr_s)_\star]^\xi}$$

- A new parameter to control the shift

$$N_{\rm eff}^{\delta\phi}$$

$$\Delta\phi(k,z;N_{\rm eff}^{\delta\phi},N_{\rm eff}) = \left[A(N_{\rm eff}^{\delta\phi}) - A(N_{\rm eff})\right] f_\phi(k,z)$$

$$S_X(k,z) \to S_X(k + \Delta\phi/r_s, z)$$



It results in the observed neutrino induced **multipole** shift in the CMB spectra

GM, Wallisch & Freese (2025)

Measuring the Phase Shift with CMB Data

Two Independent Methods

A new parameter that *only* controls the size of the phase shift.

Spectrum-Based Method

Directly in the CMB spectra

$$N_{\rm eff}^{\delta\ell}$$

→ Imprints **multipole** shift

$$\Delta\ell(N_{\rm eff}^{\delta\ell}, N_{\rm eff}) = \left[A(N_{\rm eff}^{\delta\ell}) - A(N_{\rm eff})\right] f_\ell$$

in the spectra:

$$\mathcal{C}_\ell \to \mathcal{C}_{\ell+\Delta\ell}$$

Perturbation-Based Method

Directly in the perturbations

$$N_{\rm eff}^{\delta\phi}$$

→ Imprints **k-mode** shift

$$\Delta\phi(k, z; N_{\rm eff}^{\delta\phi}, N_{\rm eff}) = \left[A(N_{\rm eff}^{\delta\phi}) - A(N_{\rm eff})\right] f_\phi(k, z)$$

in the source functions:

$$S_X(k, z) \to S_X(k + \Delta\phi/r_s, z)$$

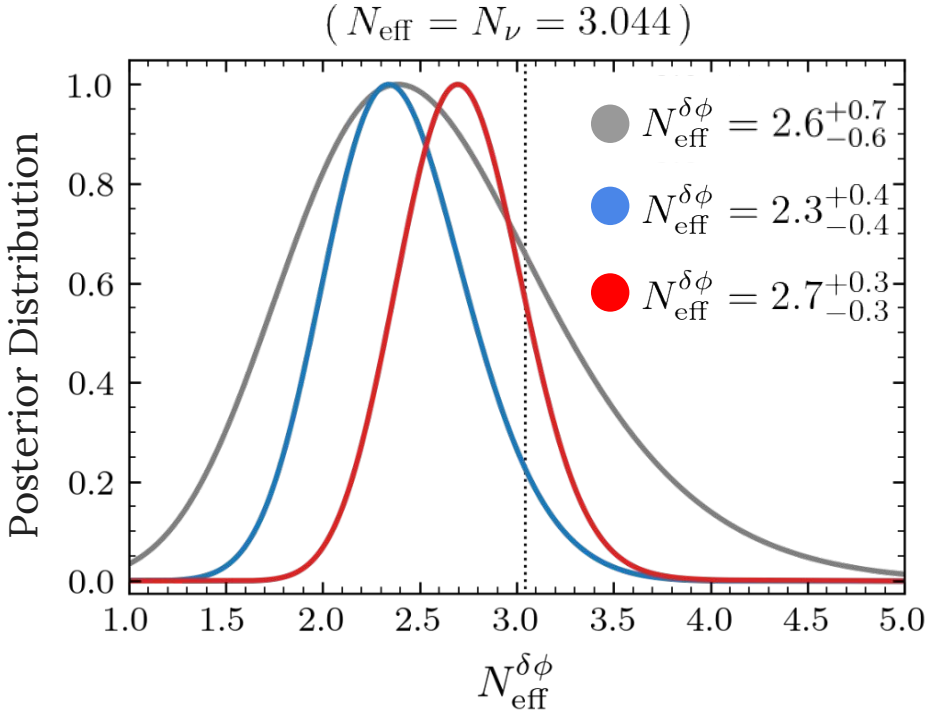
→ Results in the observed signature in the CMB spectra.

GM, Wallisch & Freese (2025)

Phase Shift Constraints from CMB Data

Perturbation-Based Analysis

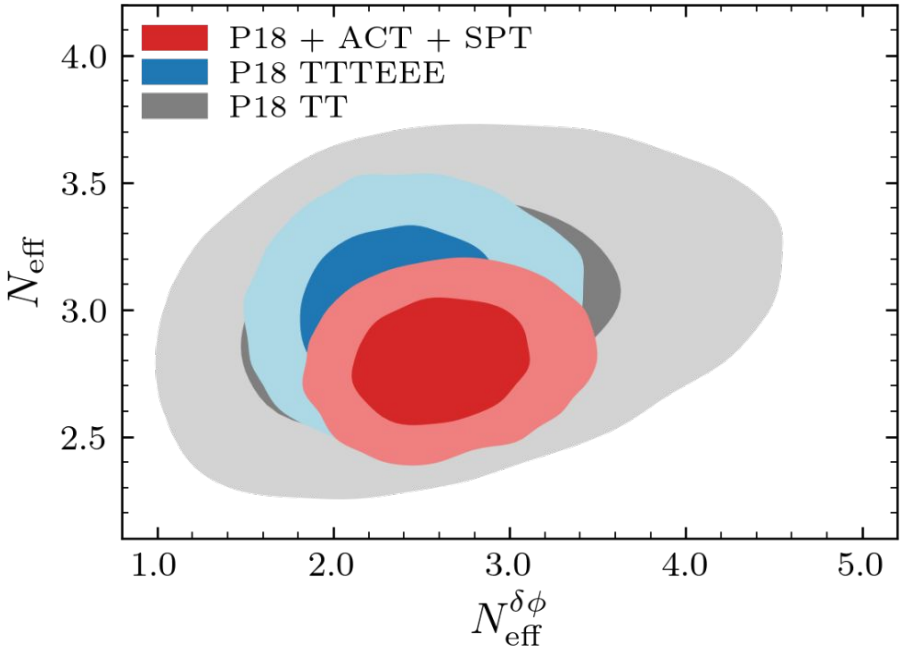
P18 + ACT + SPT
P18 TTTEEE
P18 TT

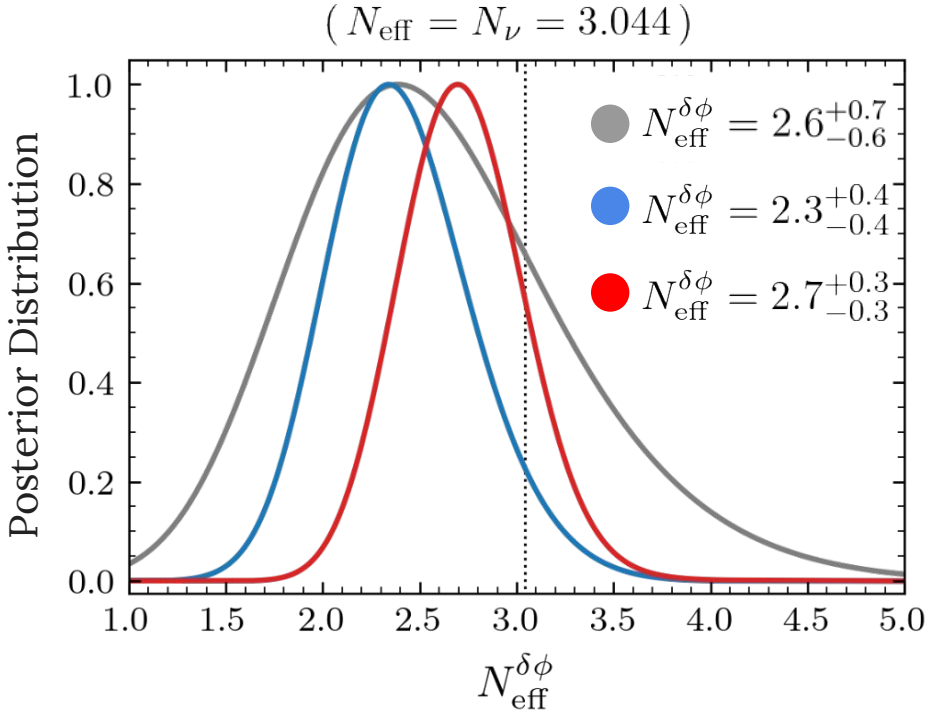


GM, Wallisch & Freese (2025)

Phase Shift Constraints from CMB Data

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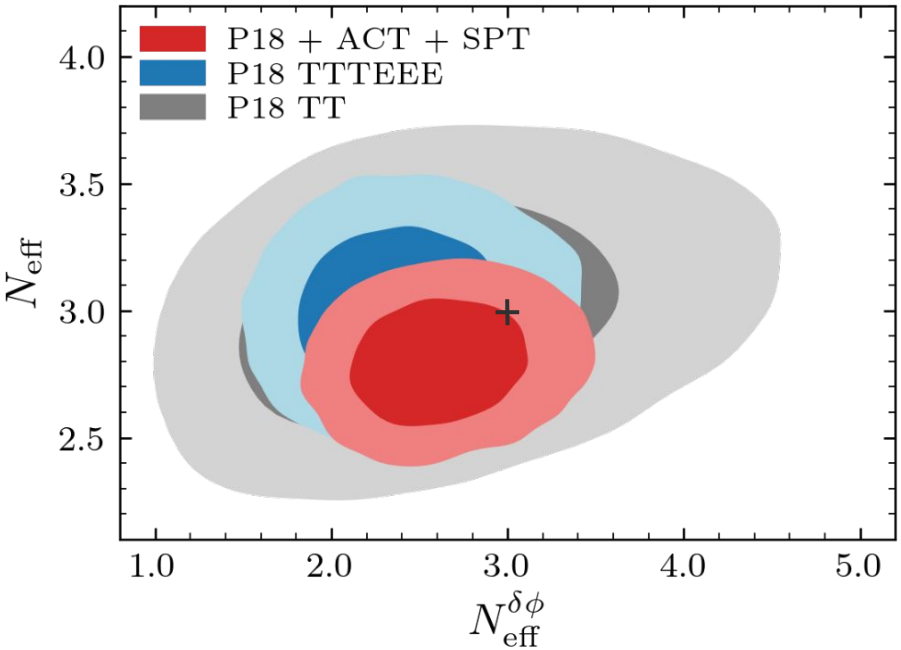




GM, Wallisch & Freese (2025)

Phase Shift Constraints from CMB Data

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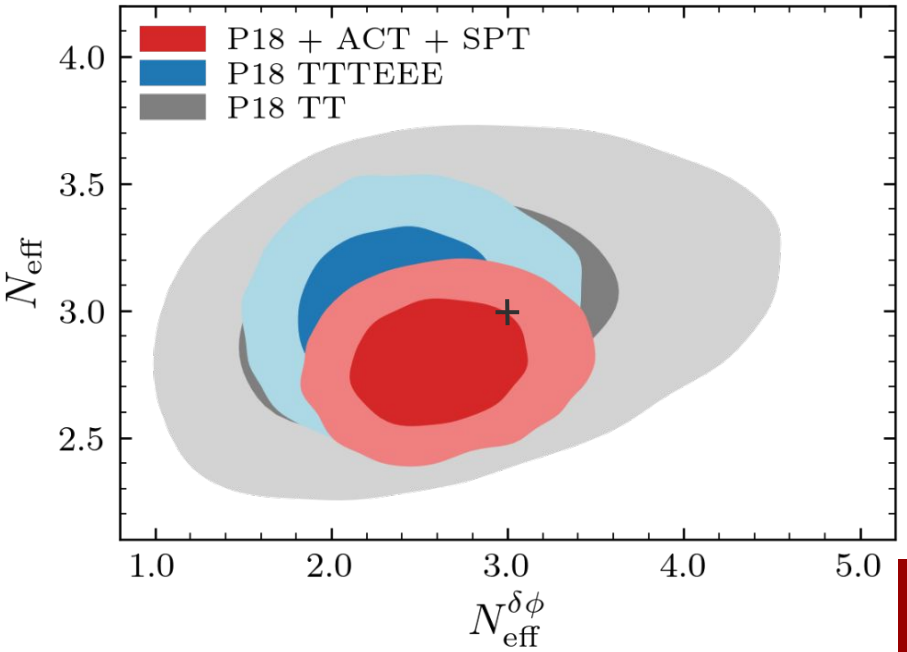


- Strong evidence of **free-streaming nature** of neutrinos
- Current CMB data are **compatible** with the three **Standard Model** neutrinos
  - Implications for physics beyond the Standard Model

GM, Wallisch & Freese (2025)

Phase Shift Constraints from CMB Data

Perturbation-Based Analysis



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Can we use these phase-shift measurements to directly probe BSM neutrino physics?

GM, Wallisch & Freese (2025)

Neutrino Self-Interactions
A Representative Example

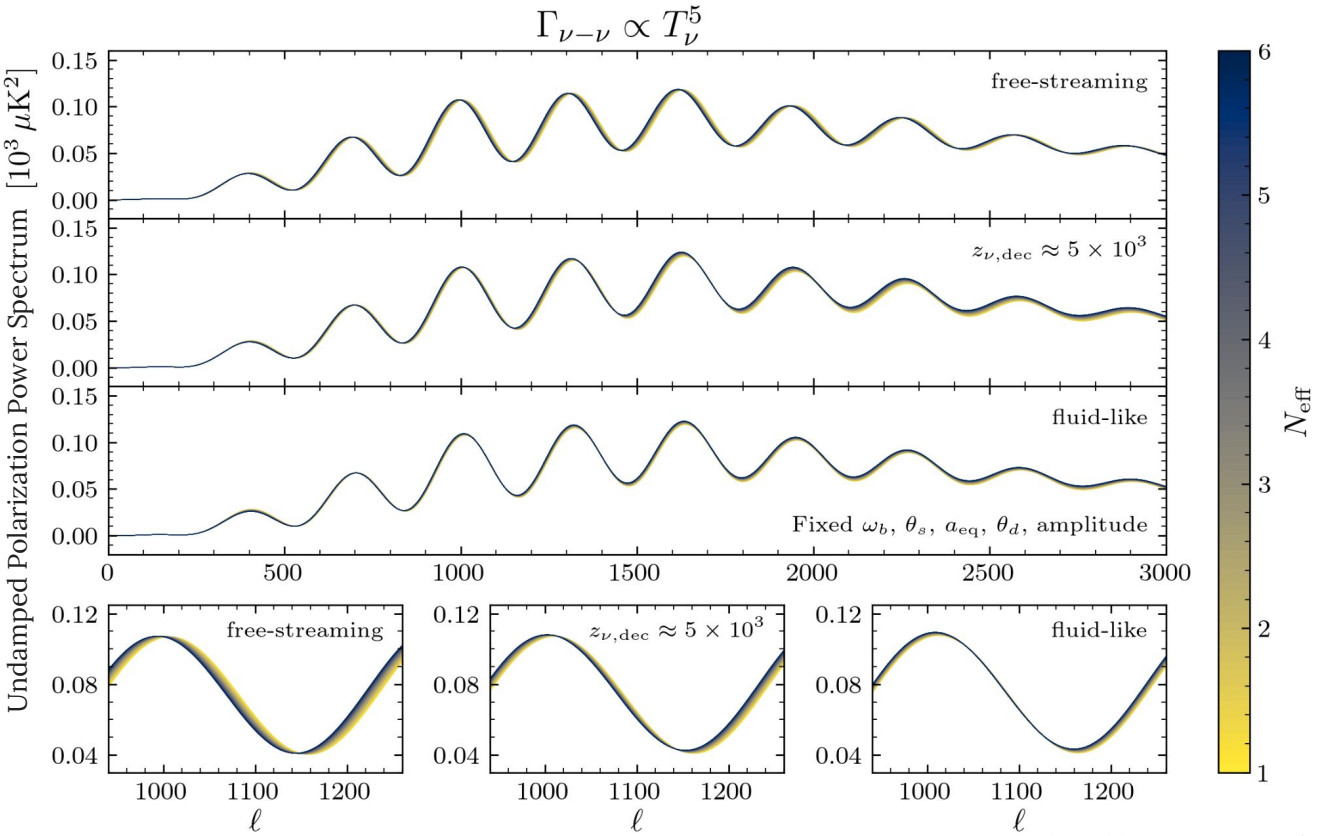
$$\Gamma_{\nu-\nu} = G_{\rm eff}^2 T_\nu^5$$

Kreisch, Cyr-Racine & Dor´ (2019)



Delaying neutrino decoupling **decreases** the size of the imprinted phase shift

$$\Gamma_{\nu-\nu}(G_{\rm eff}, z_{\nu,\rm dec})/H(z_{\nu,\rm dec}) = 1$$



GM, Ghosh, Boddy, Ho & Tsai 2025

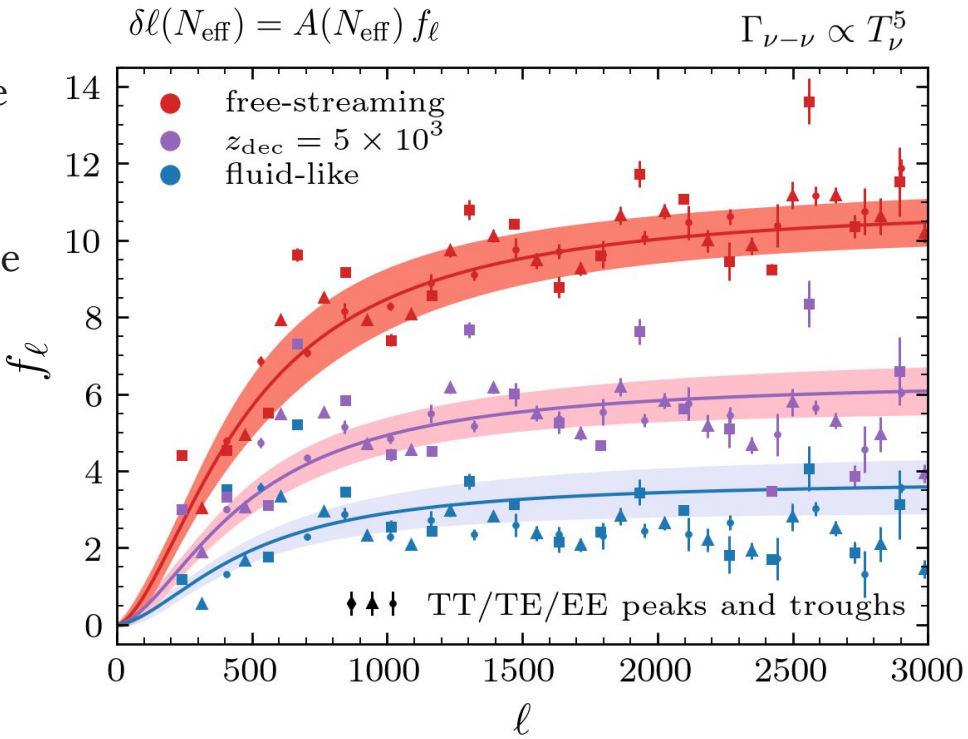
Neutrino Self-Interactions
A Representative Example

Delaying neutrino decoupling **decreases** the size of the imprinted phase shift

- Well approximated by an **amplitude offset only** relative to the free-streaming template

$$f_\ell = \frac{\ell_\infty}{1 + (\ell/\ell_\star)^\xi} = f_\ell^{\mathrm{SM}\nu} \times \mathcal{A}_\infty$$

$$\mathcal{A}_\infty \equiv \ell_\infty / \ell_\infty^{\mathrm{SM}\nu}$$ (Asymptotic Amplitude Ratio)



GM, Ghosh, Boddy, Ho & Tsai 2025

Neutrino Self-Interactions
A Representative Example

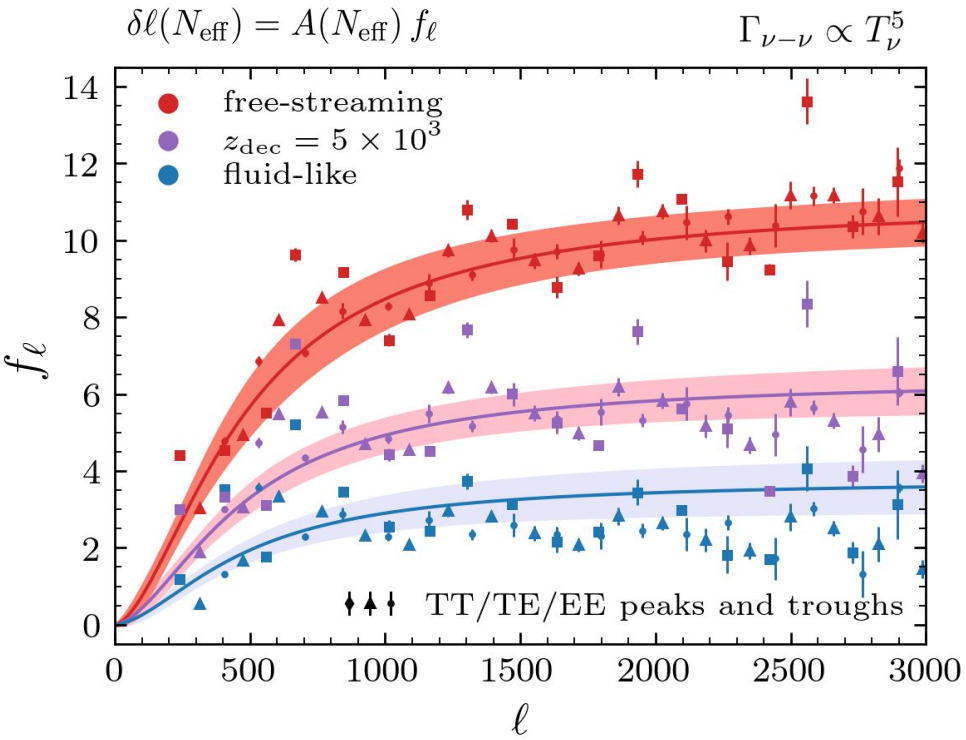
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- The **stronger** the interaction, the **smaller** the **propagation speed** of neutrino perturbations and the induced phase shift

$$c_{s,\mathrm{int}\nu} \in [c/\sqrt{3}, c]$$

$$c_{s,\mathrm{int}\nu} > c_{s,\gamma} \equiv c/\sqrt{3\,[1+3\rho_b/(4\rho_\gamma)]}$$



GM, Ghosh, Boddy, Ho & Tsai 2025

Neutrino Self-Interactions
A Representative Example

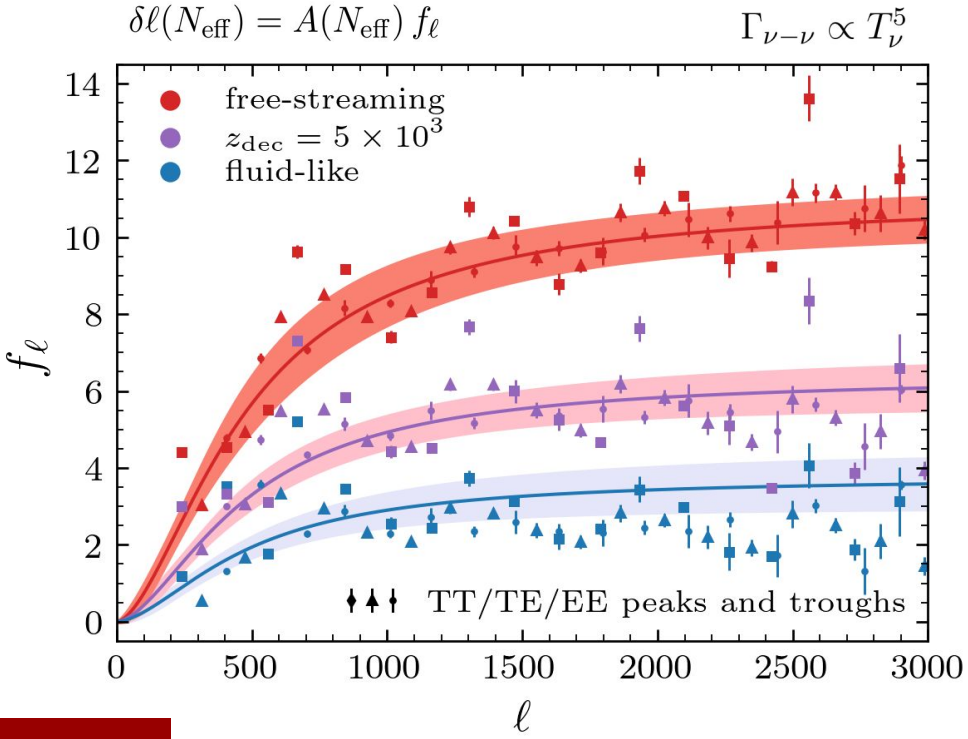
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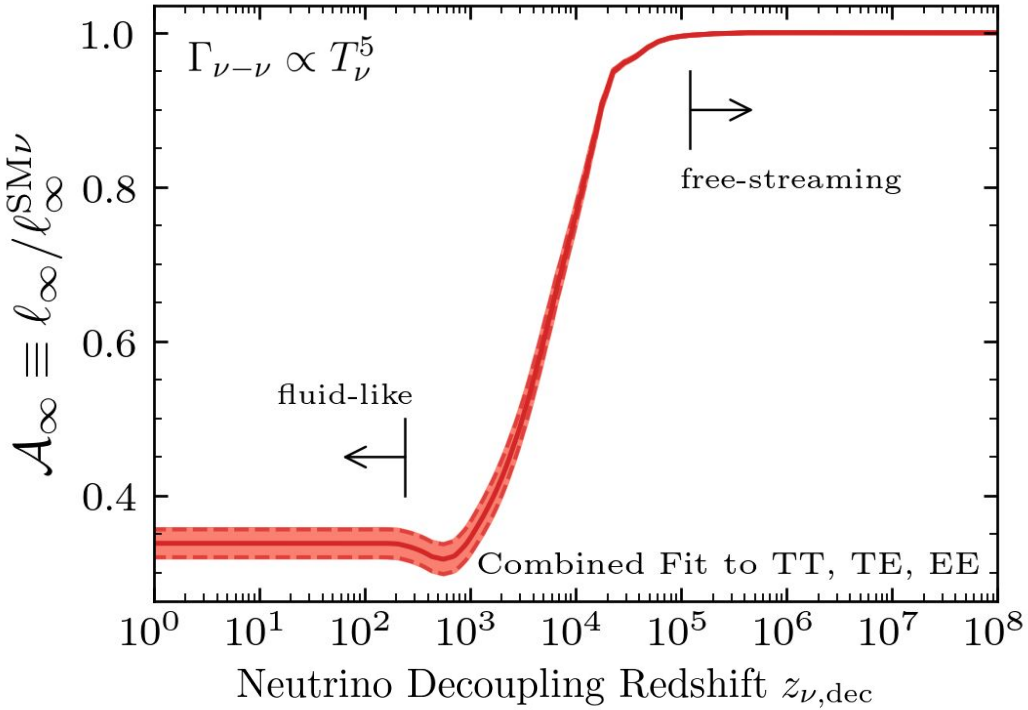
Even fluid-like neutrinos induce a **non-negligible** phase shift in the CMB spectra

GM, Ghosh, Boddy, Ho & Tsai 2025

Neutrino Self-Interactions
Multipole Shift Template

Direct mapping from phase-shift amplitude to **decoupling redshift**

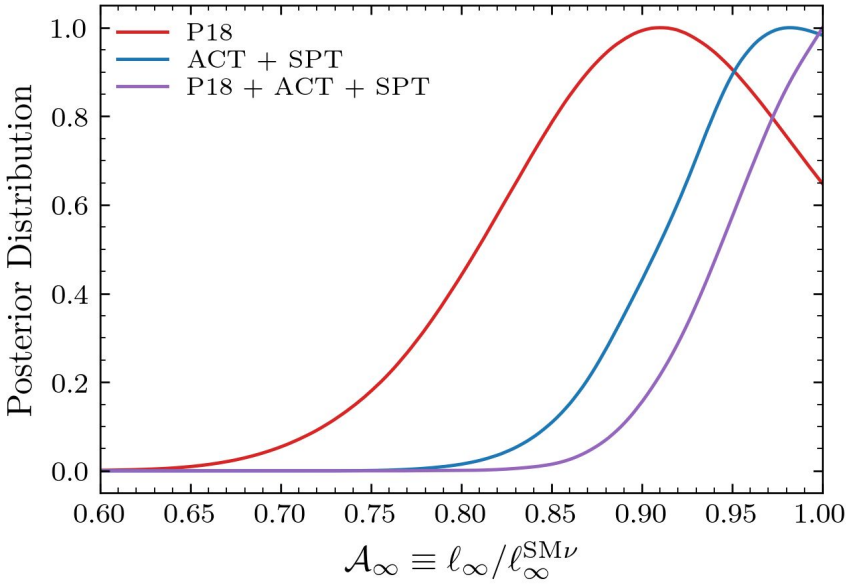
- fluid-like limit ~30% of (SM) free-streaming limit



GM, Ghosh, Boddy, Ho & Tsai 2025

Neutrino Self-Interactions
Direct Phase-Shift Constraints from CMB data

$(N_{\rm eff} = N_\nu = 3.044)$



- A new parameter to control the shift

$$\mathcal{A}_\infty$$

$$\Delta\ell(\mathcal{A}_\infty, N_{\rm eff}) \equiv (\mathcal{A}_\infty - 1) \times A(N_{\rm eff}) f_\ell^{\rm SM\nu}$$

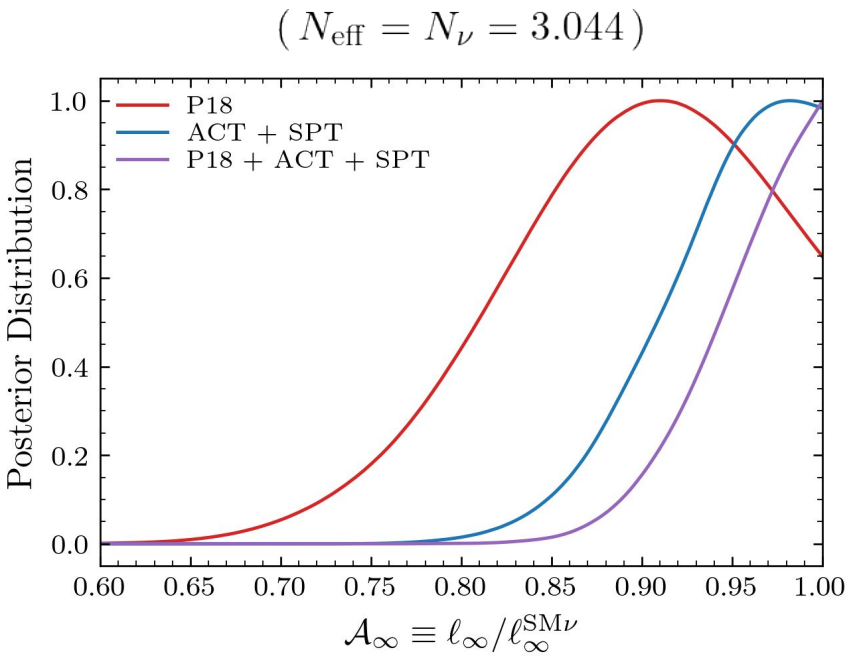
$$\mathcal{C}_\ell \to \mathcal{C}_{\ell+\Delta\ell}$$

- Equivalent to the **spectrum-based analysis**

$$\mathcal{A}_\infty = \epsilon(N_{\rm eff}^{\delta\ell})/\epsilon(3.044)$$

GM, Ghosh, Boddy, Ho & Tsai 2025

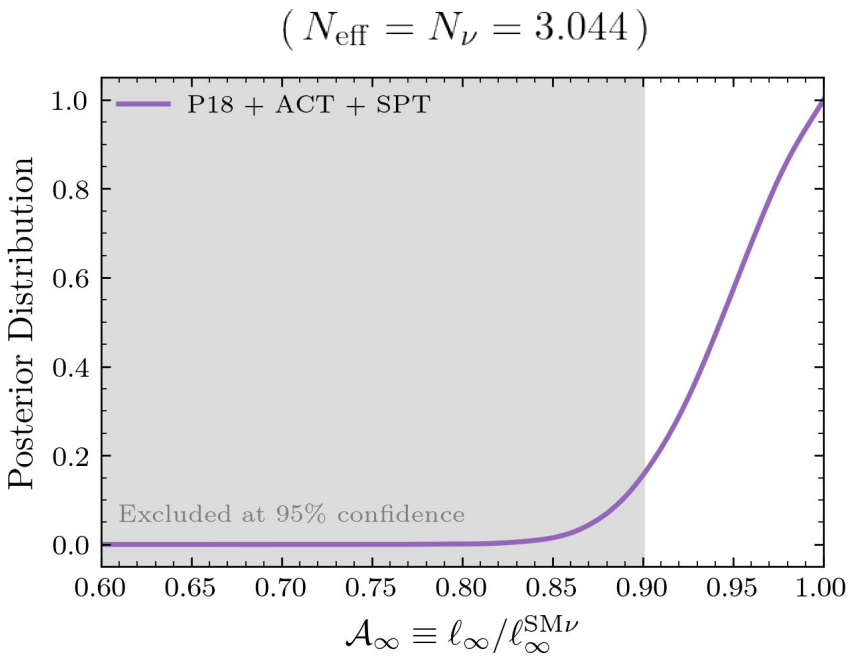
Neutrino Self-Interactions
Direct Phase-Shift Constraints from CMB data



- CMB data remain **compatible** with the three **Standard Model** neutrinos
- Ground-based experiments **surpassed** Planck's sensitivity to the phase shift!

GM, Ghosh, Boddy, Ho & Tsai 2025

Neutrino Self-Interactions
Direct Phase-Shift Constraints from CMB data



- CMB data remain **compatible** with the three **Standard Model** neutrinos

- Ground-based experiments **surpassed** Planck's sensitivity to the phase shift!

$$\mathcal{A}_\infty > 0.90$$

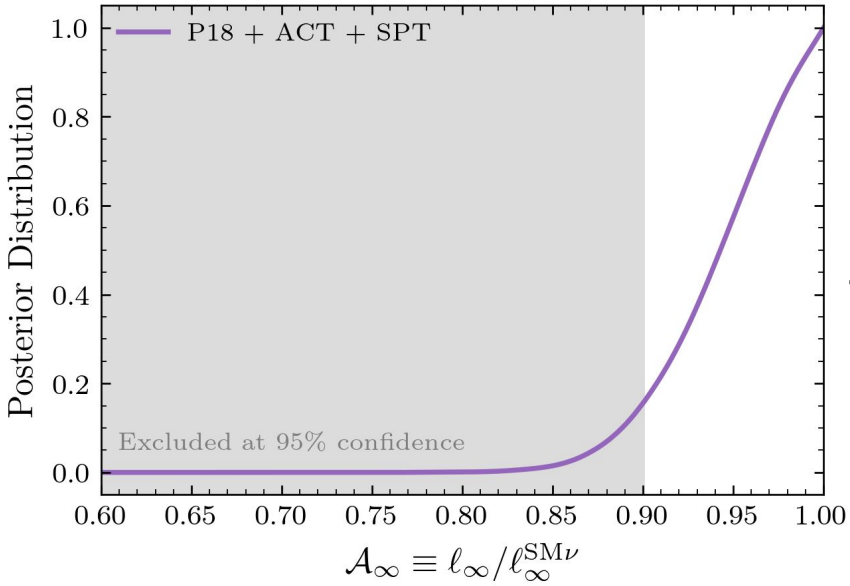
GM, Ghosh, Boddy, Ho & Tsai 2025

Neutrino Self-Interactions

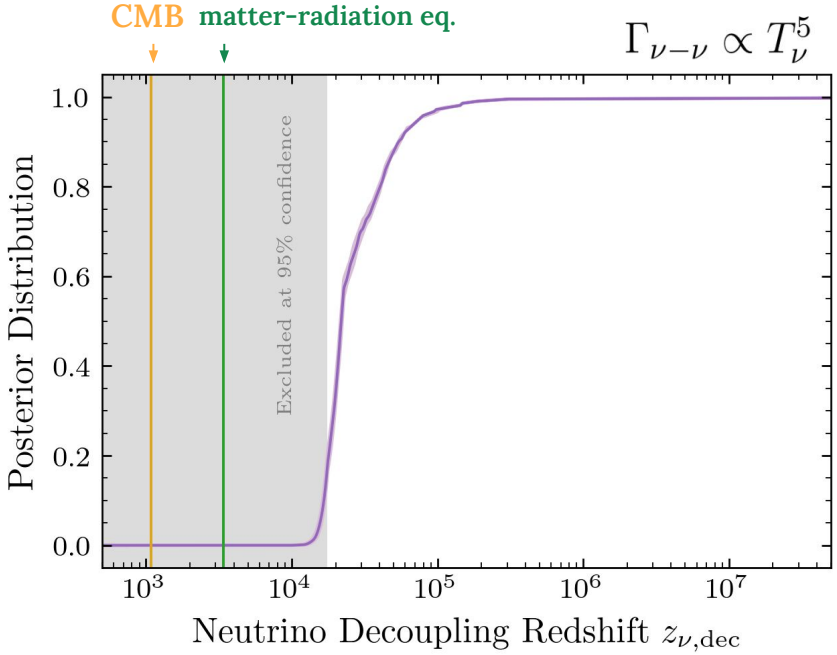
Direct Phase-Shift Constraints from CMB data

$$\Delta\ell(\mathcal{A}_\infty, N_{\rm eff}) \equiv (\mathcal{A}_\infty - 1) \times A(N_{\rm eff}) f_\ell^{\rm SM\nu}$$

$$\mathcal{C}_\ell \to \mathcal{C}_{\ell+\Delta\ell}$$



$\Gamma_{\nu-\nu} \propto T_\nu^5$



GM, Ghosh, Boddy, Ho & Tsai 2025

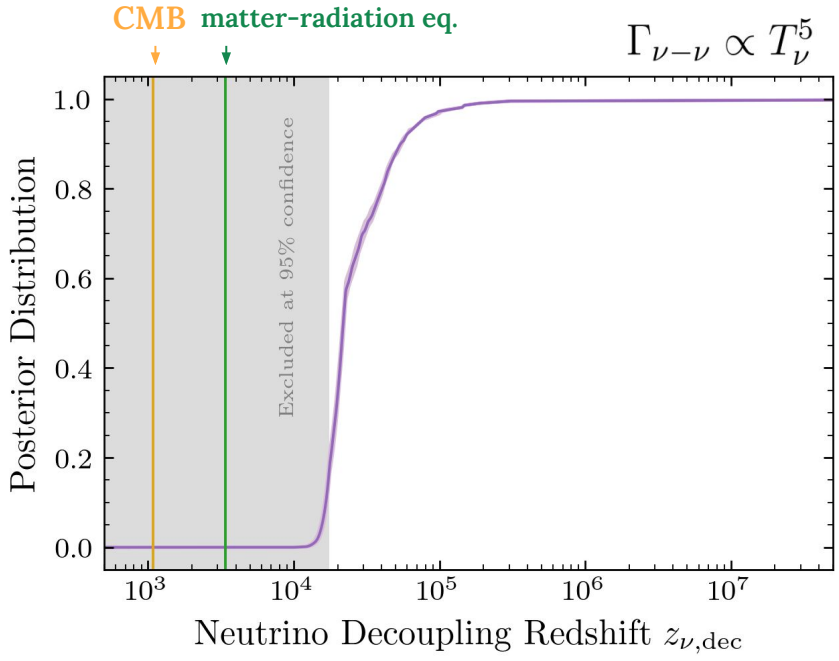
Neutrino Self-Interactions
Direct Phase-Shift Constraints from CMB data

- Neutrinos must be already **free-streaming deep** within the **radiation-dominated** epoch
- Strong BSM physics constraint from **single robust signature**!



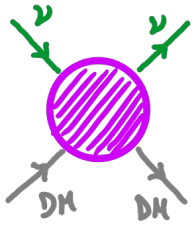
$$z_{\nu,\mathrm{dec}} > 1.7 \times 10^4$$

$$G_{\mathrm{eff}} < 9.4 \times 10^{-4}\,\mathrm{MeV}^{-2}$$



GM, Ghosh, Boddy, Ho & Tsai 2025

Additional Realistic Neutrino-Interactions
Multipole Shift Template



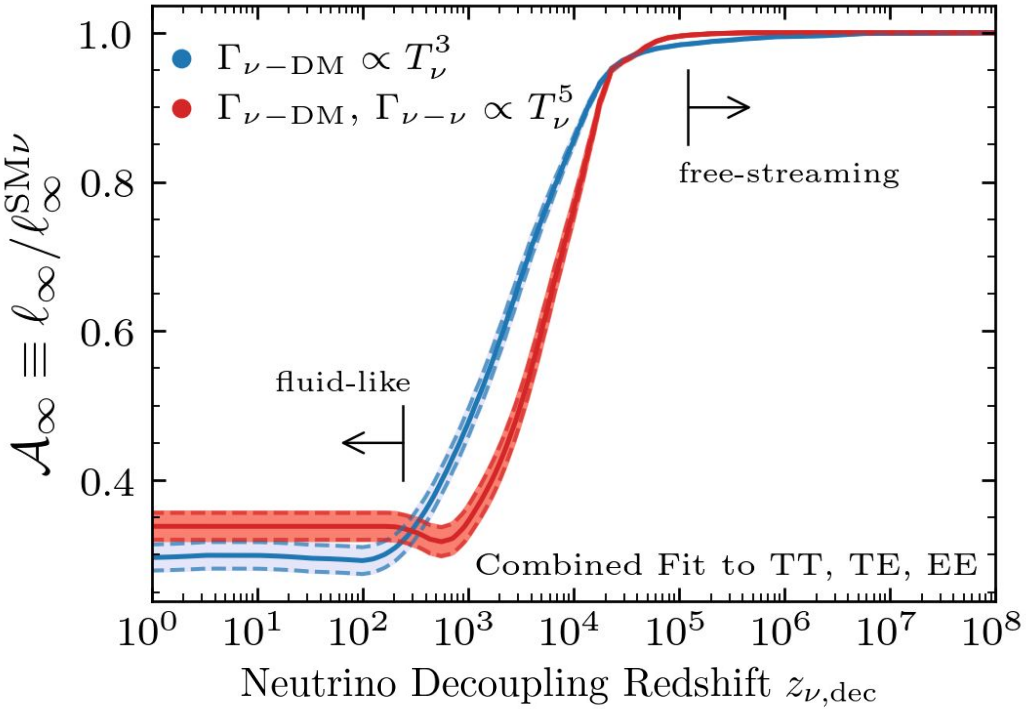
$$\Gamma_{\nu-\mathrm{DM}} \propto T_\nu^3 \longrightarrow \text{const. } \sigma_{\nu-\mathrm{DM}}$$

(Compton scattering like)

$$\Gamma_{\nu-\mathrm{DM}} \propto T_\nu^5 \longrightarrow \sigma_{\nu-\mathrm{DM}} \propto T_\nu^2$$

(electron-neutrino scattering like)

Wilkinson, Bœhm & Lesgourgues (2014)
Ghosh, Khatri & Tuhin S. Roy (2017)



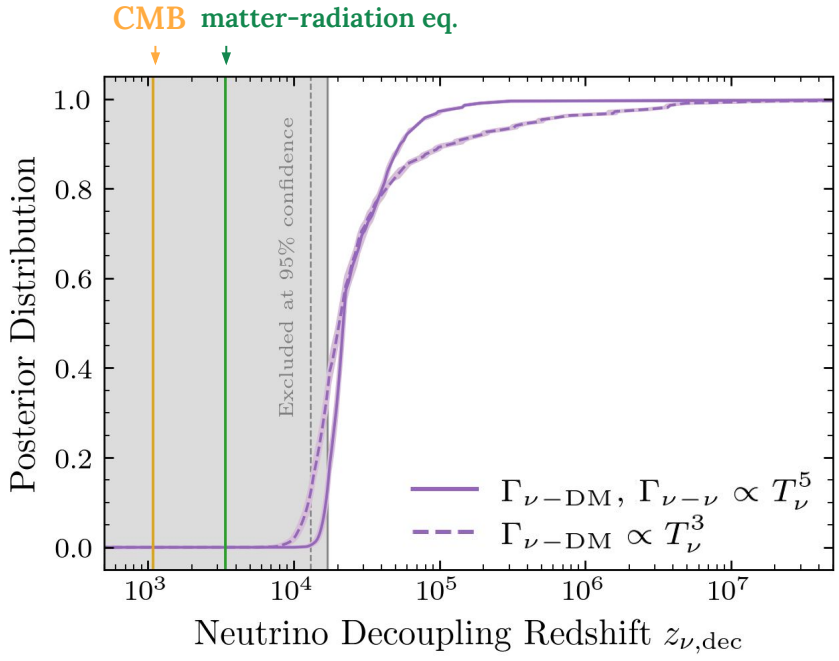
GM, Ghosh, Boddy, Ho & Tsai 2025

Neutrino Self-Interactions
Direct Phase-Shift Constraints from CMB data

- **Multipole BSM** neutrino-physics constraints from **single robust signature**
- Neutrinos are already **free-streaming deep** within the **radiation-dominated** epoch

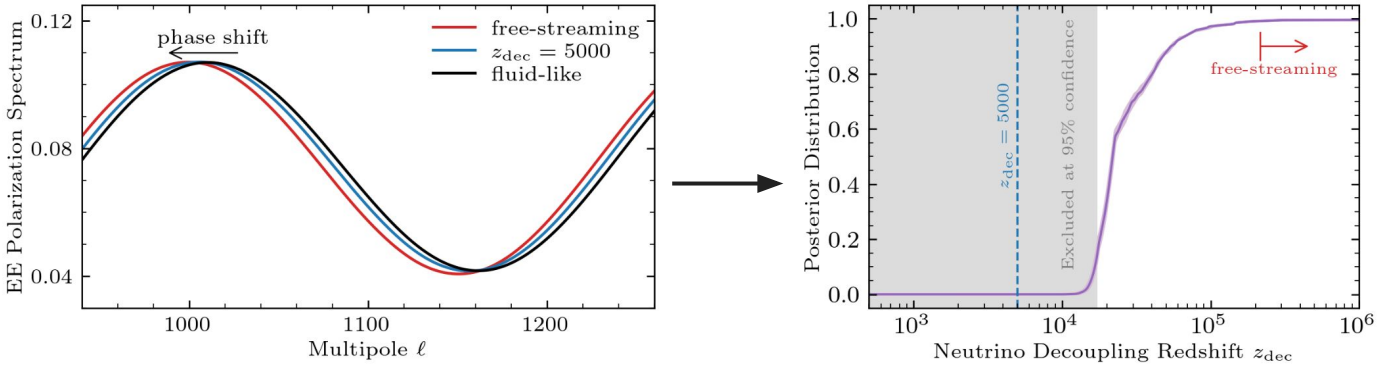
$\Gamma_{\nu-\mathrm{DM}},\ \Gamma_{\nu-\nu} \propto T_\nu^5$ $\quad$ $\Gamma_{\nu-\mathrm{DM}} \propto T_\nu^3$

$z_{\nu,\mathrm{dec}} > 1.7 \times 10^4$ $\quad$ $z_{\nu,\mathrm{dec}} > 1.3 \times 10^4$



GM, Ghosh, Boddy, Ho & Tsai 2025

Summary



- The **phase shift** is a robust probe of **free-streaming relativistic species**
- Two different model-agnostic and signature-driven approaches:
  - Spectrum-based Template
  - Perturbation-based Template
- **High-significance detection** of free-streaming neutrinos with current data **compatible** with the three **Standard Model** neutrinos
- The measured phase shift can be **directly mapped** to neutrino interactions strength, allowing competitive **constraints on BSM scenarios** using a **single robust signature**.

Summary

Signature-driven approach to connect theory and data
And probe new physics with cosmological observations

DM-SM signatures

Boddy, Freese, **GM**, Shams 2024

GM, Elor, Boddy & Bellomo 2025

GM, Hooper, Freese, Kelso, Kuhnel & Sandick 2025

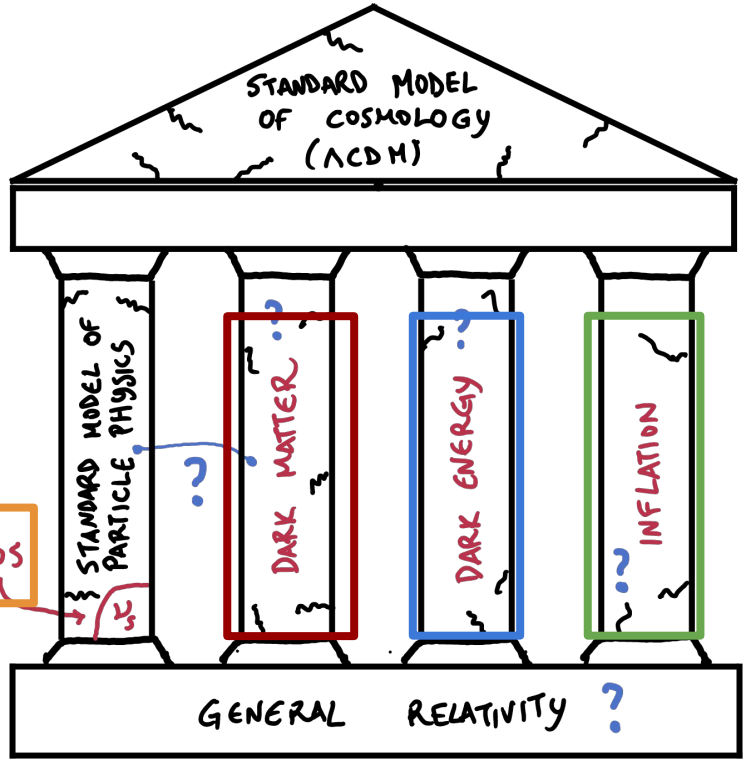
GM, Bellomo & Boddy 2026

Neutrino phase-shift

GM, Wallisch & Freese 2025

GM, Ghosh, Boddy, Ho & Tsai 2025

GM, Cruz, Muñoz, Kovetz & Kamionkowski (2025)



Warm inflation

GM, Aragam, Visinelli & Freese 2022a

GM, Aragam, Visinelli & Freese 2022b

GM, Aragam, Visinelli & Freese 2023

GM, Ramos, Vicente & Freese 2023

Freese, **GM** & Shams 2024

GM, Shams, Xu & Freese 2025

Evidence for DE

Toomey, **GM**, McDonough & Freese 2025

GM & Stiskalek 2026

Summary

Grazie per l'attenzione!

Signature-driven approach to connect theory and data
And probe new physics with cosmological observations

DM-SM signatures

Boddy, Freese, **GM**, Shams 2024

GM, Elor, Boddy & Bellomo 2025

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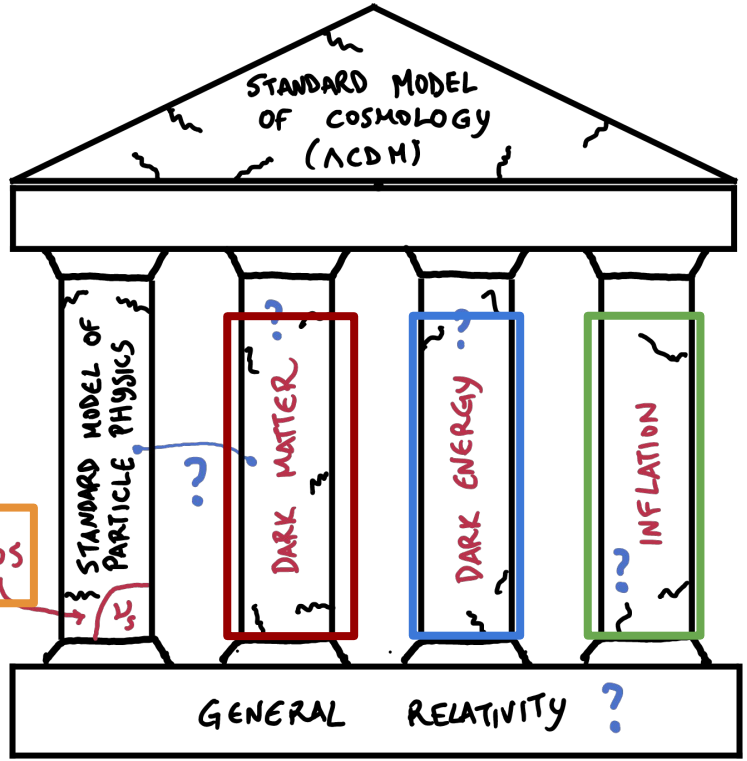
GM, Bellomo & Boddy 2026

Neutrino phase-shift

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GM, Ghosh, Boddy, Ho & Tsai 2025

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GM & Stiskalek 2026

Back-up Slides

Gabriele Montefalcone

Weinberg Institute for Theoretical Physics, University of Texas at Austin

The Phase Shift Beyond the CMB
BAO from Galaxy Surveys

The same phase shift is naturally imprinted in the spectrum of **baryon acoustic oscillations (BAOs)** extracted from galaxy surveys

- After recombination, baryons no longer feel any pressure, and their **sound wave stalls**.

At Decoupling

$$\delta_\gamma \sim \delta_b \sim \cos(kr_s + \phi)$$



The **BAO** scale

Credit: Murali Saravanan

The Phase Shift Beyond the CMB
BAO from Galaxy Surveys

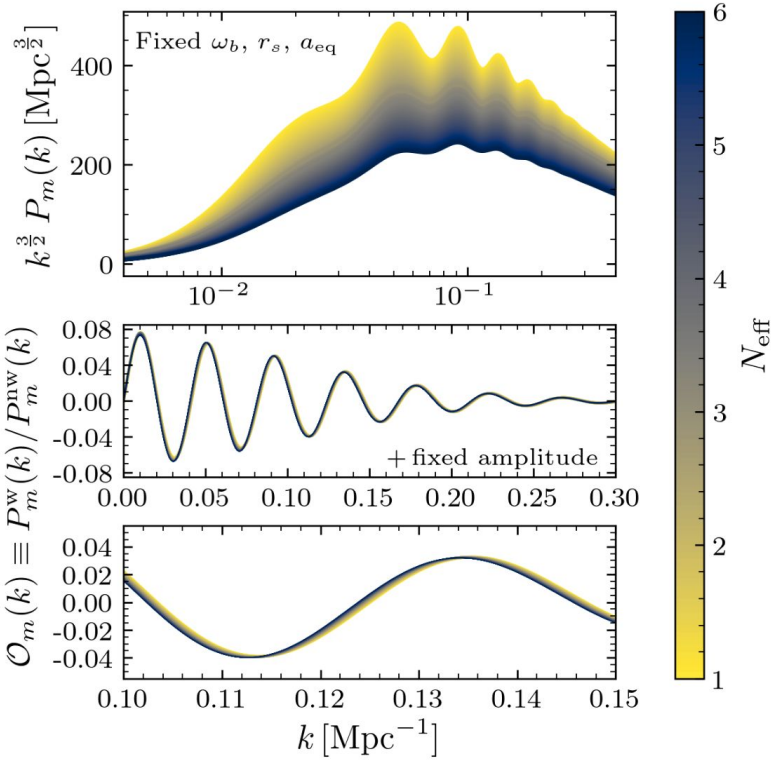
The same phase shift is naturally imprinted in the spectrum of **baryon acoustic oscillations (BAOs)** extracted from galaxy surveys

- Extracted in BOSS and DESI data providing independent evidence for the free-streaming nature of neutrinos.

Baumann, Green & Zaldarriaga (2017)
Baumann, Green & Wallisch (2017)
Baumann et. al. (2019)
Whitford et. al. (2024)

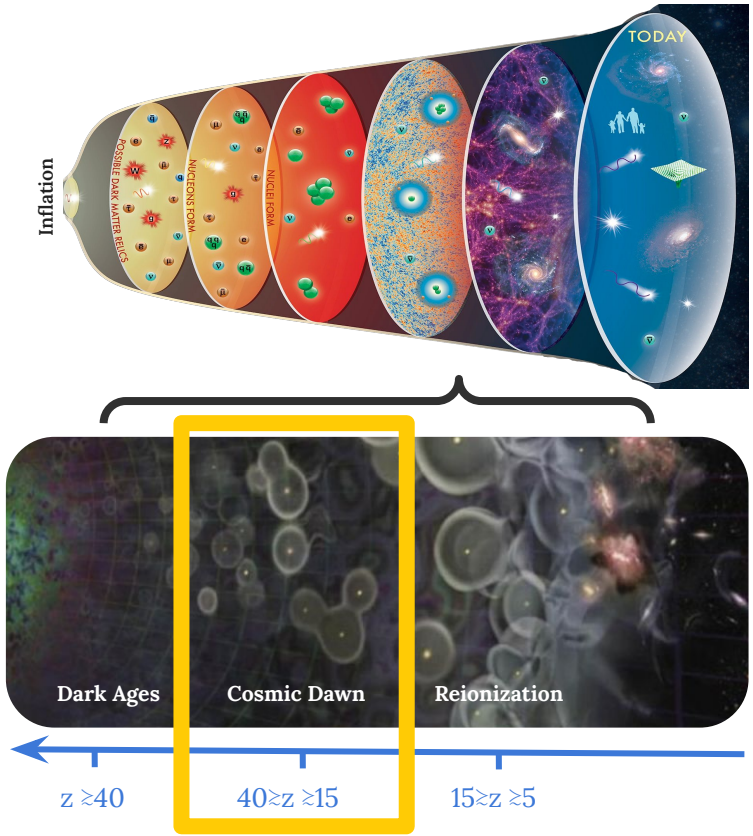
$$\beta \equiv \epsilon(N_{\rm eff}^{\delta\phi})/\epsilon(3.044)$$

$$\sigma_\beta \sim 0.5$$



The Phase Shift Beyond the CMB

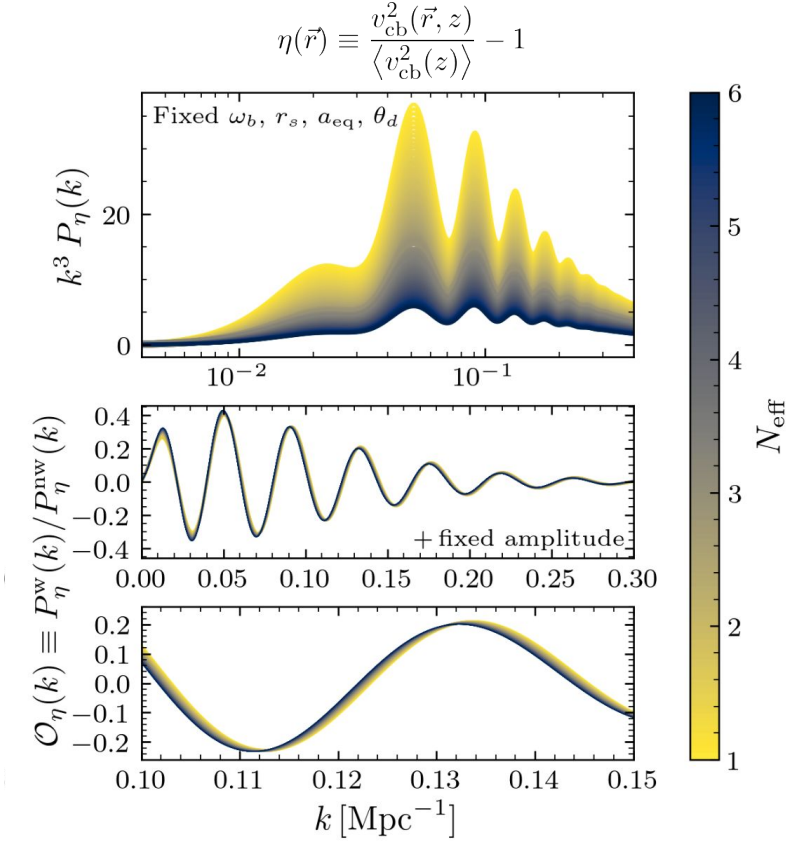
The 21-cm line from Cosmic Dawn



The Phase Shift Beyond the CMB
Velocity Acoustic Oscillations

The phase shift will also appear in the 21-cm spectrum at cosmic dawn!

- **Distinct** redshift- and mode-dependence from that observed in the CMB and BAO spectra, due to the presence of **velocity acoustic oscillations** (VAOs) sourced by DM-baryon relative velocities.

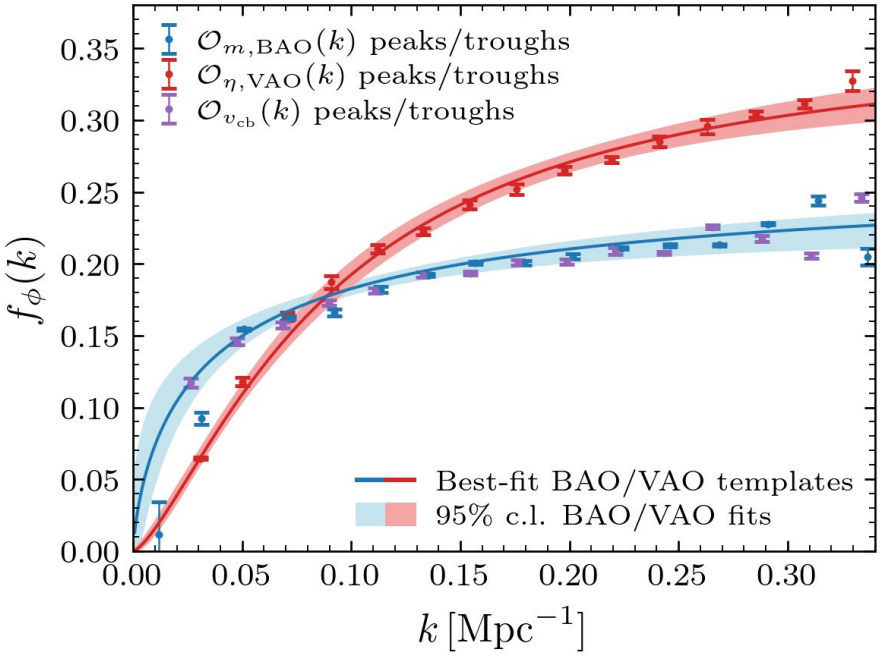


GM, Cruz, Munoz, Kovetz, & Kamionkowski (2025)

The Phase Shift Beyond the CMB
BAO vs VAO Phase Template

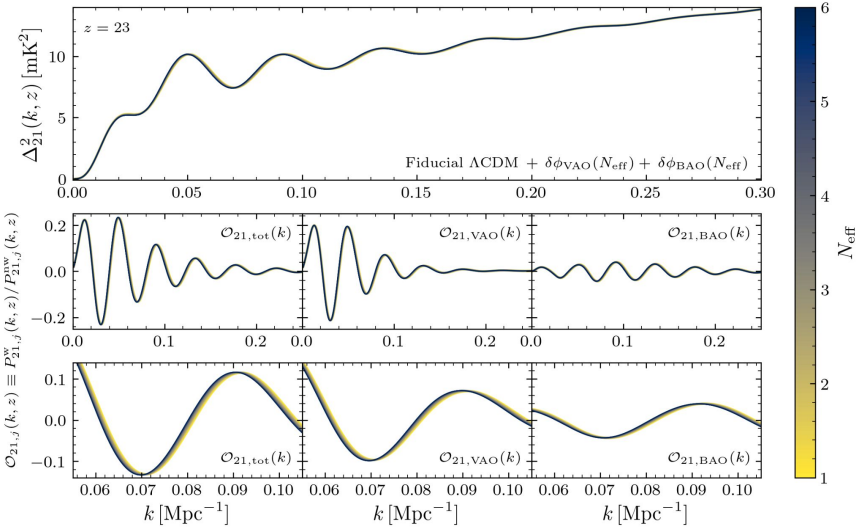
The phase shift will also appear in the 21-cm spectrum at cosmic dawn!

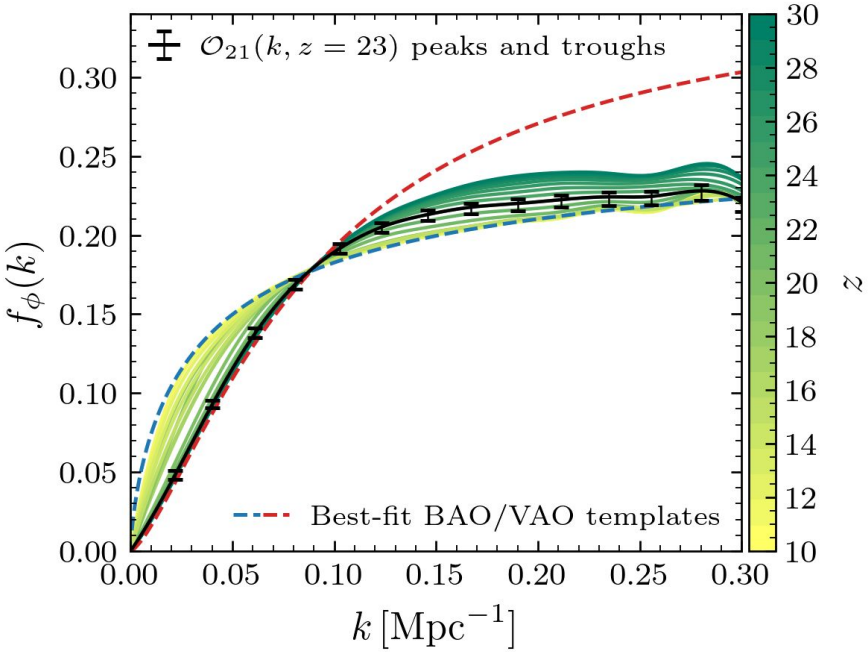
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GM, Cruz, Munoz, Kovetz, & Kamionkowski (2025)

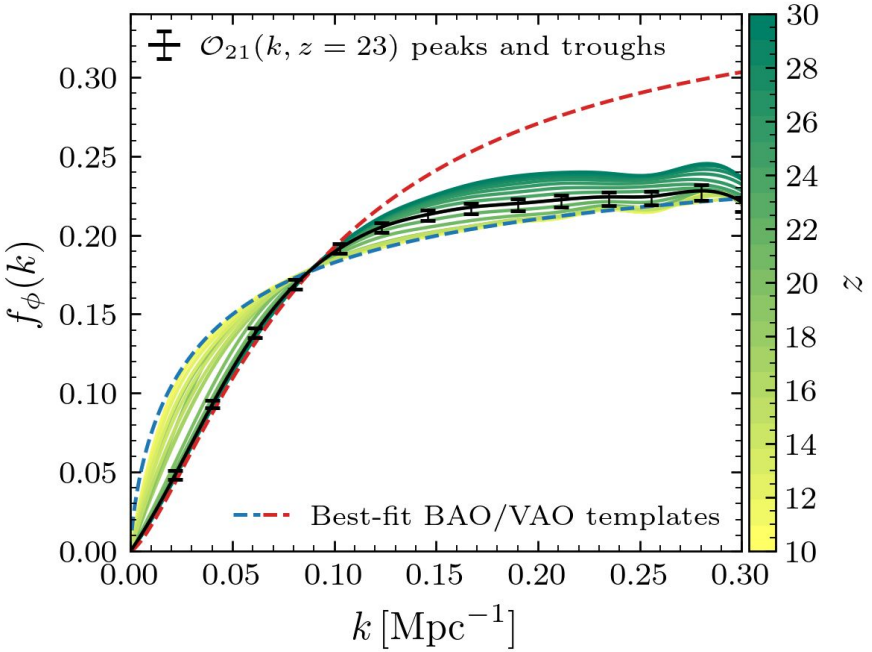
The Phase Shift Beyond the CMB
The Phase in the 21-cm Spectrum





GM, Cruz, Munoz, Kovetz, & Kamionkowski (2025)

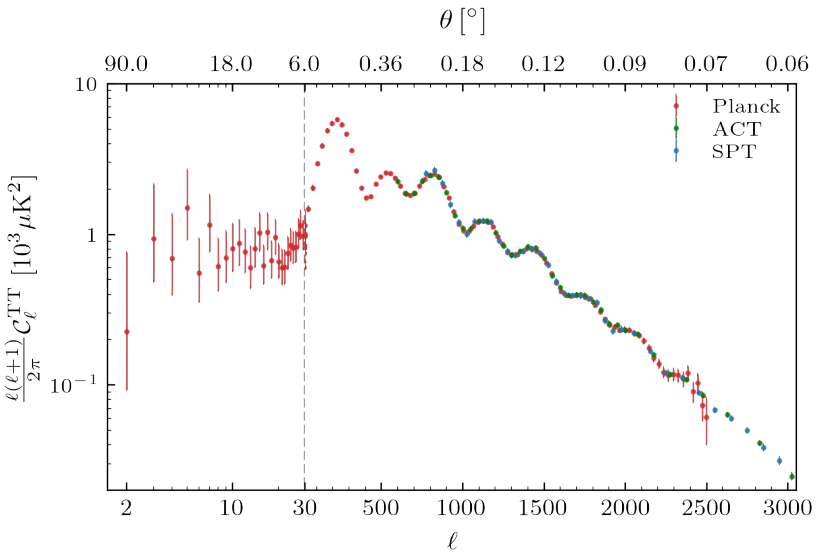
The Phase Shift Beyond the CMB
The Phase in the 21-cm Spectrum



GM, Cruz, Munoz, Kovetz, & Kamionkowski (2025)

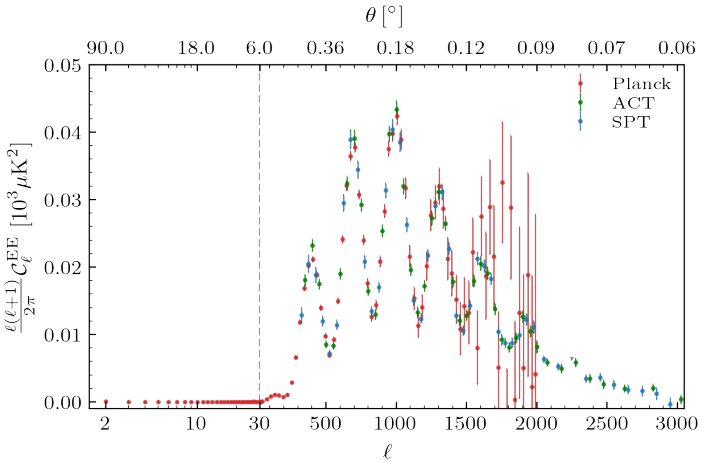
The CMB Power Spectra

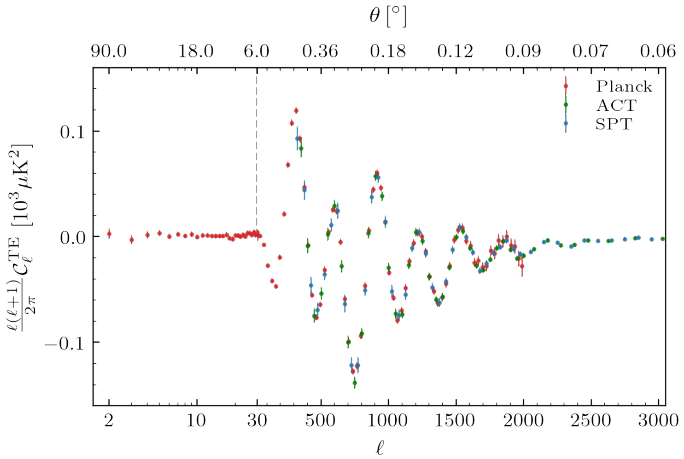
Temperature spectrum traces **density** perturbations, roughly the gravitational potential



TE spectrum roughly tells us how the plasma is moving into the gravitational potential wells

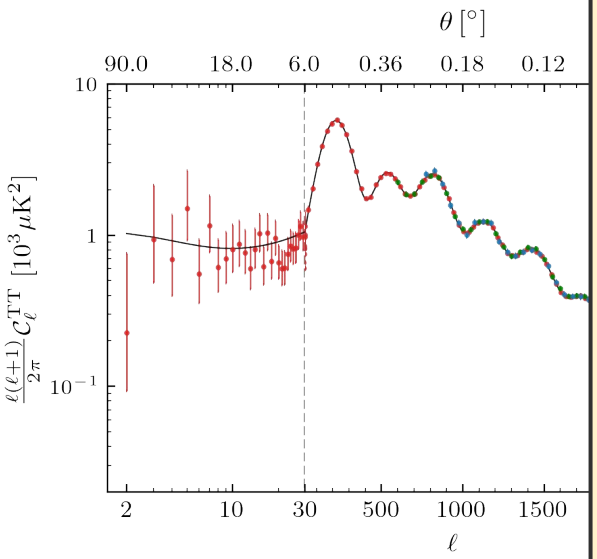
Polarization spectrum traces **velocity** perturbations



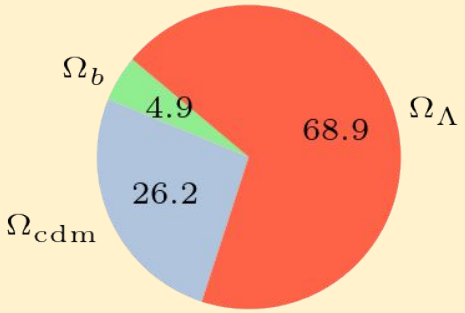


The CMB Power Spectra

Temperature spectrum traces **density**
roughly the gravitational potential



TE spectrum roughly
moving into the gravi



$\ln(10^{10} A_s) = 3.045$

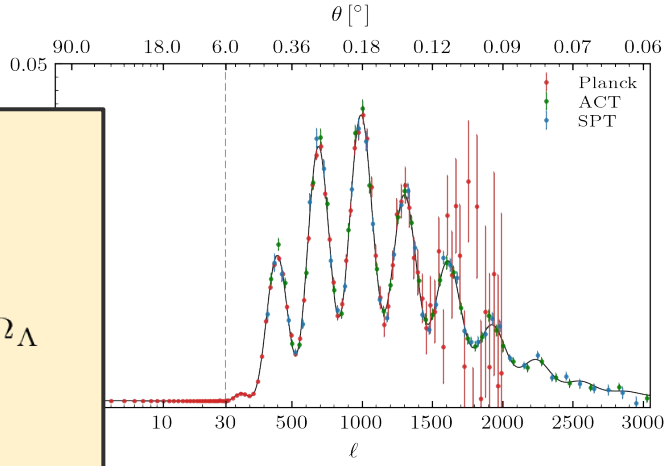
$n_s = 0.965$

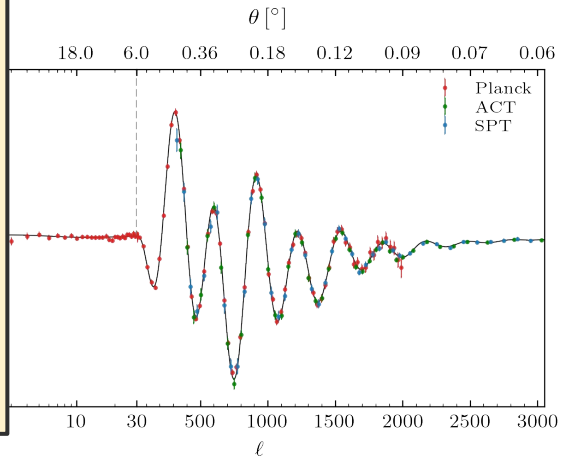
$\tau_{reio} = 0.054$

+ SM neutrinos $N_{eff} = 3.044$

+ ...

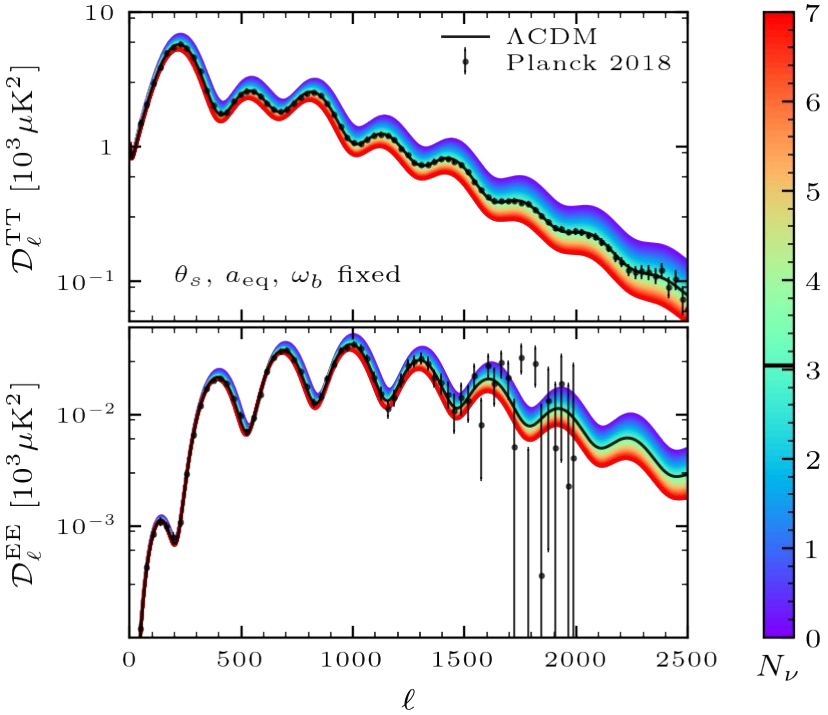
Polarization spectrum traces **velocity** perturbations

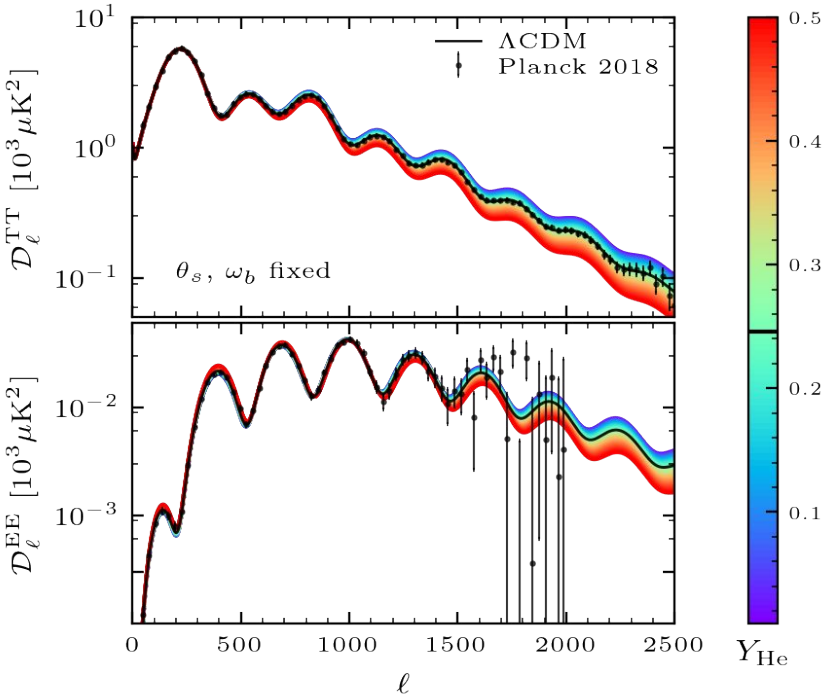




Cosmic neutrinos in the CMB
energy density

$$\theta_d \propto (H/n_e)^{1/2}\theta_s$$





Degeneracy with primordial Helium fraction $\mathbf{Y_{He}}$ via $\mathbf{n_e}$

Cosmic Neutrinos in the CMB
Extraction of the Multipole Shift Template

$$\delta\ell(N_{\rm eff}) = A(N_{\rm eff})\, f_\ell$$

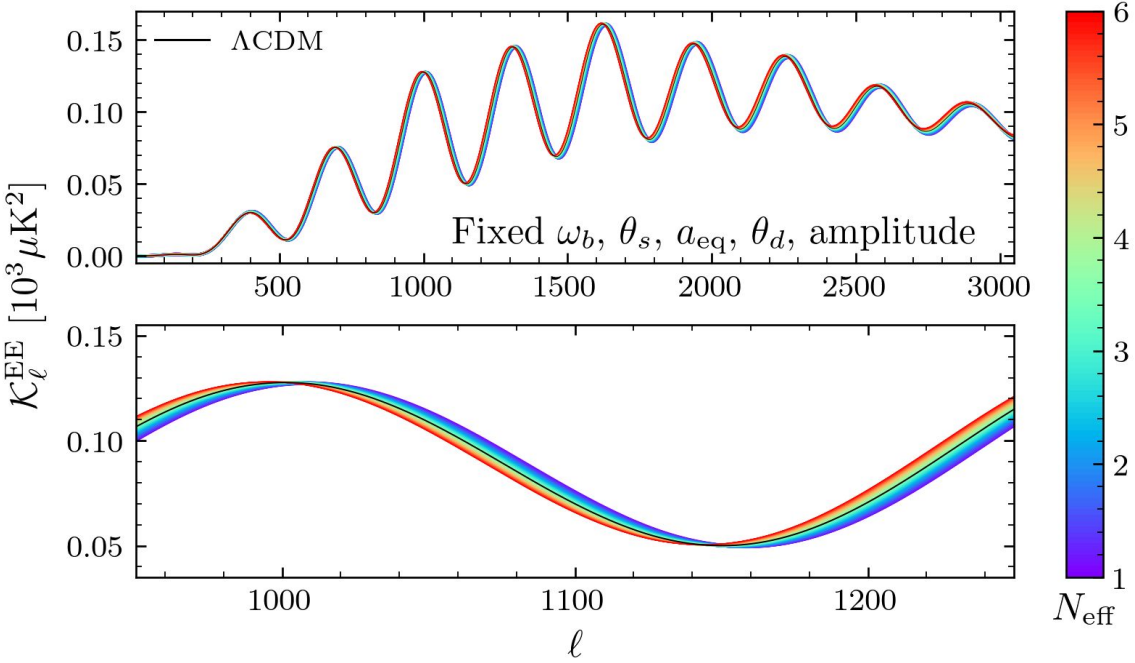
Multipole Dependence

$$f_\ell = \frac{\ell_\infty}{1 + (\ell/\ell_\star)^\xi}$$

Normalized Amplitude

$$A(N_{\rm eff}) \equiv \frac{\epsilon(N_{\rm eff}) - \epsilon(3.044)}{\epsilon(1) - \epsilon(3.044)}$$

$$\mathcal{K}_\ell^{XY} \equiv \frac{\ell(\ell+1)}{2\pi}\mathcal{C}_\ell^{XY} \exp\left\{a(\ell\theta_d)^\kappa\right\}$$



Following Follin, Knox, Millea & Pan (2015)

Cosmic Neutrinos in the CMB
Extraction of the Multipole Shift Template

$$\delta\ell(N_{\rm eff}) = A(N_{\rm eff})\, f_\ell$$

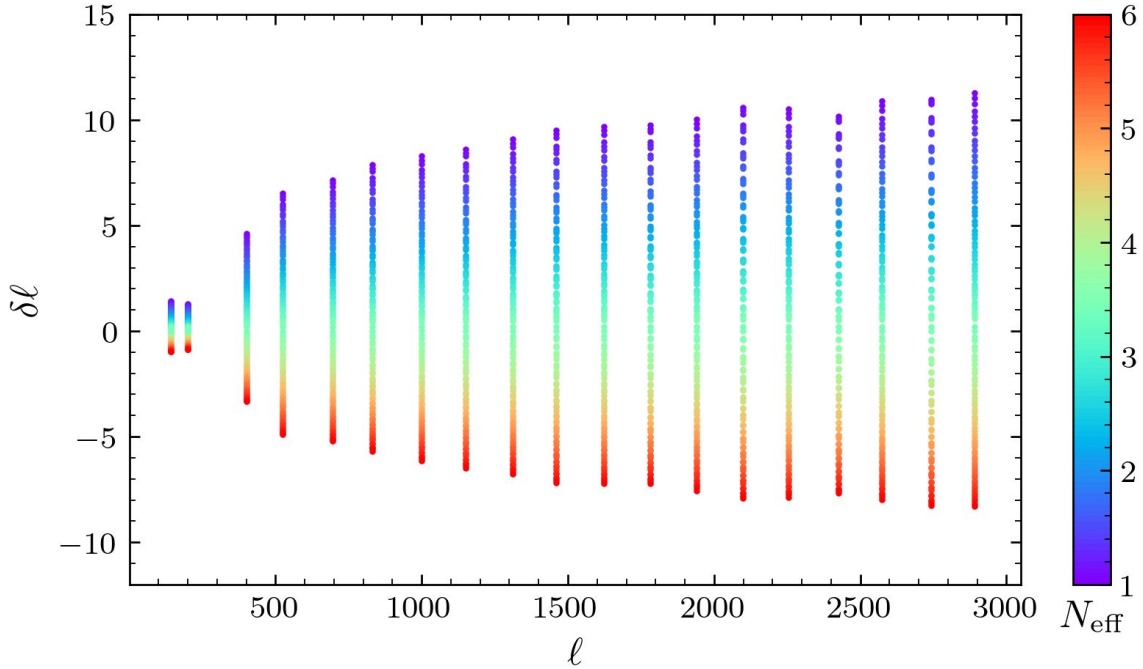
Multipole Dependence

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Normalized Amplitude

$$A(N_{\rm eff}) \equiv \frac{\epsilon(N_{\rm eff}) - \epsilon(3.044)}{\epsilon(1) - \epsilon(3.044)}$$

1. Compute the multipole shifts w.r.t. the fiducial cosmology.



Following Follin, Knox, Millea & Pan (2015)

Cosmic Neutrinos in the CMB
Extraction of the Multipole Shift Template

$$\delta\ell(N_{\rm eff}) = A(N_{\rm eff})\, f_\ell$$

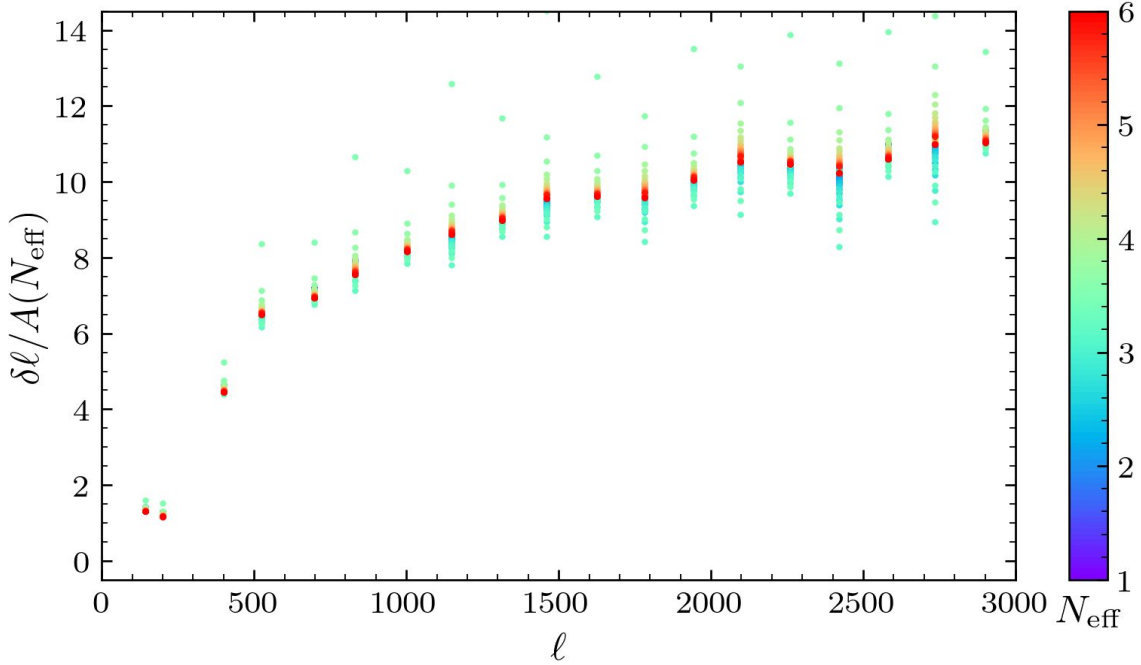
Multipole Dependence

$$f_\ell = \frac{\ell_\infty}{1 + (\ell/\ell_\star)^\xi}$$

Normalized Amplitude

$$A(N_{\rm eff}) \equiv \frac{\epsilon(N_{\rm eff}) - \epsilon(3.044)}{\epsilon(1) - \epsilon(3.044)}$$

2. Divide by the respective phase shift amplitude.



Following Follin, Knox, Millea & Pan (2015)

Cosmic Neutrinos in the CMB
Extraction of the Multipole Shift Template

$$\delta\ell(N_{\rm eff}) = A(N_{\rm eff})\, f_\ell$$

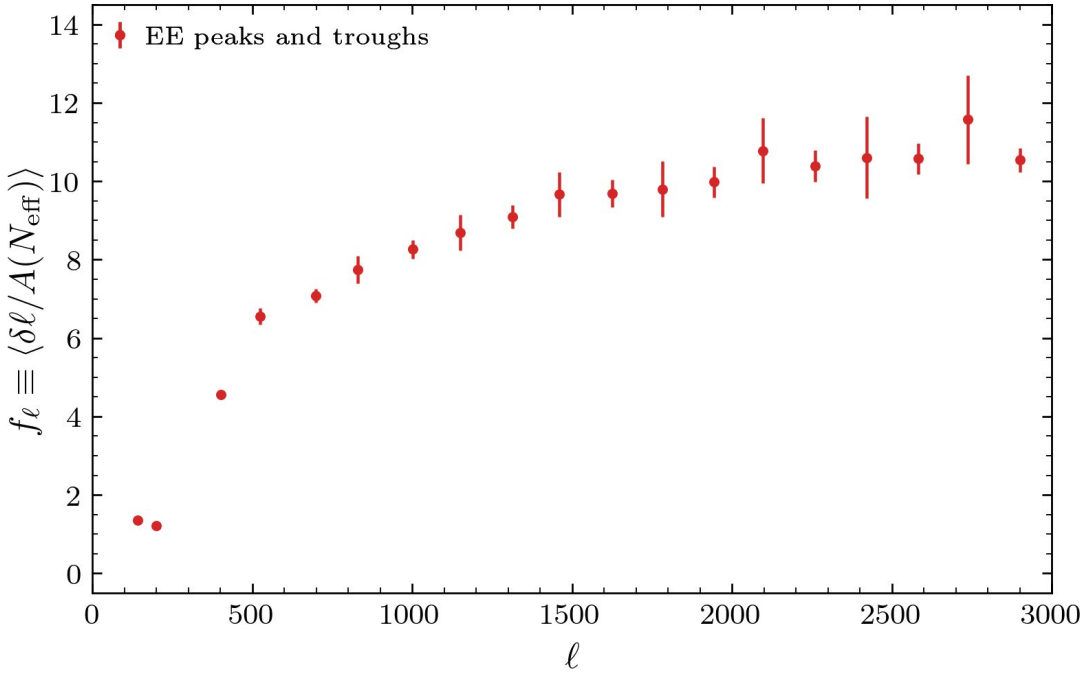
Multipole Dependence

$$f_\ell = \frac{\ell_\infty}{1 + (\ell/\ell_\star)^\xi}$$

Normalized Amplitude

$$A(N_{\rm eff}) \equiv \frac{\epsilon(N_{\rm eff}) - \epsilon(3.044)}{\epsilon(1) - \epsilon(3.044)}$$

3. Compute the mean/standard deviation per peak/trough.



Following Follin, Knox, Millea & Pan (2015)

Cosmic Neutrinos in the CMB
Extraction of the Multipole Shift Template

$$\delta\ell(N_{\rm eff}) = A(N_{\rm eff})\, f_\ell$$

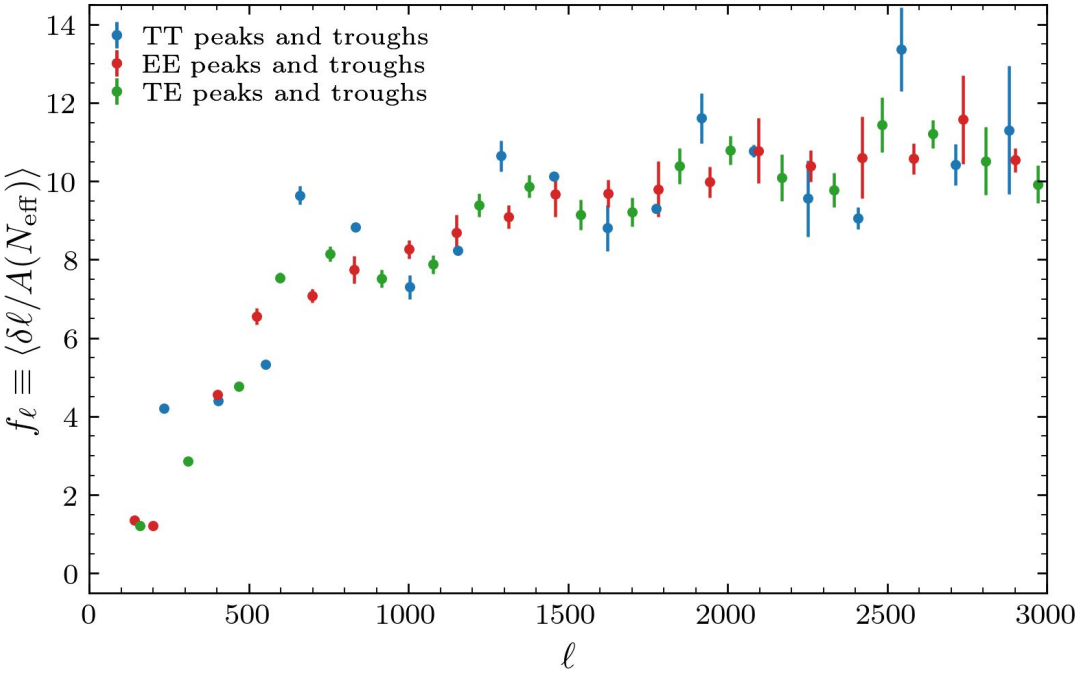
Multipole Dependence

$$f_\ell = \frac{\ell_\infty}{1 + (\ell/\ell_\star)^\xi}$$

Normalized Amplitude

$$A(N_{\rm eff}) \equiv \frac{\epsilon(N_{\rm eff}) - \epsilon(3.044)}{\epsilon(1) - \epsilon(3.044)}$$

4. Repeat the same process for the TT and TE spectra.



Following Follin, Knox, Millea & Pan (2015)

Cosmic Neutrinos in the CMB
Extraction of the Multipole Shift Template

$$\delta\ell(N_{\rm eff}) = A(N_{\rm eff})\, f_\ell$$

Multipole Dependence

$$f_\ell = \frac{\ell_\infty}{1 + (\ell/\ell_\star)^\xi}$$

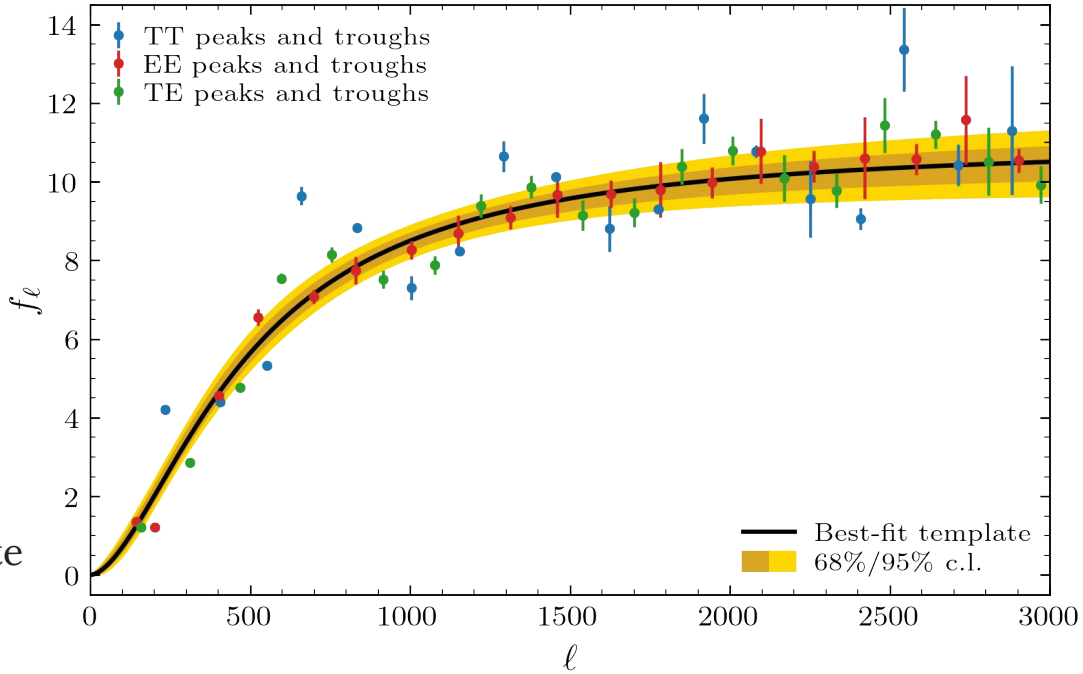
Normalized Amplitude

$$A(N_{\rm eff}) \equiv \frac{\epsilon(N_{\rm eff}) - \epsilon(3.044)}{\epsilon(1) - \epsilon(3.044)}$$

Our Contributions:

- A new analytic form of the template
- Used both temperature and polarization spectra

4. Determine the best-fit parameters



Following Follin, Knox, Millea & Pan (2015)

The Phase Shift in the Perturbations

- Extract the **phase shift** at the **perturbations** level, instead of the CMB power spectra
  - Cleaner signature: avoids projection and smearing effects

X Physical Source Term

$$S_X(k,\eta)$$

$$\begin{cases} S_T(k,\eta) = g_\gamma\,(\delta_\gamma/4 + \Psi) + \dots \\ S_E(k,\eta) = \sqrt{6}/2\,\cdot g_\gamma\,\Pi \end{cases}$$

X Anisotropies

$$\Delta_\ell^X = \int_0^{\eta_0} d\eta\, S_X(k,\eta)\, P_{X\ell}(k[\eta_0 - \eta])$$

Projection

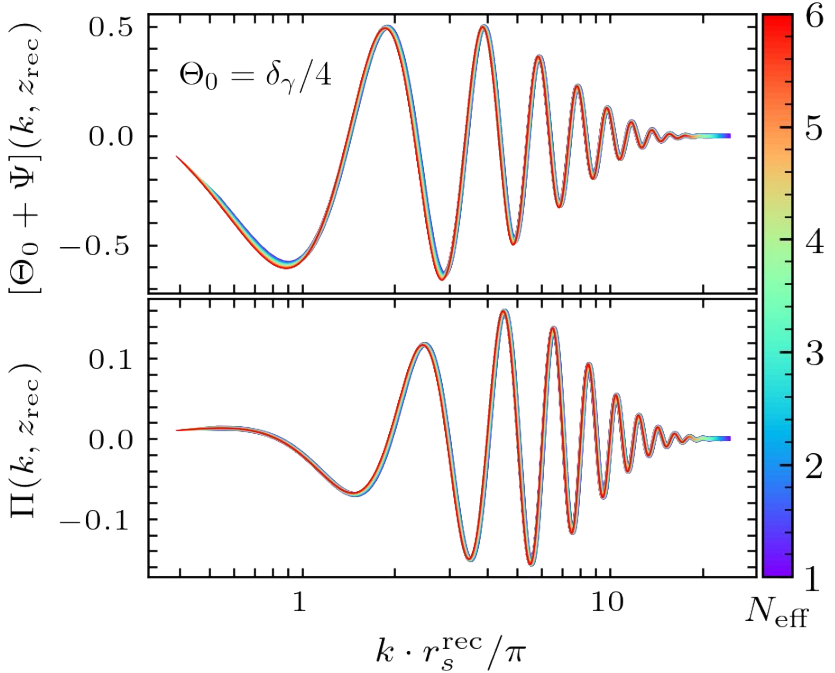
XY Power Spectra

$$\mathcal{C}_\ell^{XY} = \frac{2}{\pi}\int k^2 dk\, \Delta^2_{\mathcal{R}}(k)\, \Delta_\ell^X(k)\Delta_\ell^Y(k)$$

Primordial Spectrum

$$\delta_\gamma(\vec{k}) \approx A(\vec{k})\cos(kr_s + \phi)$$

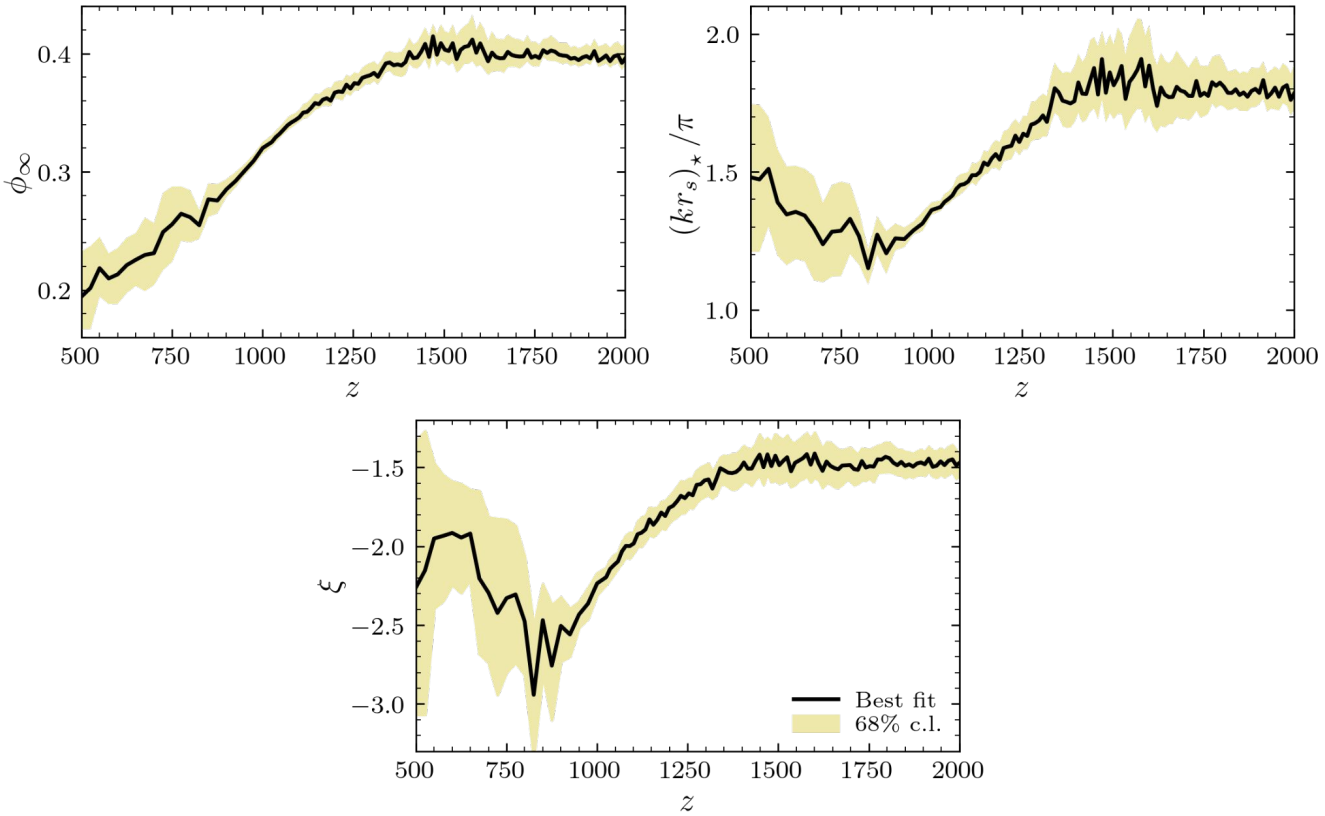
$$\Pi(\vec{k}) \sim A(\vec{k})\frac{c_s}{\dot{\tau}}\sin(kr_s + \phi)$$



GM, Wallisch & Freese (2025)

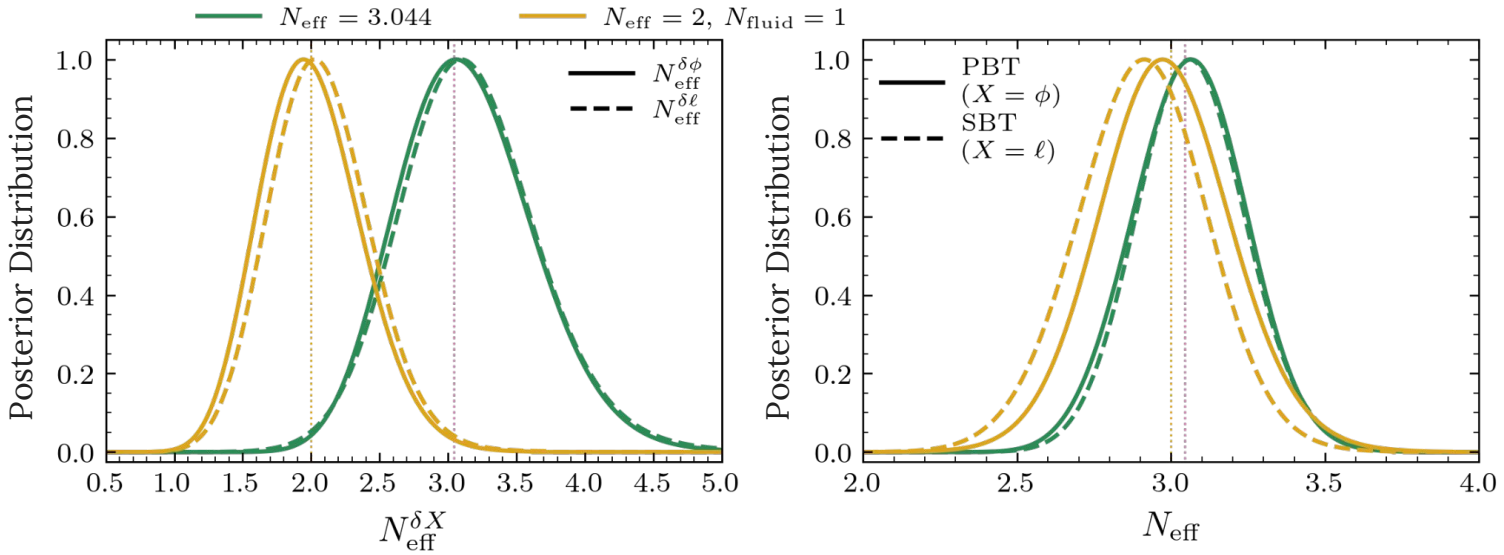
The Phase shift in the Perturbations

Details on the template extraction



Validation of Phase Shift Extraction
Warm Up with Planck-like Mock Data

- We recover the imprinted phase shift by standard model neutrinos.
- We can *robustly isolate* the induced phase shift by *free-streaming radiation*.

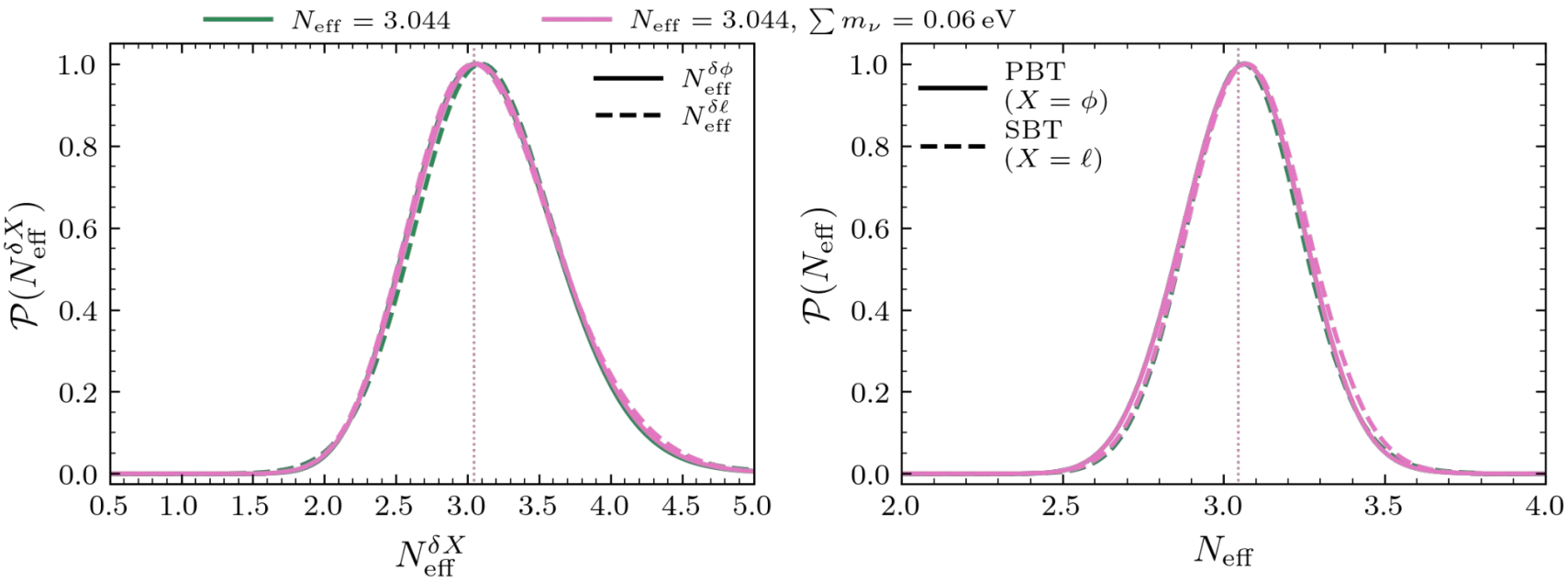


The spectrum and perturbation based approach yield consistent results

GM, Wallisch & Freese (2025)

Validation of Phase Shift Extraction

Planck-like Mock Data Analysis

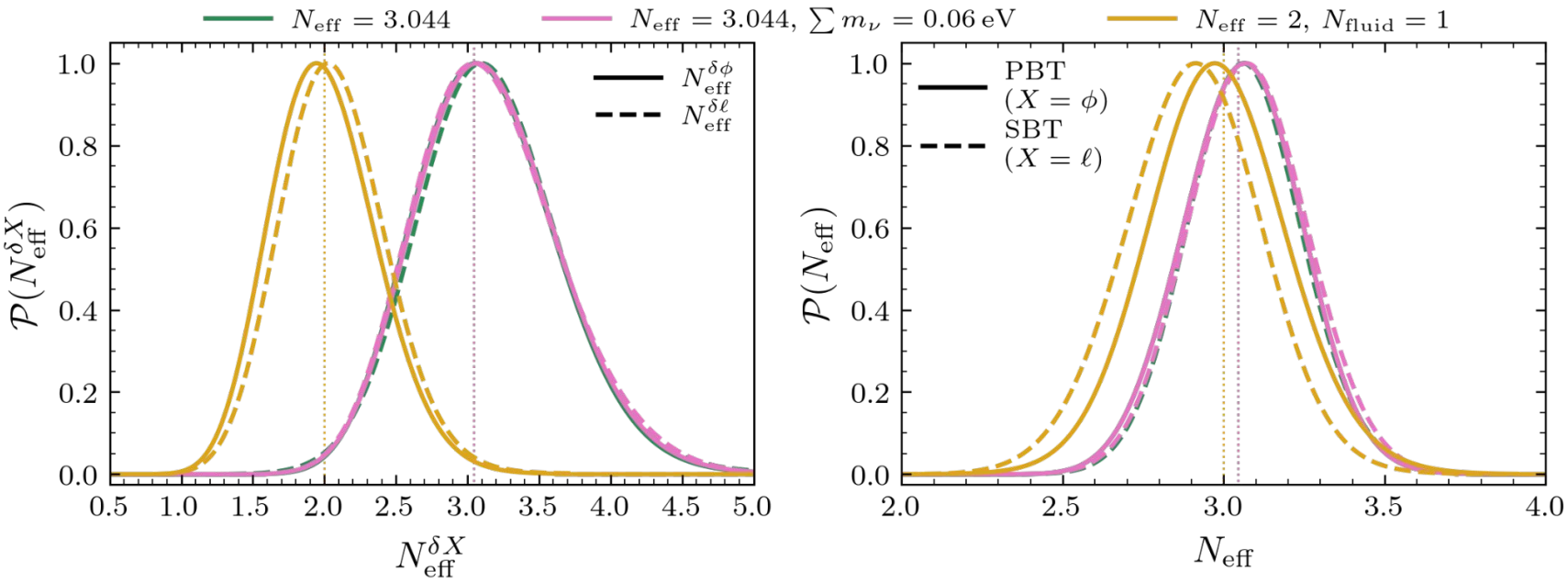


Neutrino masses have negligible impact on the phase shift extraction

GM, Wallisch & Freese (2025)

Validation of Phase Shift Extraction

Planck-like Mock Data Analysis



Both the spectrum and perturbation based approach successfully recover the imprinted phase shift by free-streaming radiation

GM, Wallisch & Freese (2025)

Phase Shift Constraints from CMB Data

Spectrum-Based Analysis

- Based on Planck 2013 temperature only:

$$N_{\rm eff}^{\delta\ell} = 2.3^{+1.1}_{-0.4}$$ Follin et al. (2015)

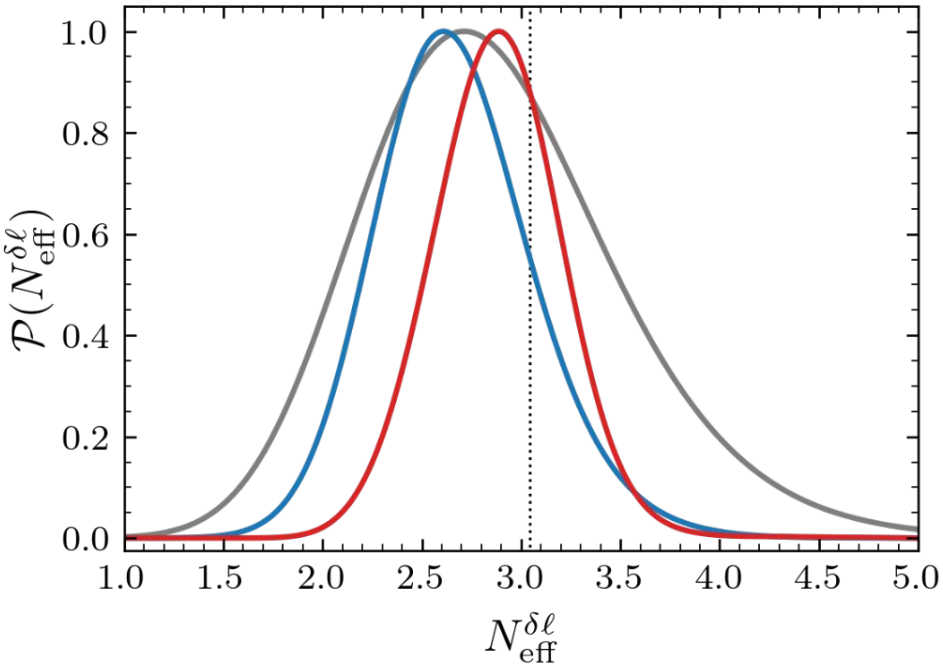
$$(N_{\rm eff} = N_\nu = 3.044)$$

- Current Data:
  - **Planck 2018 TT only:** $N_{\rm eff}^{\delta\ell} = 2.9^{+0.7}_{-0.6}$
  - Including polarization:

    Planck 2018: $N_{\rm eff}^{\delta\ell} = 2.7^{+0.5}_{-0.4}$

    + ACT + SPT: $N_{\rm eff}^{\delta\ell} = 2.9^{+0.3}_{-0.3}$

$$(N_{\rm eff} = N_\nu = 3.044)$$

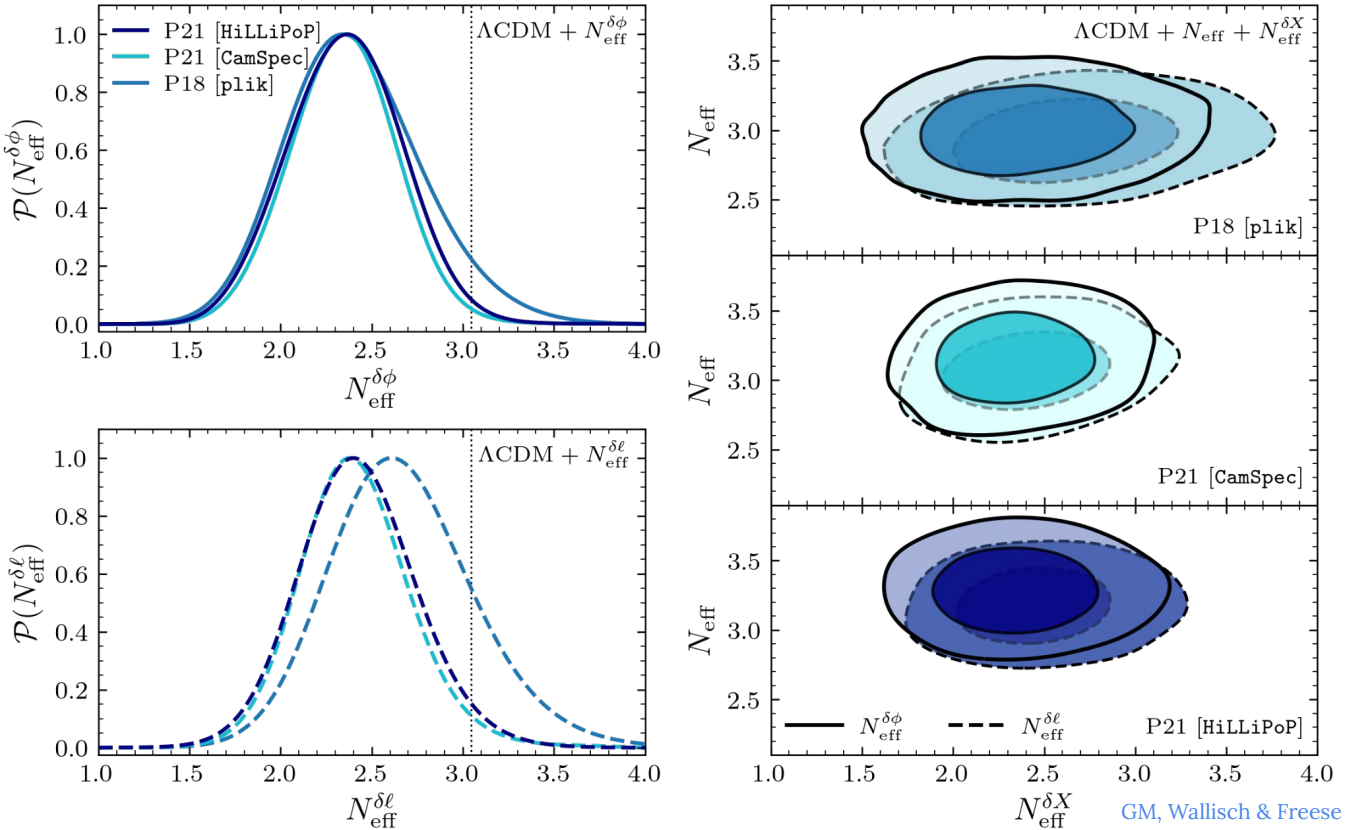


GM, Wallisch & Freese (2025)

Planck 2018 vs Planck 2021

Comparison of Spectrum- and Perturbation-Based Approach

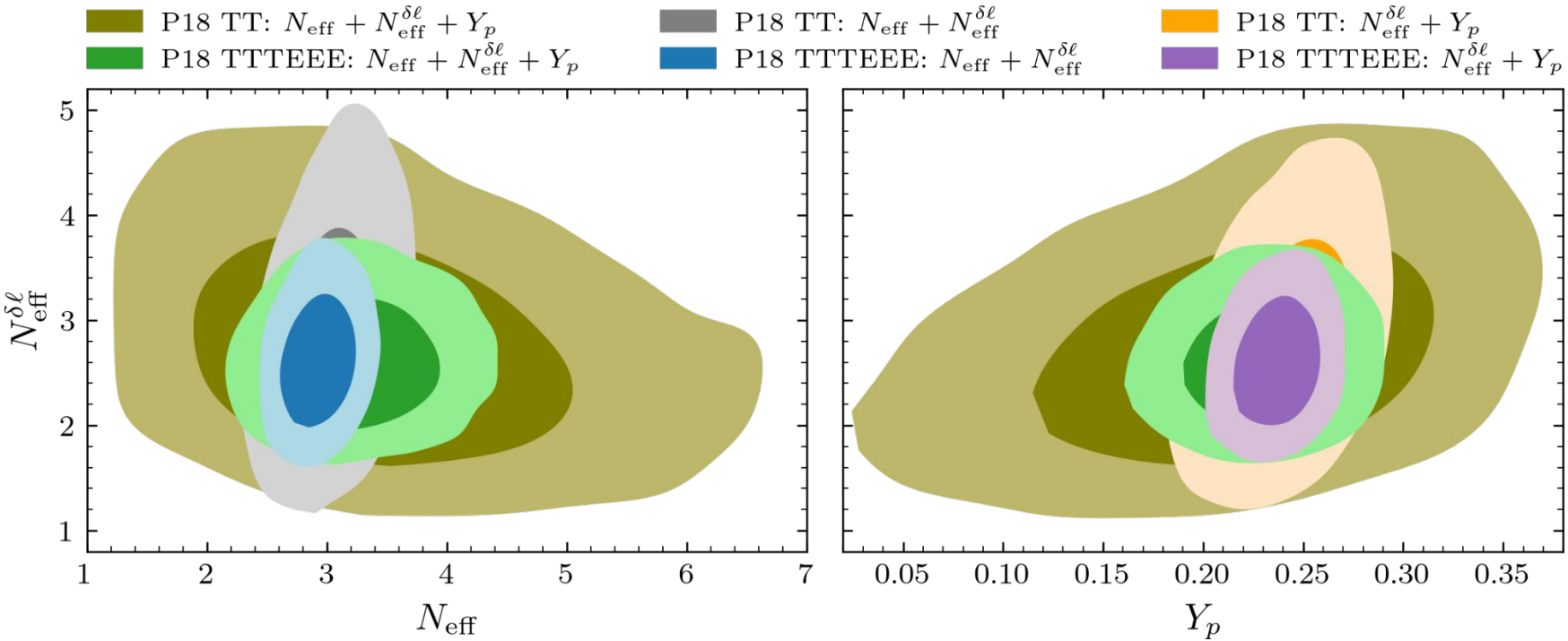
The **perturbation-based method** is a more **robust** approach to extract the phase shift



GM, Wallisch & Freese (2025)

Phase Shift Constraints from CMB Data

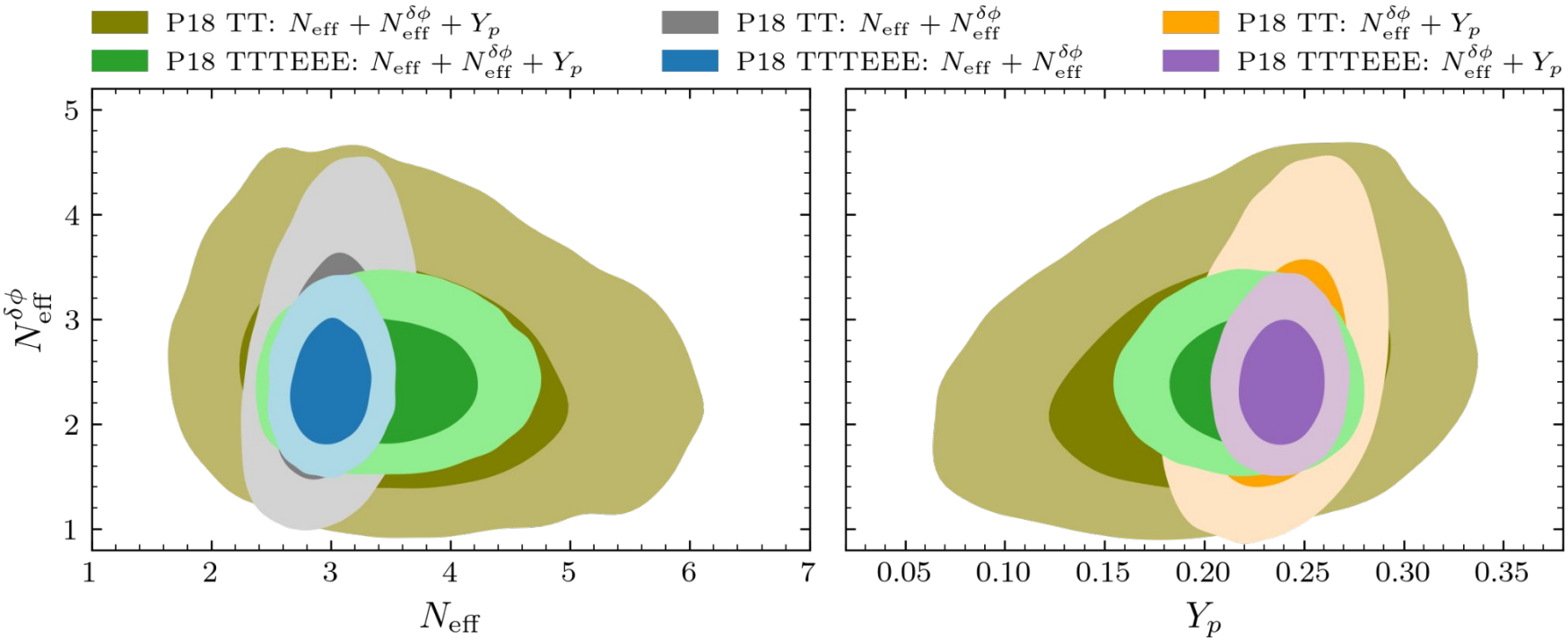
Spectrum-Based Analysis



GM, Wallisch & Freese (2025)

Phase Shift Constraints from CMB Data

Perturbation-Based Analysis



GM, Wallisch & Freese (2025)

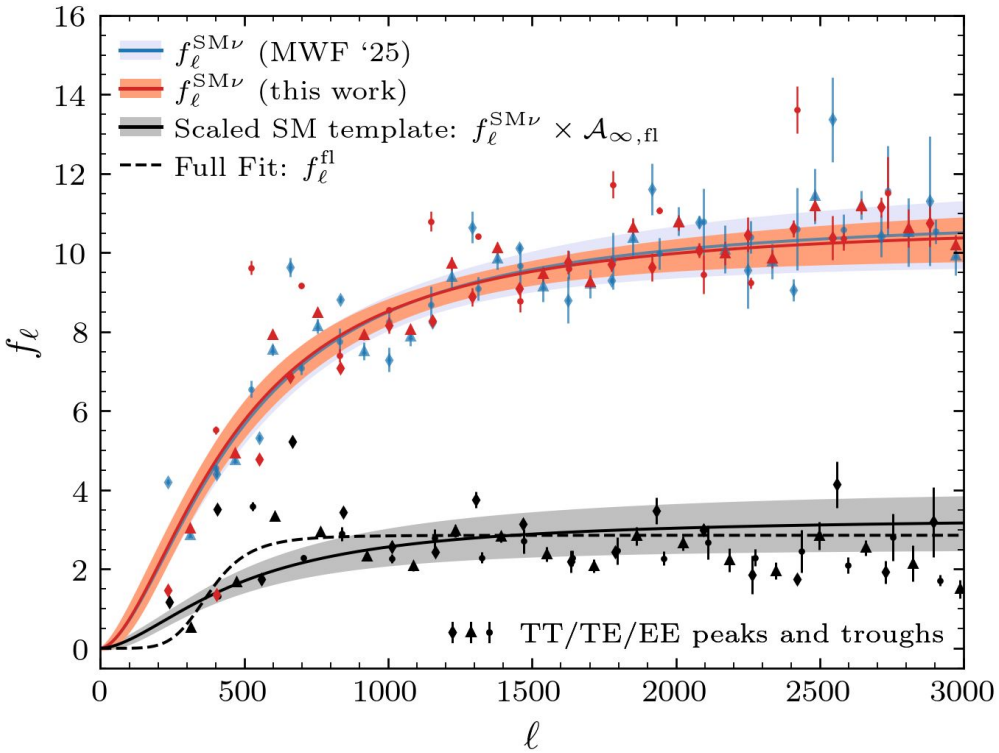
Phase Shift Forecasts

	$\Lambda\mathrm{CDM} + N_{\mathrm{eff}}$	$\Lambda\mathrm{CDM} + N_{\mathrm{eff}}^{\delta\phi}$	$\Lambda\mathrm{CDM} + N_{\mathrm{eff}} + N_{\mathrm{eff}}^{\delta\phi}$		$\Lambda\mathrm{CDM} + N_{\mathrm{eff}} + N_{\mathrm{eff}}^{\delta\phi} + Y_p$	
	$\sigma(N_{\mathrm{eff}})$	$\sigma(N_{\mathrm{eff}}^{\delta\phi})$	$\sigma(N_{\mathrm{eff}})$	$\sigma(N_{\mathrm{eff}}^{\delta\phi})$	$\sigma(N_{\mathrm{eff}})$	$\sigma(N_{\mathrm{eff}}^{\delta\phi})$
SO	0.054	0.13	0.054	0.14	0.29	0.14
CMB-S4	0.030	0.078	0.031	0.080	0.20	0.080
CVL	0.012	0.046	0.012	0.048	0.094	0.048

- Delensed (Fisher) forecasts.
- SO: Baseline configuration with 40% of the sky.
- CMB-S4: Restricted to S4-wide survey with noise curves for 62% of the sky .
- CVL: $f_{\mathrm{sky}} = 0.75$ and $\ell \leq 5000$

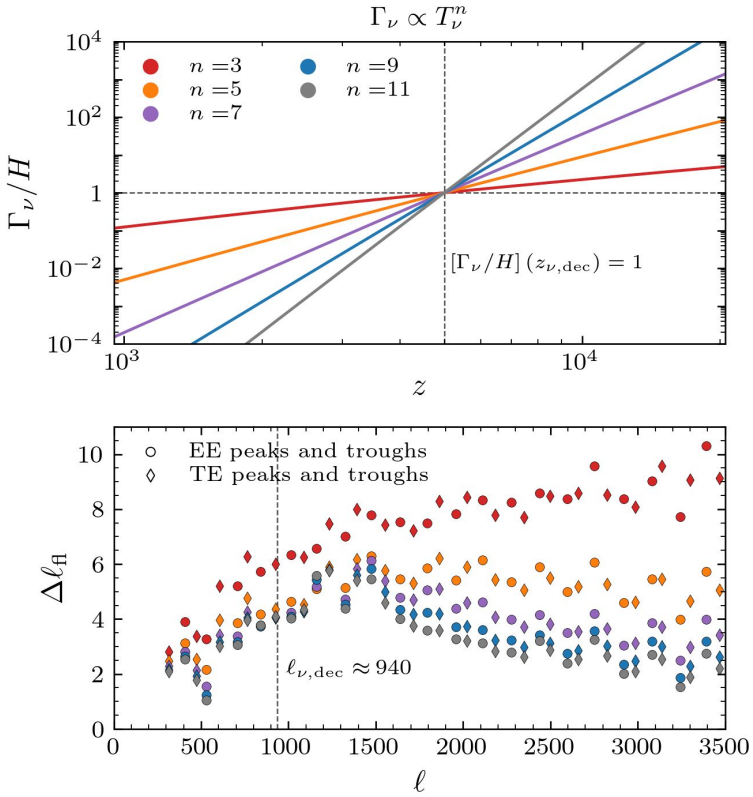
GM, Wallisch & Freese (2025)

Realistic Neutrino Interactions
Generalized Multipole Shift Template



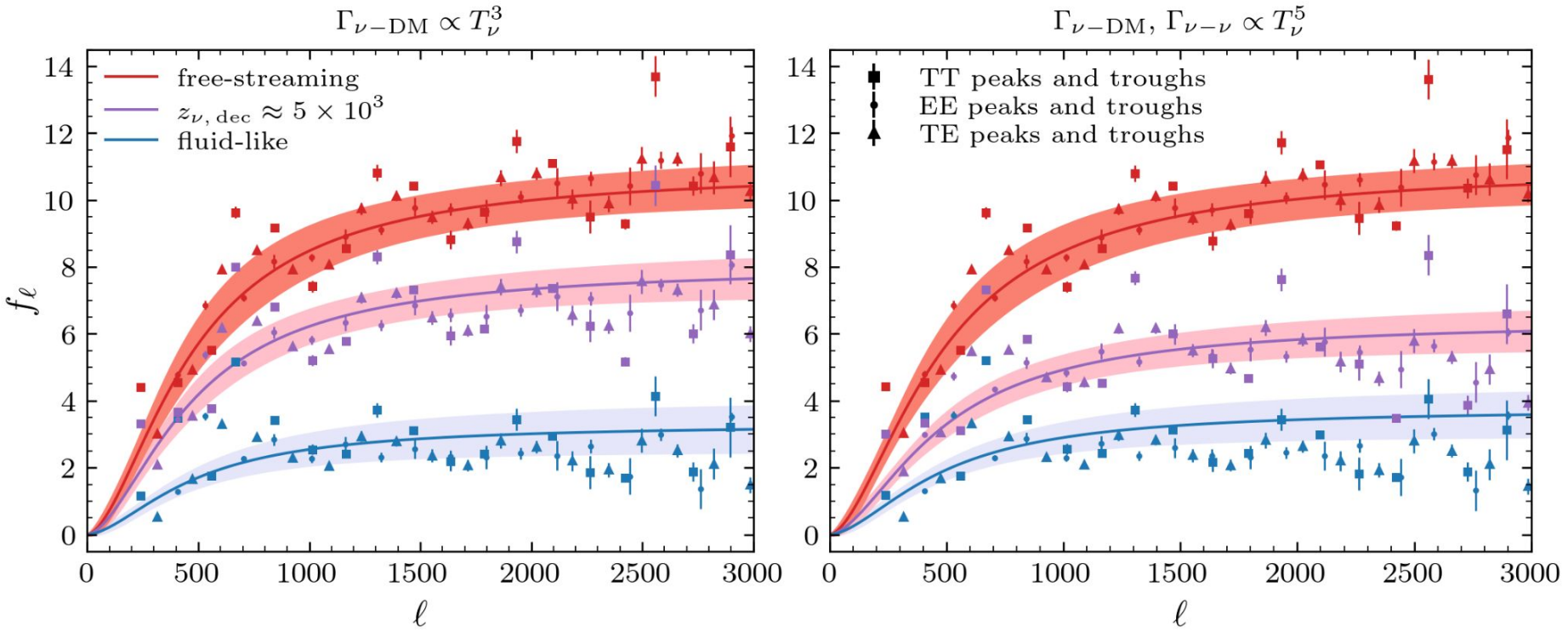
GM, Ghosh, Boddy, Ho & Tsai 2025

Realistic Neutrino Interactions
Generalized Multipole Shift Template



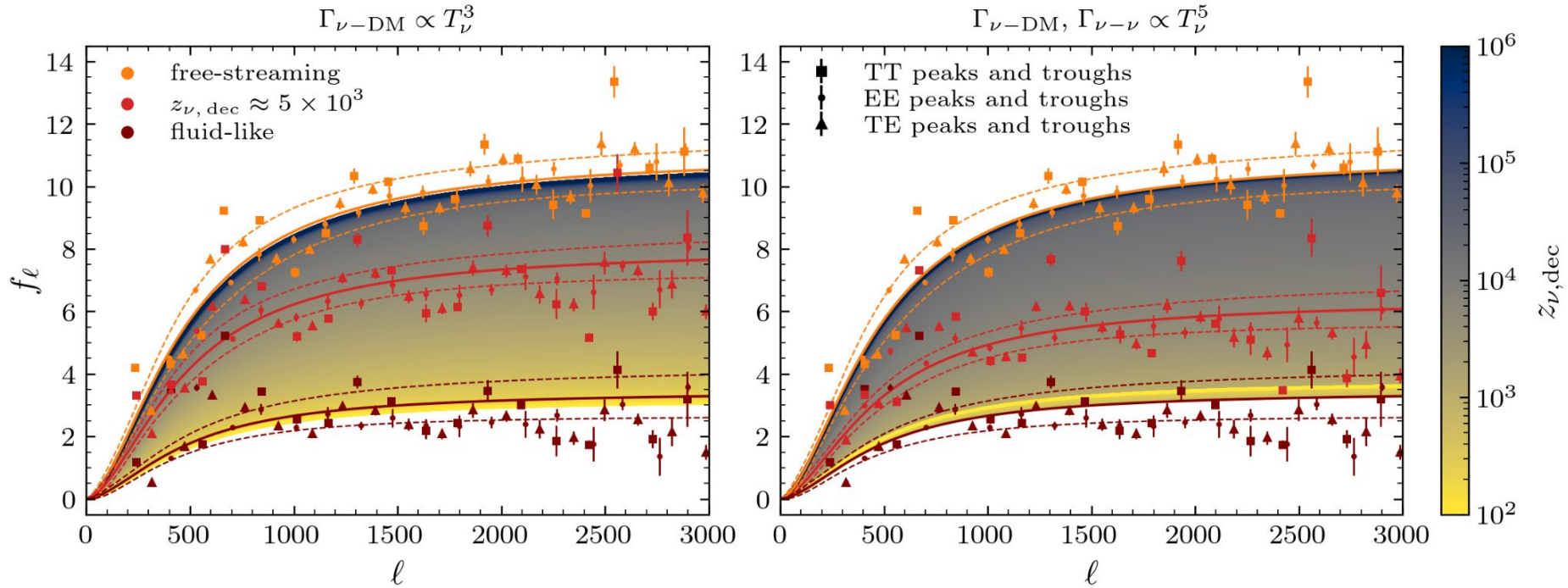
GM, Ghosh, Boddy, Ho & Tsai 2025

Realistic Neutrino Interactions
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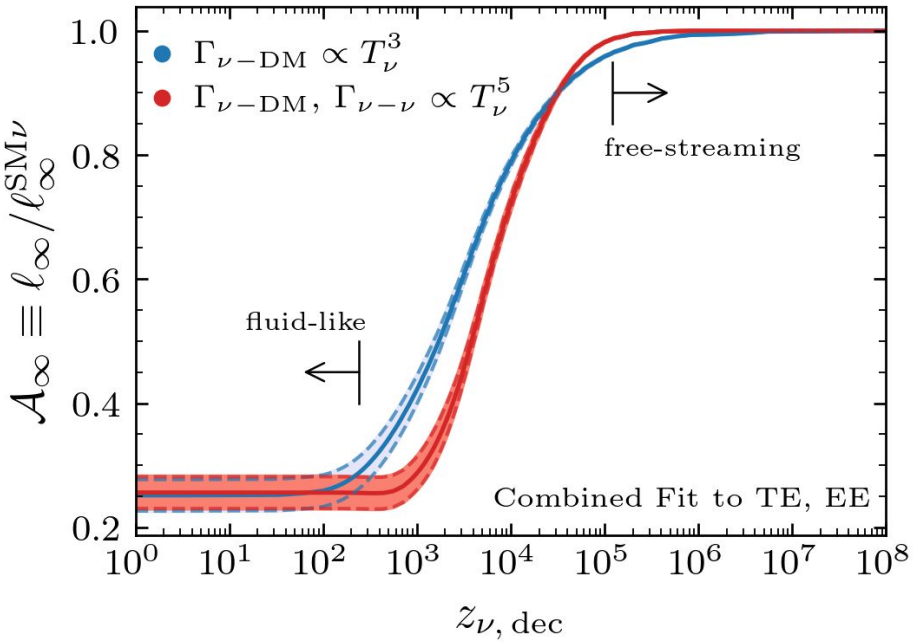
GM, Ghosh, Boddy, Ho & Tsai 2025

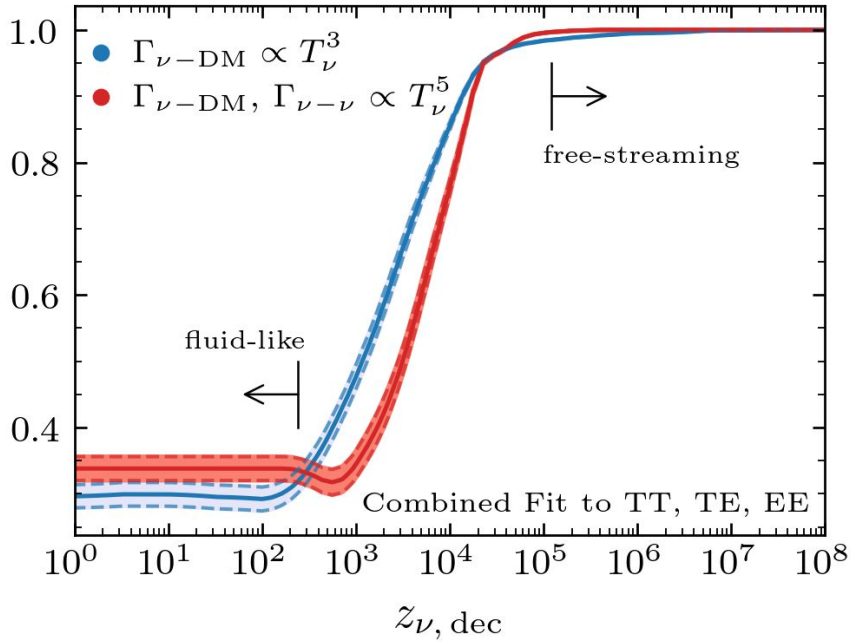
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GM, Ghosh, Boddy, Ho & Tsai 2025

Realistic Neutrino Interactions
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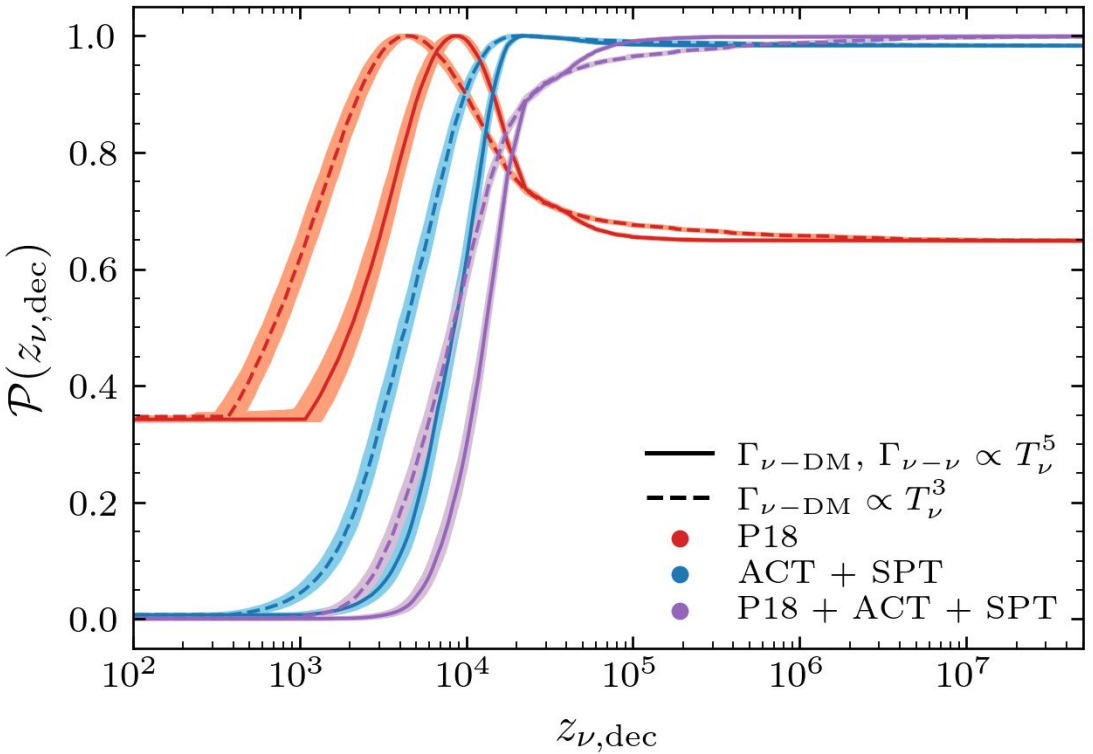




GM, Ghosh, Boddy, Ho & Tsai 2025

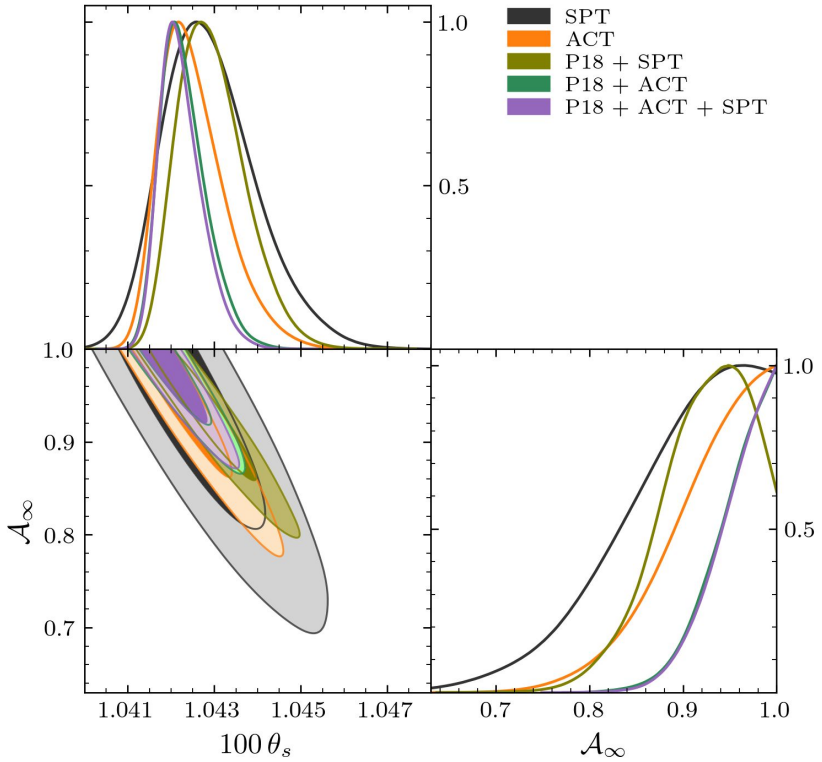
Realistic Neutrino Interactions
Direct Phase-Shift Constraints from CMB data

Only **one** interacting species



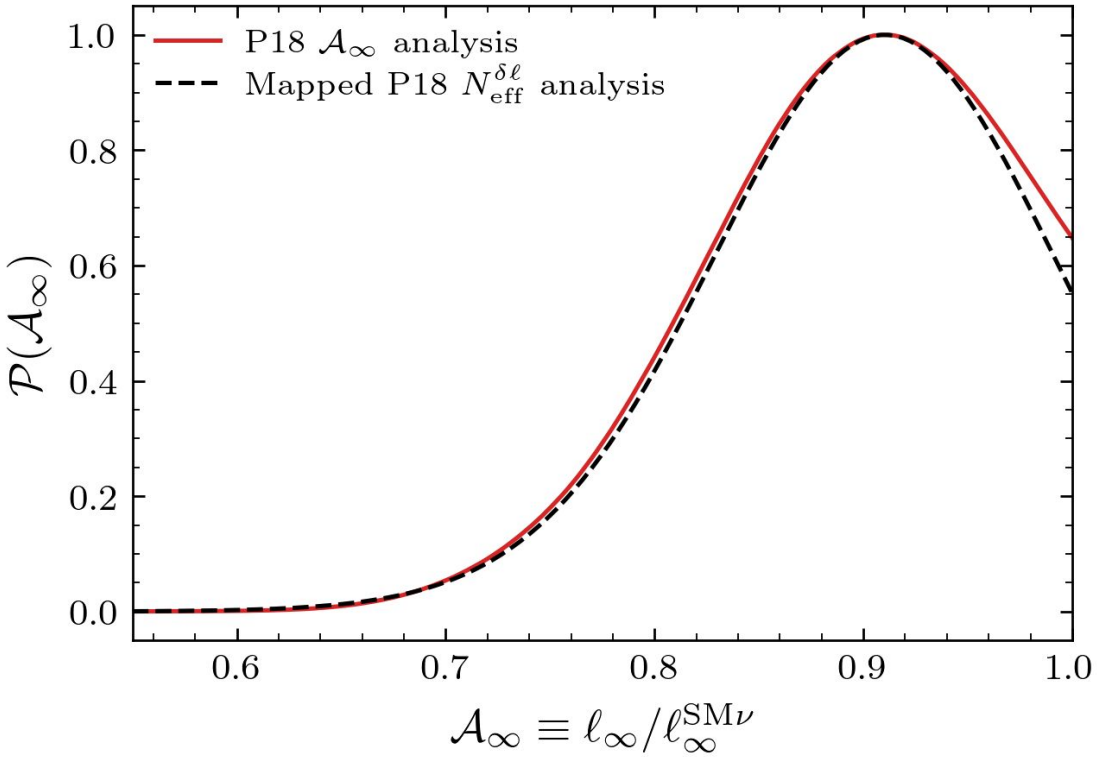
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Realistic Neutrino Interactions
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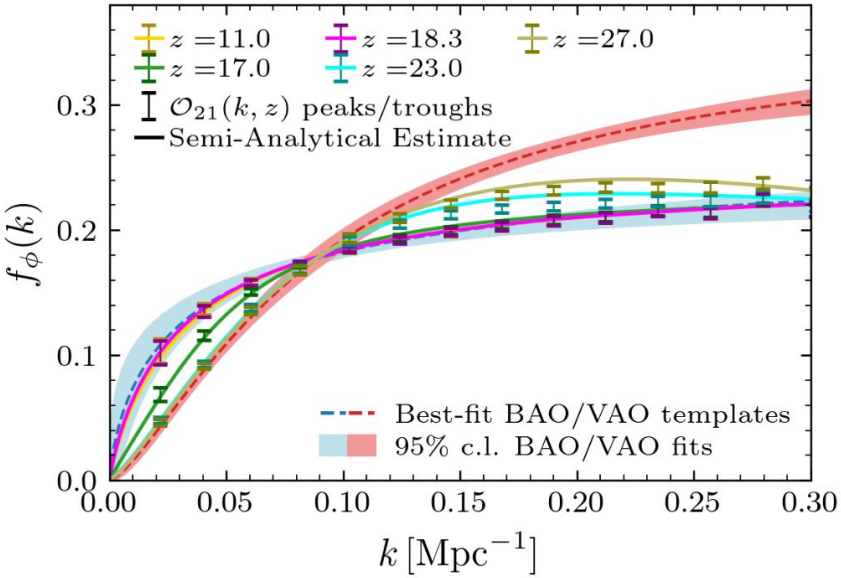
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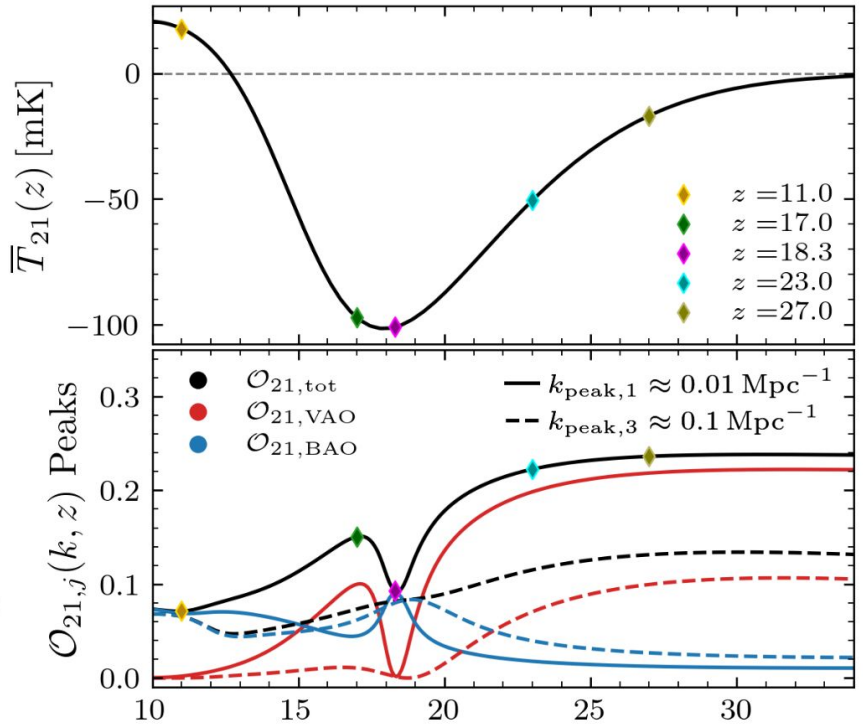


GM, Ghosh, Boddy, Ho & Tsai 2025

The Phase Shift in the 21-cm Spectrum

A distinct redshift- and mode-dependent Signature





GM, Cruz, Munoz, Kovetz, & Kamionkowski (2025)